

Spring Final Exam Review

Date _____ Period _____

Solve each equation. Remember to check for extraneous solutions.

1) $\sqrt{-4 - 2b} = \sqrt{1 - b}$
 $\{-5\}$

2) $5 = \sqrt{1 - 12b}$
 $\{-2\}$

3) $\sqrt{-40 + 14v} = v$
 $\{10, 4\}$

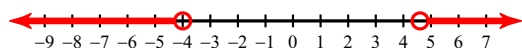
Solve each equation.

4) $|4 - 10x| = 0$ $\left\{\frac{2}{5}\right\}$

5) $|5a + 1| = 46$ $\left\{9, -\frac{47}{5}\right\}$

Solve each inequality and graph its solution.

6) $|3 - 10m| + 9 > 52$



$m < -4 \text{ or } m > \frac{23}{5}$

Simplify each and state the excluded values.

7) $\frac{10a - 60}{a^2 - 4a - 32}$
 $\frac{10(a - 6)}{(a - 8)(a + 4)}; \{8, -4\}$

Simplify each expression.

8) $\frac{x + 5}{x + 3} \div \frac{x^2 + 2x - 15}{x^2 - 8x + 15}$ $\frac{x - 5}{x + 3}$

9) $\frac{p^2 - 14p + 49}{3p} \cdot \frac{1}{p - 7}$ $\frac{p - 7}{3p}$

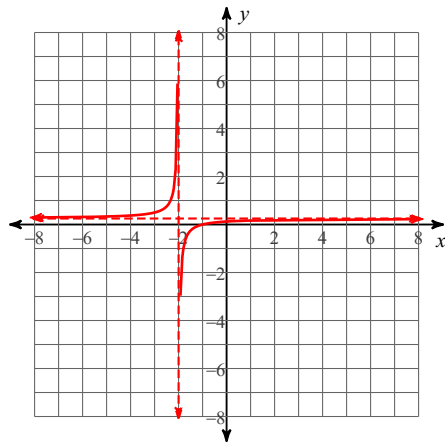
10) $\frac{x + 8}{5x - 5} - \frac{4}{7x}$
 $\frac{7x^2 + 36x + 20}{35x(x - 1)}$

11) $\frac{5}{2m^2} + \frac{m - 5}{7m + 8}$
 $\frac{35m + 40 + 2m^3 - 10m^2}{2m^2(7m + 8)}$

12) $\frac{\frac{3x - 1}{4}}{\frac{3x - 1}{x^2} + \frac{3x - 1}{4}}$ $\frac{x^2}{x^2 + 4}$

Identify the holes, vertical asymptotes, x-intercepts, horizontal asymptote, and domain of each. Then sketch the graph.

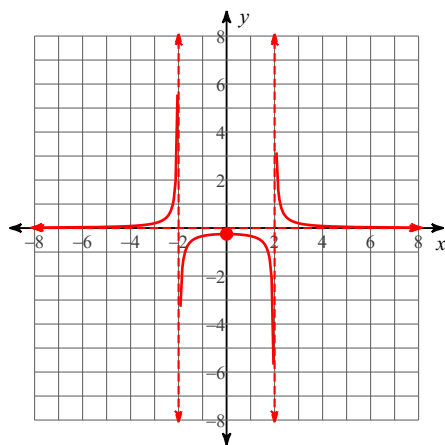
13) $f(x) = \frac{x+1}{4x+8}$



Vertical Asym.: $x = -2$
 Holes: None
 Horz. Asym.: $y = \frac{1}{4}$
 X-intercepts: -1
 Domain:
 All reals except -2

Identify the holes of each. Then sketch the graph.

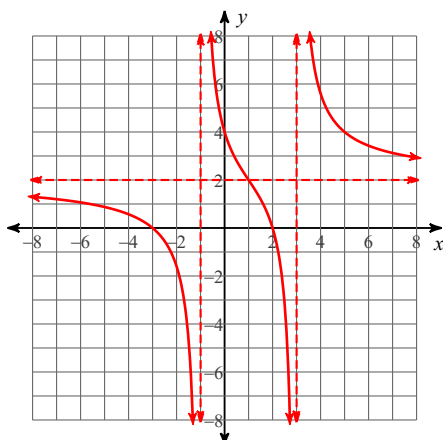
14) $f(x) = \frac{x}{x^3 - 4x}$



Holes: $x = 0$

Graph each function.

$$15) f(x) = \frac{2x^2 + 2x - 12}{x^2 - 2x - 3}$$



Simplify.

$$16) (n^8)^{-\frac{3}{4}} \frac{1}{n^6}$$

Simplify. Your answer should contain only positive exponents with no fractional exponents in the denominator.

$$17) 4x \cdot 2x^{\frac{1}{3}} y^{\frac{1}{3}} 8x^{\frac{4}{3}} y^{\frac{1}{3}}$$

$$18) u^{\frac{1}{2}} v^{-\frac{3}{2}} \cdot 3u \frac{3v^{\frac{1}{2}} u^{\frac{3}{2}}}{v^2}$$

Write each expression in radical form.

$$19) 10^{\frac{6}{5}} \\ (\sqrt[5]{10})^6$$

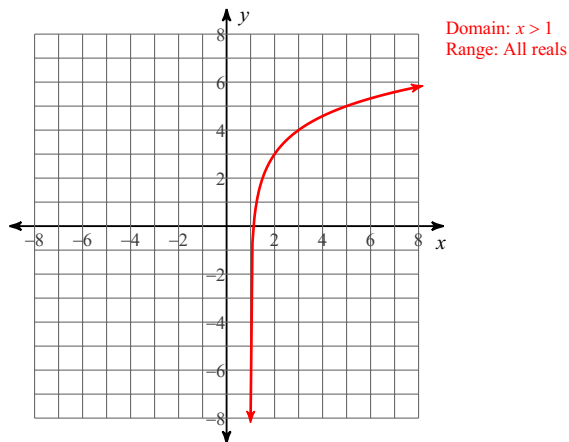
Find the inverse of each function.

$$20) y = 10^x - 10 \\ y = \log(x + 10)$$

$$21) y = \log_3(-4x) \\ y = -\frac{3^x}{4}$$

Identify the domain and range of each. Then sketch the graph.

22) $y = \log_2 (x - 1) + 3$



Identify the domain and range of each.

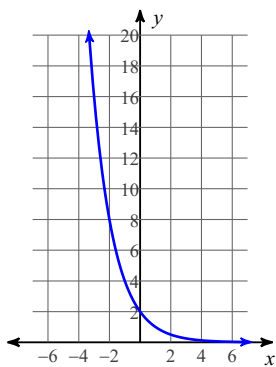
23) $y = \ln (x - 2) + 2$

Domain: $x > 2$
Range: All reals

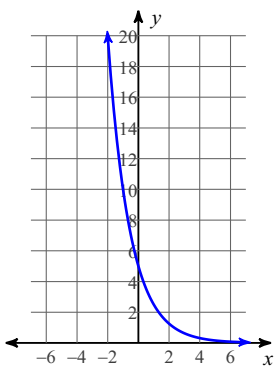
Sketch the graph of each function.

24) $y = \frac{1}{3} \cdot \left(\frac{1}{7}\right)^x$

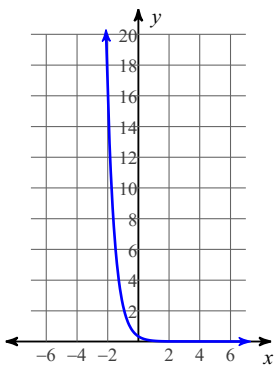
A)



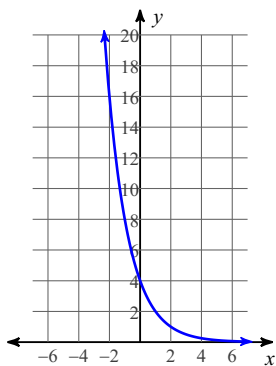
B)



*C)



D)



Solve each equation.

25) $\log_5 10 + \log_5 (x + 4) = 2$ $\left\{-\frac{3}{2}\right\}$

26) $\log_7 4 - \log_7 4x = 1$ $\left\{\frac{1}{7}\right\}$

$$27) \log_5 3 - \log_5 (x - 10) = 1 \quad \left\{ \frac{53}{5} \right\}$$

$$28) \log_2 (x - 4) - \log_2 7 = 3$$

$$\{60\}$$

$$29) 2^{2v} = 2^{-v}$$

$$\{0\}$$

Solve each equation. Round your answers to the nearest ten-thousandth.

$$30) 3^{k+9} - 7 = 84$$

$$-4.894$$

Evaluate each expression.

$$31) \log_4 \frac{1}{16}$$

$$-2$$

$$32) \text{ If you deposit \$5,000 into a bank account which earns 4.25\% compounded continuously, how long will it take for your money to triple?}$$

$$25.85 \text{ years}$$

$$33) \text{ If your boat depreciates at a rate of 5\% each year, how much is it worth in 6 years if the initial value of your boat was \$16,000?}$$

$$\$11,761.47$$

Algebra 2: Themes for the Big Final Exam

Final will cover the whole year, focusing on the big main ideas.

Graphing:

Overall: x and y intercepts, fct vs relation, fct vs inverse, x , y and origin symmetries, types of roots (single, double, complex) for polynomials

Transformations: $y=a(f(b(x+c)))+d$, what does each letter do to the graph

Specific types: linear, conics, polynomials, exponential, logarithmic, sin/cos, **other trig for honors, rational for honors**

Functions:

domain vs range, meaning of each in formula, notation, fct vs relation, fct vs inverse, interpreting meaning in word problems, notation with and without calculator

Solve:

Linear, quadratic, rational, radical, polynomial degree 3+, exponential, logarithmic, trig, linear ineq, poly ineq, absolute value equation, absolute value inequality, system of equations, system of inequalities

Rules/laws: radical, exponent, logarithmic, trig (SOH-CAH-TOA, unit circle, special angles)

Word Problems: making equation to match situation, using given equation, interpreting results

Key skills:

Simplify, graph, use formula, make formula, solve, calculator

Algebra 2

Emphasis in C1 to C3:

- Exponentials, Rational exponents
- Factoring: general, including $A^2 - B^2$ formula
- Quadratic equations: factoring and quad formula, including emphasis on substitution
- Complex numbers: add, subtract, multiply, divide
- Add, subtract, multiply, divide inequalities
- Graphing by plotting points
- Equation of a circle
- Symmetry along x, y origin axes
- Graphing by calculator
- Definition of function: domain, range, vertical line test
- Transformations of functions, reflection
- One to one functions and their inverses: including horizontal line test

Algebra 2: Year in Review

Solving:

The first step to solving an equation is to recognize the general structure of the equation. Is there anything in common that can be pulled out (reverse distribute)? Is it linear? Is it quadratic? A higher order polynomial? A key idea is to see an expression as the variable and not just the x .

Examples: $e^{2x} - e^x - 6 = 0$ is quadratic in e^x . $4\sin x = 2$ is linear in $\sin x$. $4+3|x-6| = 22$ is linear in $|x-6|$.

Linear equations you solve by isolating the variable expression. Basic equations (see 1-1) also follow this approach. Quadratic equations you solve by factoring or using the quadratic equation (see 1-3 with factoring reviewed in P6 or the cookbook). Higher order polynomials (degree 3 or higher) rarely have a variable expression involved, just an x . We solve them by factoring by grouping (p. 46 or p. 340 Ex 1) or testing/ using rational roots with synthetic division (see 4-3 and 4-4).

There are also some special cases like fractions where you need to multiply through by the LCD to remove all the denominators (section 1-1) and square roots (section 1-5) where you square both sides. In both of these cases, you are multiplying both sides of the equation by a variable so you need to check your answers in the original to make sure they are not **extraneous solutions**. (p. 112-113) You never want to divide by a variable (this will lose solutions) but factor out what is common instead.

Once the variable expression is isolated, you can solve for x by either using the definition of the expression ($|x-2|=6$ means $x-2 = 6$ or -6 or $\sin x = \frac{1}{2}$ means $x = \frac{\pi}{3}$ or $\frac{2\pi}{3}$) or rewriting the expression using its inverse ($\log_2 x = 4$ becomes $2^4 = x$ or $\sqrt{x} = 6$ becomes $x = 6^2$).

We also solved two equations with 2 variables (you will not see 3 equations with 3 variables on the final). Your options for this are substitution, elimination or graphing.

If you can solve an equation, you can also solve the corresponding inequality. The answers to $f(x) = 0$ gives us the places where $f(x)$ potentially changes sign from positive to negative or vice versa. We can then check what type of region each piece (all points between two zeroes is a piece) is (positive or negative) by plugging in one point from this region. We can then keep the regions that we want. (ex 3 on p. 122) We saw this in section 1-6 with polynomials and also in 1-7 with absolute values. We can also use the fact that we know what the graph of many functions look like to figure out our solutions. For example, $(x-4)(x+5) > 0$ will have a solution outside the x intercepts because this is a parabola opening up. A common mistake is to treat an inequality polynomial problem exactly the same as a polynomial equation. Be able to use appropriate notation like interval notation, a number line or inequality format (section P-1, don't worry about union and intersection)

To graph an inequality with 2 variables (or a system of inequalities), you first graph the equality version and then shade on one side or the other of the graph. If it is a system, the final answer is the region that is shaded from all the graphs. You can figure out where to shade by testing points or thinking about the inequality.

Functions:

The idea of **functions** and working with functions was a major theme of this class. (chapter 3) A function is a "machine" that takes certain numbers (the domain) and turns them into another set of numbers (the range). If a machine has more than one output for any one input it is not a function but a **relation**. Functions can be given four ways (this is sometimes called the "rule of four"). A function can be given as

an equation, a table of values, a graph or given through words. Working from an equation may be the most common but the other three show up at times as well. The notation (way of writing) of functions is important to understand. $f(2) = 8$ means that an input of 2 produces an output of 8 using the rule called f . More complicated expressions are used also at times. For example: $f(g(-2)) = 7$ or $(f+g)(x) = 4$.

Understanding the connection between f and f^{-1} is important (section 3-7). The **inverse** (f^{-1}) of f , is just looking at the f rule in the opposite order. The input and outputs are switched. If $f(8) = -3$ then $f^{-1}(-3) = 8$. This switching of x and y (the input and output) causes the graphs to be reflection of each other over the line $y = x$. To find the formula for the inverse, you take the formula of f , switch the x and y , and then solve. Not all functions have an inverse. To have an inverse, a function must have each output coming from only one input. When looking at a graph of a function, we can check for it having an inverse by using the **horizontal line test**. In a similar way, we can check to see if the original is really a function by using the **vertical line test** on it. (p. 223). If you can understand that the horizontal line test is just the vertical line test with the domain and range of the original switched, you are doing great! (if not, you will survive)

The last function idea is the one of **transformations** (section 3-4). This idea can be applied to any type of function. The most general model is $y = a + bf(c(x - d))$ where a , b , c and d are constants (numbers) and f is the name of the function we are using ($f(x)$ could be a trig fct or exponential or log or ...) The changes outside the f effect the y or vertical part of the graph. (a and b). The changes inside the function effect the x or the horizontal part of the graph. Numbers that multiply (b and c) stretch or compress. Numbers that add/subtract (a or d) shift. Finally, the numbers inside the function with the x (c and d) effect things opposite of what you think they should while numbers outside (a and b) effect things as you expect. For example $y = f(x - 2)$ slides the f graph 2 to the right while $y = f(x) - 2$ slides the graph two down.

Graphing:

We graphed many different types of functions this year. Some general facts are finding the x and y intercepts, the general shape of each type and shifting the basic shape of the graphs using transformations. You may see problems asking to make the graph, give the equation of the graph, find out facts about the graph or give the equation/ graph matching certain information.

Linear- graph is a line, slope is an important characteristic (section 2-4), know standard, point-slope and slope-intercept forms

Quadratic – graph is a parabola, vertex and axis of symmetry are important, equation will look like $y = a(x - h)^2 + k$ or $y = ax^2 + bx + c$, be able to work with either one. Complete the square and $-\frac{b}{2a}$ are helpful tools (section 1-3)

Circles – find the center and radius and use them to graph (section 2-2)

Polynomial – degree 3 or higher, find roots and type (single/double), is leading term positive or negative, think about general shape (also called the end behavior) (section 4-1) and how complex roots effect things (section 4-4)

Exponential- general shape, passes through (0,1), D: $(-\infty, \infty)$ R: $(0, \infty)$ (section 5-1)

Logarithmic- general shape, passes through (1,0), D: $(0, \infty)$ R: $(-\infty, \infty)$ (section 5-2)

Sine/ Cosine – general shape, amplitude, average value, period, phase shift, connection between the “k” and the period (section 7-3)

Absolute Value – graph looks like a “V”, it mimics $y = x^2$, except it is straight and not curved

General things to know:

Rules of exponents/ radicals- See P3/ P4 or Night of Chapter 7 HW Worksheet (online)

Factoring- See P6 or factoring cookbook (online)

Complex numbers – key fact is “ i ” terms are only like terms with “ i ” terms, $i^2 = \sqrt{-1}$ so replace negatives under a square root with an i out front $i^{35} = (i^2)^{17} i = (-1)^{17} i = -i$, see 1-4 for more info

Log Facts –they work as exponentials in reverse (they are the inverse of exponentials) , see 5-3 and 5-4 for rules/properties, formula will be given in word problems, you will not have to make it

Trigonometry – understand radian locations on the unit circle (first quadrant will be given), know $x^2 + y^2 = 1$, know all 6 trig definitions in terms of x and y and each other (chapter 7)

Graphing calculator skills – use as tool to graph (input Y= correctly, set appropriate window), use to solve equations (using intercept or zero), use to solve inequalities (see where one is larger than the other on the graph), connect what you are seeing on the graph to what you are being asked for in a word problem

Word Problems – The two contexts in which you will have to make and solve a word problem completely (an equation is not given) are $D=RT$ and mixture problems. A good guide to doing word problems is understand (read,read, read, write down key facts, make pictures), translate (choose a variable, define it and get everything possible in terms of it), equation (make and solve the equation), reflect (is the answer reasonable, units, did you answer the question)

Honors ideas:

Some honors problems will just be harder versions of the above concepts. However there were a few ideas that were exclusively honors.

Graphing rational functions- vertical asymptotes, horizontal asymptotes, general shape, zeroes

Graphing Tan/Cot/Sec/Csc – how to adapt sine/cosine to get these

Partial fractions- the idea from 10-8

No Ch 12 or Ch 13 on the test. Test will be two-thirds second semester (Ch 4, 5, 6/7, 8, 10) and one-third the first semester.

Algebra 2 Practice Final Exam (2005 – 2006 2nd Semester)

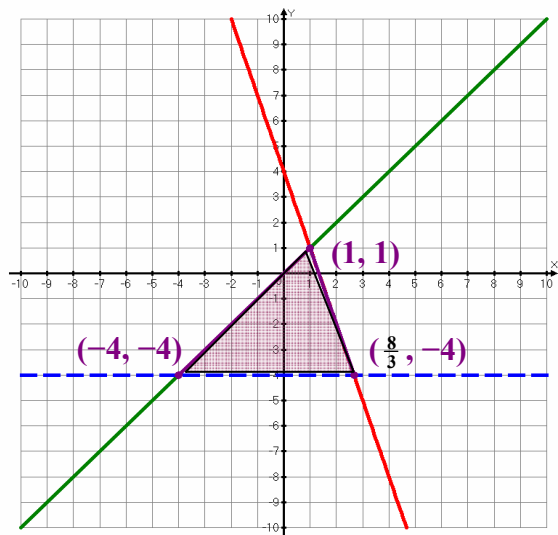
1. Graph the system, $\begin{cases} x - y \geq 0 \\ 3x + y \leq 4 \\ y > -4 \end{cases}$, and find its vertices:
2. Solve $\begin{cases} 2x + 3y = 4 \\ 5x + 4y = 3 \end{cases}$
3. If $f(x) = x^2 + 2$ and $g(x) = 1 - x$, find:
a. $f(3)$ b. $g(4x - 1)$ c. $f(g(-2))$ d. $f^{-1}(x)$
4. Simplify $\left(\frac{16x^4y^7}{9y^4}\right)\left(\frac{81y^2}{4x^5}\right)$ using positive exponents only.
5. Simplify the expression $\frac{4x^2 - 144}{2x^2 - 24x + 72} \times \frac{5x - 30}{10x^2 + 80x + 120}$
6. Simplify the following expressions:
a. $(2\sqrt{3} + 3\sqrt{27})^2$ b. $\sqrt{3} - \frac{1}{2\sqrt{3}}$ c. $\frac{2 + 3i}{4 - 5i}$
7. Solve for x for the following equations.
a. $3x - 5\sqrt{x} = 2$ b. $4x^2 = 81$ c. $(x - 5)^2 - 25 = 11$
8. Find all solutions to the equation, $x^4 - 3x^3 - 7x^2 + 27x - 18 = 0$.
9. Find the equation of the circle whose center is $(4, 4)$ and contains the origin on its path.
10. Solve the system of equations, $\begin{cases} 16x^2 + 4y^2 = 16 \\ 4x - y = -4 \end{cases}$
11. For the points $(-7, 5)$ and $(-9, -21)$, find the midpoint and distance between the two.
12. Solve for x . Give exact and approximate answers (if applicable):
a. $\log x = 0.8531$ b. $25^{(x+2)} = 5^{(3x-3)}$ c. $10^{5x} = 2$
13. Over the past twenty years, the city of San Jose has grown by an average of 1.4% per year, compounded yearly.
a. If the population in 1995 was 980,000, what is the estimated population in 2010?
b. Estimate the year the population was 500,000.
14. Coach Bookin can line the soccer field by himself faster than Coach D can. In fact, he can complete the job by himself ten minutes faster than Coach D. If they work together, they can finish the job in 20 minutes. If Coach Bookin breaks something and can't help, how long would it take Coach D to do the job himself?
15. A plane whose air speed is 200 mph flew from San Francisco to San Diego in 2 hours with a tail wind. On the return trip, against the same wind, the plane was still 50 miles away from San Francisco after 2 hours of flying. Find the wind speed and the distance from San Francisco to San Diego.

16. Sketch $y = -3\sin(2x)$ in the range $-\pi < x < \pi$.

17. Sketch $y = -5\cos[2(x - \pi)] - 4$ over two periods.

Answers to Algebra 2 Practice Final Exam (2005 – 2006 2nd Semester):

1.



2. $(-1, 2)$

3. a. 11 b. $-4x + 2$ c. 11 d. $\sqrt{x-2}$

4. $\frac{36y^5}{x}$ 5. $\frac{1}{(x+2)}$

6. a. 363 b. $\frac{5\sqrt{3}}{6}$ c. $\frac{-7+22i}{41}$

7. a. 4 (only answer that verifies)

b. $\pm \frac{9}{2}$ c. 11 and -1

8. Using synthetic division: $x = 1, 2, 3$ and -3

9. $(x-4)^2 + (y-4)^2 = 32$

10. $(-1, 0)$ and $(-\frac{3}{5}, \frac{8}{5})$

11. $M = (-8, -8)$; $d = 2\sqrt{170}$

12. a. 7.13

b. 7

c. 0.0602

13. a. 1,207,244

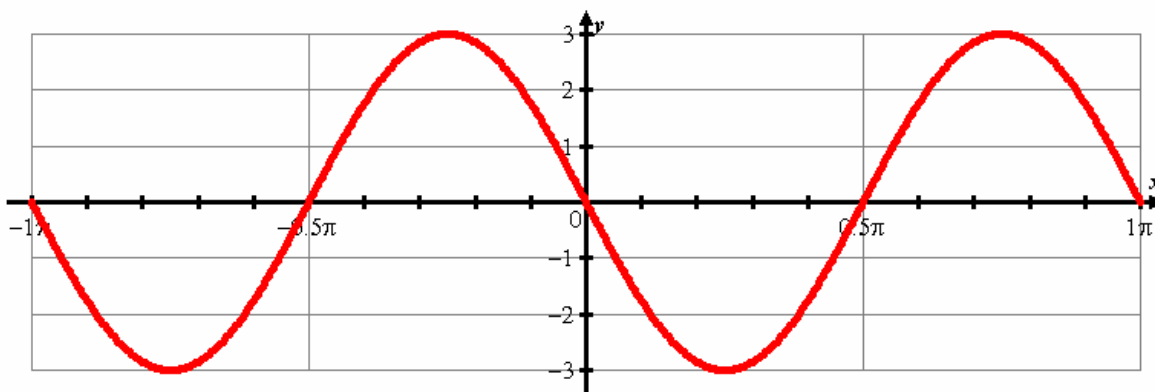
b. 1947

14. 45.62 min

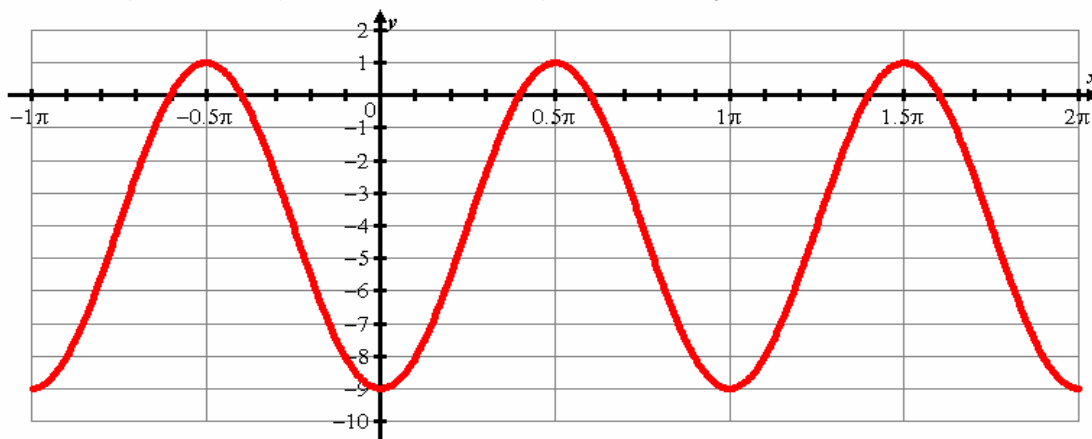
15. wind speed = 12.5 miles/hr; $d = 400$ miles

$(1, 1)$

16. Amplitude = 3; Period = π ; No Phase Shift; No Vertical Displacement



17. Amplitude = 5; Period = π ; Phase Shift = π left; Vertical Displacement = -4



1. Simplify the expression.

$$\left(\frac{a^2 b^{5/3}}{a^{1/3} b^{2/3}} \right)^6$$

2. Find all real solutions of the equation.

$$\frac{x+2}{x-2} = \frac{3x}{3x-6}$$

3. Two points P and Q are given.

$$P(0, -8), \quad Q(-11, -8)$$

- (a) Find the distance from P to Q .
 (b) Find the midpoint of the line segment PQ .

4. Find an equation for the line that passes through the point $(5, 1)$ and is perpendicular to the line $x - 3y + 16 = 0$.

5. In a certain city, the property tax collected for a home is varies directly to the valuation of the property. The tax collected on a \$105,000 home is \$2,846 per year. What is the value of a home if the tax collected is \$1,735 ?

6. Find the domain of the function.

$$f(x) = 0.5x - \frac{2}{\sqrt{x+1}}$$

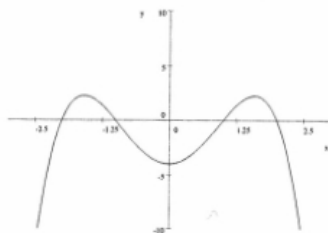
7. Find the maximum or minimum value of the function $f(x) = x^2 - 4x + 2$.

8. Evaluate $f(-2)$, $f(-1)$, $f(0)$, $f(1)$, and $f(5)$ for the piecewise-defined function.

$$f(x) = \begin{cases} 3x^2 & \text{if } x < 0 \\ 2x+1 & \text{if } x \geq 0 \end{cases}$$

9. Use $f(x) = 3x - 2$ and $g(x) = 3 + 2x^2$ to evaluate the expression $(f \circ g)(2)$.

10. Determine whether the function in the figure is even, odd, or neither.



Stewart/Redlin/Watson - Algebra and Trigonometry 2e Final Exam Form A

11. Find the inverse of the function.

$$f(x) = \sqrt{25 - x^2}, \quad 0 \leq x \leq 5$$

12. Use a graphing device to graph the polynomial $y = 2x^5 + 3x^2 - 2x + 1$ and ~~describe its end behavior.~~ *find its real zeroes*

13. Find the quotient and remainder.

$$\frac{x^3 + 3x - 2}{x - 1}$$

14. Use the Factor Theorem to show that $(x - 4)$ is a factor of the polynomial.

$$P(x) = x^5 + x^4 - 36x^3 - 16x^2 + 320x$$

15. Find all rational, irrational, and complex zeros (and state their multiplicities).

$$P(x) = x^4 + 10x^2 - 2x^3 - 18x + 9$$

16. Graph the rational function. Show clearly all x - and y -intercepts and asymptotes.

$$y = \frac{2x + 4}{x - 2}$$

HONORS ONLY

17. Sketch the graph of the function. State the domain, range, and asymptote.

$$f(x) = 2 + \log_2(x + 4)$$

18. Expand the logarithmic expression.

$$\log \left(\frac{4x^3}{y(x-2)^6} \right)$$

19. Find the solution of the equation $3^{x+2} = 5^{2x}$, correct to four decimal places.

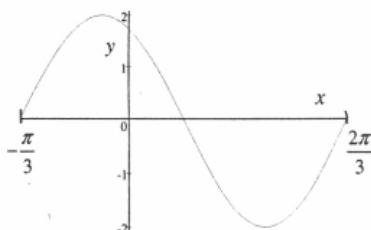
20. A culture contains 680 bacteria. After just 30 minutes the count is 3250. Find the number of bacteria after 3.5 hours. After how many minutes will the number of bacteria triple?

21. Suppose that the terminal point determined by t is the point $\left(\frac{5}{13}, \frac{12}{13}\right)$ on the unit circle. Find the terminal point determined by $\pi + t$.

22. If the terminal point determined by t is $\left(\frac{12}{13}, -\frac{5}{13}\right)$, find $\sin t$, $\cos t$, and $\tan t$.

23. If $\cos t = \frac{3}{5}$ and the terminal point for t is in quadrant IV, find $\tan t + \sec t$.

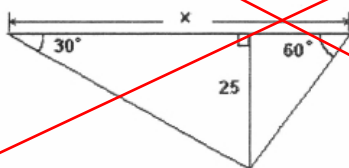
24. The graph shown is one period of a function of the form $y = a \sin k(x - b)$. Determine the function.



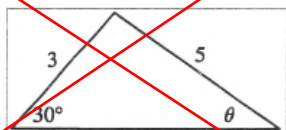
25. Find the period of the function $y = \frac{1}{2} \tan\left(x - \frac{\pi}{3}\right)$ and sketch its graph.

26. Find the measure of the central angle θ in a circle of radius 6 ft if the angle is subtended by an arc of length 7 ft.

27. Solve for x correct to one decimal place.



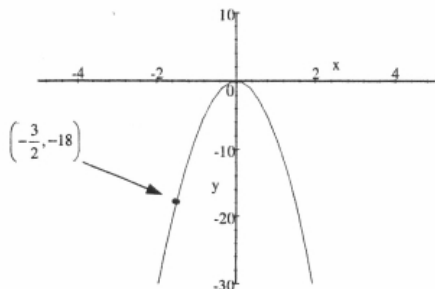
28. Find the indicated angle θ .



41. Find the partial fraction decomposition of $\frac{x+15}{x^2+3x}$.

HONORS

42. Find an equation for the conic whose graph is shown.



43. Sketch the graph of the ellipse.

$$\frac{(x-3)^2}{9} + \frac{y^2}{36} = 1$$

Cumulative Review Exercises

1. Subtract: $\frac{4x^2}{x^2 + x - 2} - \frac{3x - 2}{x + 2}$

3. Multiply: $\sqrt{2y}(\sqrt{8xy} - \sqrt{y})$

5. Solve: $5 - \sqrt{x} = \sqrt{x + 5}$

7. Solve by the addition method: $\begin{matrix} 3x - 3y = 2 \\ 6x - 4y = 5 \end{matrix}$

9. Evaluate the determinant: $\begin{vmatrix} -3 & 1 \\ 4 & 2 \end{vmatrix} = -10$

11. Solve for x : $4^x = 8^{x-1}$

13. Evaluate the series $\sum_{n=1}^7 (-1)^{n-1}(n + 2)$. **6**

15. Solve for x : $\log_6 x = 3$

17. For $g(x) = -3x + 4$, find $g(1 + h)$.

2. Factor: $2x^6 + 16$

4. Simplify: $\left(\frac{x^{-\frac{3}{4}} \cdot x^{\frac{3}{2}}}{x^{-\frac{5}{2}}} \right)^{-8}$

6. Solve: $2x^2 - x + 7 = 0$

8. Solve: $2x - 1 > 3$ or $1 - 3x > 7$

10. Write $\log_5 \sqrt{\frac{x}{y}}$ in expanded form.

12. Write the 5th and 6th terms of the sequence whose n th term is given by the formula $a_n = n(n - 1)$. **$a_5 = 20, a_6 = 30$**

14. Solve by the addition method:
 $\begin{matrix} x + 2y + z = 3 \\ 2x - y + 2z = 6 \\ 3x + y - z = 5 \end{matrix}$ **(2, 0, 1)**

16. Divide: $(4x^3 - 3x + 5) \div (2x + 1)$

18. Find the range of $f(a) = \frac{a^3 - 1}{2a + 1}$ if the domain is $\{0, 1, 2\}$.

19. Graph: $3x - 2y = -4$

20. Graph the solution set of $2x - 3y < 9$.

21. A new computer can complete a payroll in 16 min less time than it takes an older computer to complete the same payroll. Working together, both computers can complete the payroll in 15 min. How long would it take each computer, working alone, to complete the payroll?
22. A boat traveling with the current went 15 mi in 2 h. Against the current, it took 3 h to travel the same distance. Find the rate of the boat in calm water and the rate of the current.
23. An 80-milligram sample of a radioactive material decays to 55 mg in 30 days. Use the exponential decay equation $A = A_0 \left(\frac{1}{2}\right)^{\frac{t}{k}}$, where A is the amount of radioactive material present after time t , k is the half-life, and A_0 is the original amount of radioactive material, to find the half-life of the 80-milligram sample. Round to the nearest whole number.
24. A "theater in the round" has 62 seats in the first row, 74 seats in the second row, 86 seats in the third row, and so on in an arithmetic sequence. Find the total number of seats in the theater if there are 12 rows of seats.
1536 seats
25. To test the "bounce" of a ball, the ball is dropped from a height of 10 ft. The ball bounces to 80% of its previous height with each bounce. How high does the ball bounce on the fifth bounce? Round to the nearest tenth.

Final Exam

- Simplify: $12 - 8[3 - (-2)]^2 \div 5 - 3$
- Evaluate $\frac{a^2 - b^2}{a - b}$ when $a = 3$ and $b = -4$.
- Simplify: $5 - 2[3x - 7(2 - x) - 5x]$
- Solve: $\frac{3}{4}x - 2 = 4$
- Solve: $\frac{2 - 4x}{3} - \frac{x - 6}{12} = \frac{5x - 2}{6}$
- Solve: $8 - |5 - 3x| = 1$
- Solve: $|2x + 5| < 3$
- Solve: $2 - 3x < 6$ and $2x + 1 > 4$
- Find the equation of the line that contains the point $(-2, 1)$ and is perpendicular to the line $3x - 2y = 6$.
- Simplify: $2a[5 - a(2 - 3a) - 2a] + 3a^2$
- Write a quadratic equation that has integer coefficients and has solutions $-\frac{1}{2}$ and 2.
- Factor: $8 - x^3y^3$
- Factor: $x - y - x^3 + x^2y$
- Divide: $(2x^3 - 7x^2 + 4) \div (2x - 3)$
- Divide: $\frac{x^2 - 3x}{2x^2 - 3x - 5} \div \frac{4x - 12}{4x^2 - 4}$
- Subtract: $\frac{x - 2}{x + 2} - \frac{x + 3}{x - 3}$
- Simplify: $\frac{\frac{3}{x} + \frac{1}{x + 4}}{\frac{1}{x} + \frac{3}{x + 4}}$
- Solve: $\frac{5}{x - 2} - \frac{5}{x^2 - 4} = \frac{1}{x + 2}$
- Solve $a_n = a_1 + (n - 1)d$ for d . $d = \frac{a_n - a_1}{n - 1}$
- Simplify: $\left(\frac{4x^2y^{-1}}{3x^{-1}y}\right)^{-2} \left(\frac{2x^{-1}y^2}{9x^{-2}y^2}\right)^3$
- Simplify: $\left(\frac{3x^{\frac{2}{3}}y^{\frac{1}{2}}}{6x^2y^{\frac{4}{3}}}\right)^6$
- Subtract: $x\sqrt{18x^2y^3} - y\sqrt{50x^4y}$
- Simplify: $\frac{\sqrt{16x^5y^4}}{\sqrt{32xy^7}}$
- Solve by using the quadratic formula: $2x^2 - 3x - 1 = 0$
- Solve: $x^{\frac{2}{3}} - x^{\frac{1}{3}} - 6 = 0$

27. Find the equation of the line containing the points (3, -2) and (1, 4).

28. Solve: $\frac{2}{x} - \frac{2}{2x+3} = 1$

29. Solve by the addition method: $3x - 2y = 1$
 $5x - 3y = 3$

30. Evaluate the determinant: $\begin{vmatrix} 3 & 4 \\ -1 & 2 \end{vmatrix}$ 10

31. Solve for x : $\log_3 x - \log_3(x - 3) = \log_3 2$

32. Write $\sum_{i=1}^5 2y^i$ in expanded form.
 $2y + 2y^2 + 2y^3 + 2y^4 + 2y^5$

33. Find an equivalent fraction for $0.5\overline{1}$. $\frac{23}{45}$

34. Find the 3rd term in the expansion of $(x - 2y)^9$.
 $144x^7y^2$

35. Solve: $x^2 - y^2 = 4$
 $x + y = 1$

36. Find the inverse function of $f(x) = \frac{2}{3}x - 4$.

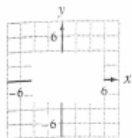
37. Write $2(\log_2 a - \log_2 b)$ as a single logarithm with a coefficient of 1.

38. Graph $2x - 3y = 9$ by using the x - and y -intercepts.

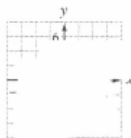
39. Graph the solution set of $3x + 2y > 6$.

40. Graph: $f(x) = -x^2 + 4$

41. Graph: $\frac{x^2}{16} + \frac{y^2}{4} = 1$



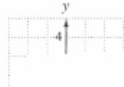
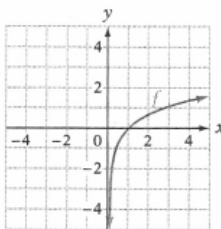
42. Graph: $f(x) = \log_2(x + 1)$



43. Graph $f(x) = x + 2^{-x}$ and approximate to the nearest tenth the values of x for which $f(x) = 2$.



44. Given the graph of $f(x) = \ln x$ shown below, graph $g(x) = \ln(x + 3)$.



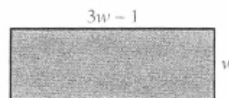
45. An average score of 70–79 in a history class receives a C grade. A student has grades of 64, 58, 82, and 77 on four history tests. Find the range of scores on the fifth test that will give the student a C grade for the course.

69 or better

46. A jogger and a cyclist set out at 8 A.M. from the same point headed in the same direction. The average speed of the cyclist is two and a half times the average speed of the jogger. In 2 h, the cyclist is 24 mi ahead of the jogger. How far did the cyclist ride? **40 mi**

47. You have a total of \$12,000 invested in two simple interest accounts. On one account, a money market fund, the annual simple interest rate is 8.5%. On the other account, a tax-free bond fund, the annual simple interest rate is 6.4%. The total annual interest earned by the two accounts is \$936. How much do you have invested in each account?

48. The length of a rectangle is 1 ft less than three times the width. The area of the rectangle is 140 ft². Find the length and width of the rectangle.



49. Three hundred shares of a utility stock earn a yearly dividend of \$486. How many additional shares of the utility stock would give a total dividend income of \$810? **200 additional shares**

50. An account executive traveled 45 mi by car and then an additional 1050 mi by plane. The rate of the plane was seven times the rate of the other car. The total time of the trip was $3\frac{1}{4}$ h. Find the rate of the plane.

51. An object is dropped from the top of a building. Find the distance the object has fallen when the speed reaches 75 ft/s. Use the equation $v = \sqrt{64d}$, where v is the speed of the object and d is the distance. Round to the nearest whole number. **88 ft**

Answers to Final Exam Form A

1. $a^{10}b^6$ 2. no solution 3. a) 11 b) $(-11/2, -8)$ 4. $y = 16 - 3x$ 5. \$64,010
6. $(-1, \infty)$ 7. -2 8. 12, 3, 1, 3, 11 9. 31 10. even
11. $g(x) = \sqrt{x^2 - 25}, 0 \leq x \leq 5$ 12. 13. $x^2 + x + 4, R2$ 14. yes, $p(4) = 0$ 15. 1, 1, $\pm 3i$
16. 17. 18. $2\log 2 + 3\log x - \log y - 6\log(x-2)$ 19. 1.0363 20. 21 minutes
21. $(-5/13, -12/13)$ 22. $-5/13, 12/13, -5/12$ 23. $1/3$ 24. $2\sin 2(x + \pi/3)$
25. 26. $7/6$ rad 27. $25\sqrt{3} + 25/3\sqrt{3}$ 28. 17.5 34. $\theta = \tan^{-1}(2/x)$
41. $5/x - 4/(x+3)$ 42. $y = -8x^2$ 43.

Answers to Cumulative Review Exercises

1. $\frac{x^2 + 5x - 2}{(x+2)(x-1)}$ 2. $2(x^2 + 2)(x^4 - 2x^2 + 4)$ 3. $4y\sqrt{x} - y\sqrt{2}$ 4. $\frac{1}{x^{26}}$
5. 4 6. $\frac{1 \pm i\sqrt{55}}{4}$ 7. $(7/6, 1/2)$ 8. $x < -2$ or $x > 2$ 10. $\frac{1}{2}(\log_5 x - \log_5 y)$
11. 3 15. 216 16. $2x^2 - x - 1, R6$ 17. $-3h + 1$ 18. -1, 0, $7/5$
19. 20. 21. 24 min new, 40 min old 22. boat 6.25 mph, current 1.25 mph
23. 55 days 25. 3.3 ft

Answers to Final exam p. 723

1. -31 2. -1 3. $33 - 10x$ 4. 8 5. $2/3$ 6. 4, $-2/3$ 7. $-4 < x < -1$ 8. $x > 3/2$
9. $y = -2/3x - 1/3$ 10. $6a^3 - 5a^2 + 10a$ 11. $6/5 - 3/5i$ 12. $2x^2 - 3x - 2 = 0$
13. $(2 - xy)(4 + 2xy + x^2y^2)$ 14. $(x-y)(1+x)(1-x)$ 15. $x^2 - 3x - 3, R-5$
16. $\frac{x(x-1)}{2x-5}$ 17. $-\frac{10x}{(x+2)(x-3)}$ 18. $\frac{x+3}{x+1}$ 19. $-7/4$ 21. $\frac{y^4}{162x^3}$
22. $\frac{1}{64x^8y^5}$ 23. $-2x^2y\sqrt{2y}$ 24. $\frac{x^2\sqrt{2y}}{2y^2}$ 25. $\frac{3 \pm \sqrt{17}}{4}$ 26. 27, -8 27. $y = -3x + 7$
28. $3/2, -2$ 29. (3,4) 31. 6 35. $(5/2, -3/2)$ 36. $f^{-1}(x) = 3/2x + 6$
37. $\log_2 \frac{a^2}{b^2}$ 38. $(9/2, 0), (0, -3)$ 43. -2, 1.7 47. \$8000, \$4000 48. 20 by 7
50. 420mph

Basic Calculator only on this part

1. Make the equation of the line (in any form) that passes through $(-2, 6)$ and is parallel to $y = 4x - 5$ (2 pts)2. Use the line, $8x - 3y = 9$, to:a) Give the x -intercept. (1 pt)

b) Give another point on the line. (1 pt)

3. Find the following values exactly. (2 pts each)

a) $\cos\left(\frac{\pi}{3}\right)$

b) $\sin\left(\frac{-3\pi}{4}\right)$

c) $\tan \pi$

d) $\cot^{-1} 1$

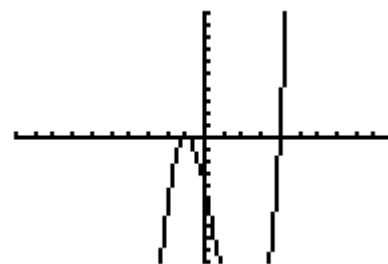
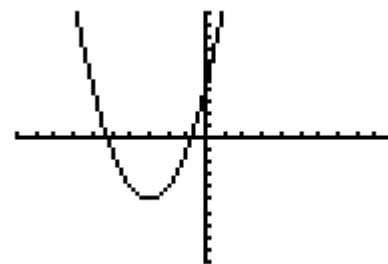
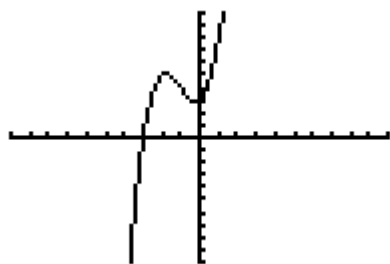
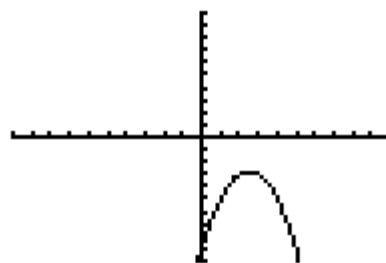
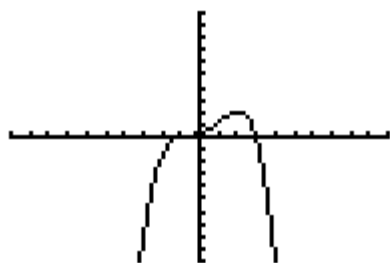
4. Use the quadratic formula to solve this equation: $4x^2 = 16x - 19$ (3 pts)

5. Write the letter of the correct description below its graph. (1 pt each)

A. quadratic with irrational roots

B. quadratic with complex roots

C. cubic with a double root

D. quintic (5^{th} power) with negative leading coefficientE. quartic (4^{th} power) with negative leading coefficientF. Cubic with a root of i 6. For the equation $2x^3 - 11x^2 + 2x + 15 = 0$

a) List all of the possible rational solutions (1 pt)

b) Show that $x = 3$ is not a solution (1 pt)c) Solve for all the solutions, using the fact that $x = 5$ is a solution (4 pts)

7. Factor completely. Use only integer coefficients and show all work. (3 pts each)

a) $x^3 - 3x^2 + 5x - 15$

b) $4y^4 - 9y^2$

8. Fill in the table of values such that the following criteria is true (1 pt each)

a) It has y -axis symmetry

b) It is a function with no inverse

c) It is not a function

x	-2	-1	0	1	2
y					

x	-2	-1	0	1	2
y					

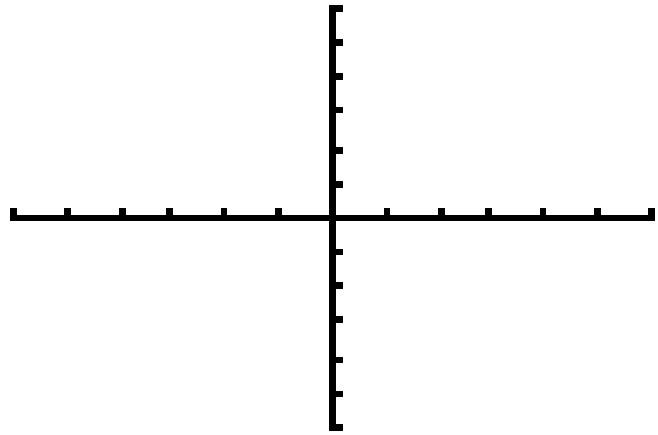
x					
y					

d) Extra Credit: Write an absolute value inequality that has $-12 \leq x \leq 4$ as a solution (1 pt)

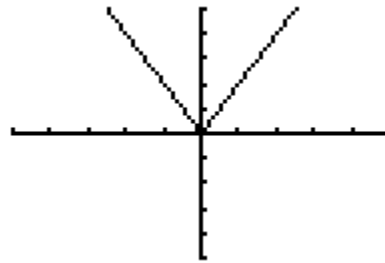
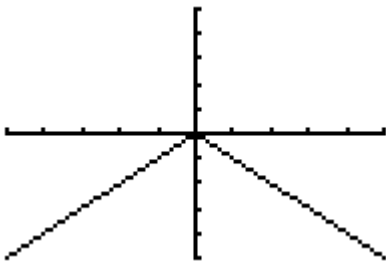
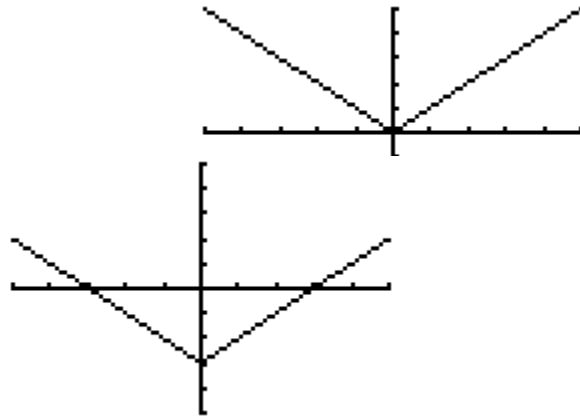
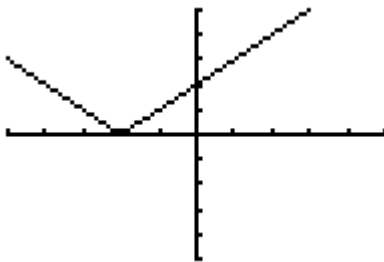
9. Graph the solution to this system of inequalities. Make your graphs neat and clear. (4 pts)

$$y - 2x > 1$$

$$x^2 + y^2 \leq 16$$



10. Given is the graph on the right of $f(x) = |x|$.
Use it to give an equation for the following:
Each tick mark is one unit. (1.5 pts each)



11. Solve these equations: (3 pts each)

a) $5^{3x-3} = 25^{x+2}$

b) $\frac{3}{x} - \frac{2}{5} = 4$

c) $\sqrt{x+7} - x = 5$

d) $\log_2 x - \log_2 3 = 2$

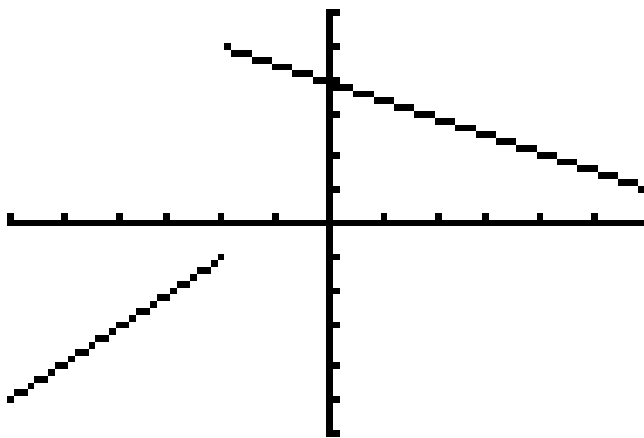
12. For the equation, give its period, average value (middle), amplitude and phase shift. Then graph it on an axis system that allows the entire graph to be seen for one period. Please label the coordinates of the first and last point of your graph. (5 pts)

$$y = -3 + 2\sin\left(2\left(x + \frac{\pi}{3}\right)\right)$$

13. An open house is happening in 18 hours and the house needs to be painted by then. Painting Company #1 can paint the house in 30 hours working alone. Painting Company #2 can do it alone in 24 hours. Company #1 and Company #2 start doing the job together. After both have worked for 5 hours, Company #1 goes on strike and Company #2 must finish alone. Will Company #2 finish in time? Justify your answer, showing all your work. (4 pts)

14. Use the graph of $f(x)$ to answer the following. All of $f(x)$ is visible. (1 pt each)

- Find $f(2)$
- Find $f(-3)$
- What is the domain of $f(x)$?
- What is the range of $f(x)$?
- Honors: Give a piece-wise defined formula for $f(x)$



2006-2007 Algebra II Second Semester Final-A

Name: _____

1. Use your graphing calculator all you can on this part, giving answers rounded to the nearest hundredth.

- Solve $x^3 + 2x^2 - 7x = -3$ (2 pts)
- Find the y- intercept of $y = x^3 + 2x^2 - 7x + 3$ (1 pt)
- Solve $x^3 + 2x^2 - 7x + 3 > 0$, writing solutions in correct mathematical notation (2 pts)
- Extra Credit: Give a polynomial with the same roots as in part c) but with a y-intercept of -5 (1 pt)

2. The cost of a can of coke was originally 10 cents in 1960 and has increased over time. The cost can be modeled by the formula, $C(t) = 0.10e^{0.057t}$ where C is the cost in dollars and t is the number of years after 1960.

- Find $C(38)$ and explain, using correct units, what your answer represents (2 pts)
- Solve $C(t) = 2.00$ and explain, using correct units, what your answer represents (2 pts)
- Extra Credit: What is the average rate of change in C between the years 1968 and 1995? Explain your answer, using correct units (1 pt)

Honors Questions for the Final:

1. Find the value of $\cos(\sin^{-1}(1/4))$. Show work to get partial credit for wrong answers (2 pts)

2. For the function $f(x) = 3 + \sqrt{x+2}$ (1 pt each)

- find $f(f(2))$
- Find the domain
- Find the range
- Give a formula for $f^{-1}(x)$
- What is the domain of the inverse?

3. Graph $y = \frac{x}{(x-1)(x+3)}$. Give the equations of all horizontal and vertical asymptotes and show them clearly on the graph. (3 pts)

Answers:

Part 1: Basic Calculator Section

1. $4 = \frac{y-6}{x+2}$ 2a. $x\text{-int} = \frac{9}{8}$ b. answer varies but fit equation $y = \frac{8}{3}x - 3$

3a. $\frac{1}{2}$ 3b. $\frac{-\sqrt{2}}{2}$ 3c. 0 3d. $\frac{\pi}{4}$ 4. $x = \frac{4 \pm i\sqrt{3}}{2}$

5. From left to right: (1st row) – E, D, B; (2nd row) – F, A, C

6a. Potential Rational Roots = $\pm 1, \pm 3, \pm 5, \pm 15, \pm \frac{1}{2}, \pm \frac{3}{2}, \pm \frac{5}{2}, \pm \frac{15}{2}$

6b. $2(3)^3 - 11(3)^2 + 2(3) + 15 \neq 0 \rightarrow -74 \neq 0$ 6c. $x = -1, \frac{3}{2}$ and 5

7a. $(x-3)(x^2+5)$ 7b. $y^2(2y-3)(2y+3)$

8a. answers may vary

x	-2	-1	0	1	2
y	b	a	any #	a	b

8b. answers may vary

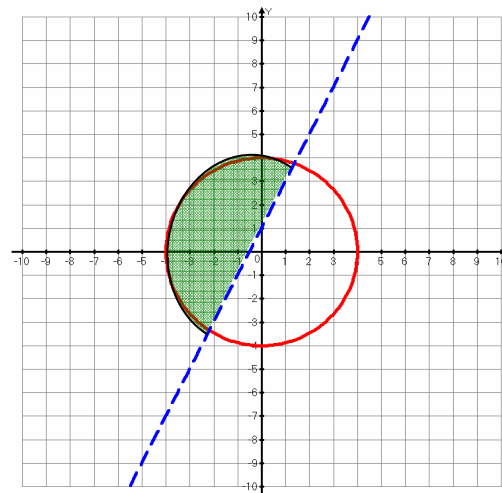
x	-2	-1	0	1	2
y	any two or more #'s being the same				

8c. answers may vary

x	-2	-1	0	1	2
y	any #'s with $\pm$ values				

8d. $|x+4| \leq 8$

9.

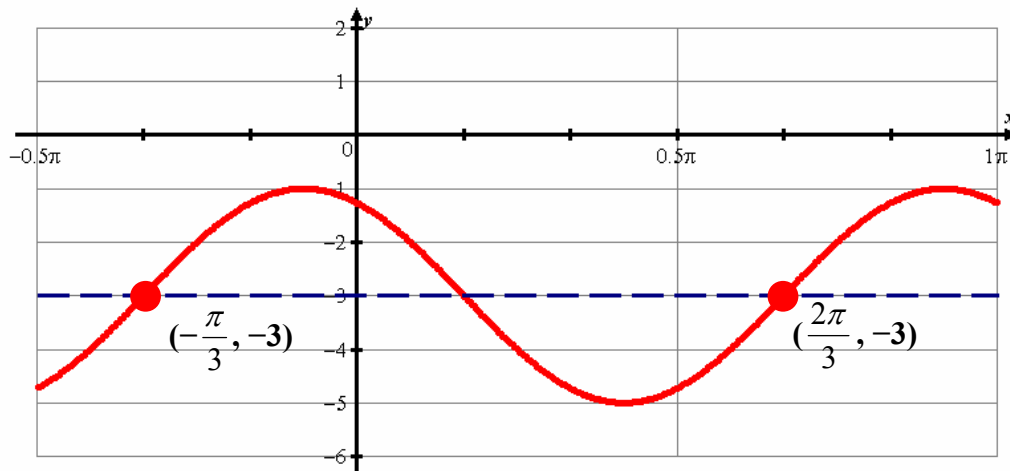


10. From left to right: (1st row) $y = |x+2|$ $y = |x| - 3$
 (2nd row) $y = -|x|$ $y = 2|x|$

11a. $x = 7$ 11b. $x = \frac{15}{22}$

11c. $x = -3$ 11d. $x = 12$

12. Period = π
 Avg (Mid) Value = -3
 Amp = 2
 Phase Shift = $\frac{\pi}{3}$ left



13. No. If they work together the whole time, it would take $13\frac{1}{3}$ hours to complete the whole thing. That means for each hour, they can only finish $0.075 = 7.5\%$ of the job. If Company #1 quits after 5 hours, there is still $0.625 = 62.5\%$ of the job to be finished. Company #2 would have to take another 15 hours to do it alone. Total Time needed would be 20 hours ($15 + 5$). Hence, it would not be finished within the 18 hours allotted.

14a. $f(2) = 3$ 14b. $f(-3) = -2$

14c. Domain: $-6 \leq x \leq 6$

14d. Range: $-5 \leq y \leq -1$ and $1 \leq y \leq 5$

14e. $f(x) = \begin{cases} -\frac{1}{2}x + 4 & \text{if } x \geq -2 \\ x + 1 & \text{if } x < -2 \end{cases}$

Part 2: Graphing Calculator Section

1a. $x = -3.96, 0.53$ and 1.43

1b. $y\text{-int} = 3$

1c. $-3.96 < x < 0.53$ or $x > 1.43$

1d. $y = \frac{-5}{3}(x^3 + 2x^2 - 7x + 3)$

2a. $C(38) = \$0.87$; This is the cost of a can of coke in 1998.

2b. In 2012

2c. $\$0.02139/\text{year}$

Honors

1. $\frac{\sqrt{15}}{4}$

2a. $3 + \sqrt{7}$

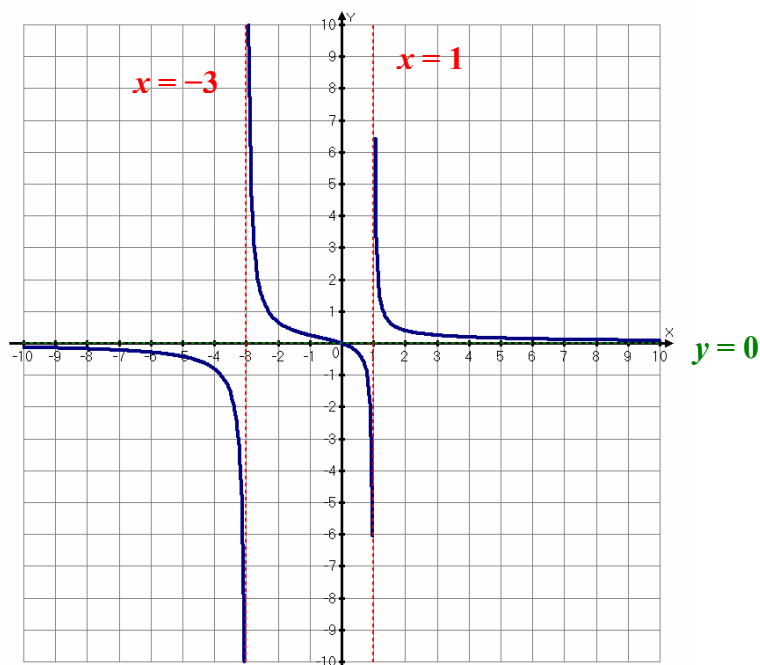
2b. Domain: $x \geq -2$

2c. Range: $y \geq 3 + \sqrt{5}$

2d. $f^{-1}(x) = (x - 3)^2 - 2$ for $x \geq 3$

2e. Domain $x \geq 3$

3. Vertical Asymptotes: $x = 1$ and $x = -3$
Horizontal Asymptote: $y = 0$



Algebra II Semester 2 Practice Exam A

1. Find the solution set of $6x^2 - 2x - 2 = 1$.

- A. $\left\{\frac{1}{6} + \frac{\sqrt{5}}{6}, \frac{1}{6} - \frac{\sqrt{5}}{6}\right\}$
B. $\left\{\frac{1}{6} + \frac{\sqrt{19}}{6}, \frac{1}{6} - \frac{\sqrt{19}}{6}\right\}$
C. $\left\{\frac{1}{3} + \frac{\sqrt{5}}{3}, \frac{1}{3} - \frac{\sqrt{5}}{3}\right\}$
D. $\left\{\frac{1}{3} + \frac{\sqrt{19}}{3}, \frac{1}{3} - \frac{\sqrt{19}}{3}\right\}$

2. What is the simplified form of the expression $\sqrt[3]{4a^6} + a\sqrt[3]{108a^3}$?

- A. $4a^2\sqrt[3]{4}$
B. $2a^2\sqrt[3]{14}$
C. $2a^2 + 3a\sqrt[3]{4}$
D. $2a^3 + 6a^2\sqrt[3]{3a}$

3. Simplify $\frac{1}{b^4} \cdot b^{\frac{1}{4}}$.

- A. b
B. $b^{-\frac{15}{4}}$
C. b^{-1}
D. 1

4. Let $f(x) = 2x - 3$ and $g(x) = 2\sqrt{x}$. Which expression is equivalent to $f(x) \cdot g(x)$?

- A. $4\sqrt{x} - 3$
B. $2\sqrt{2x - 3}$
C. $2x - 3 + 2\sqrt{x}$
D. $(4x - 6)\sqrt{x}$

5. If $h(x) = 10x^2$ and $j(x) = x + 4$, what is $h(-2) + j(3)$?

- A. 47
B. 92
C. 139
D. 407

6. Let $f(x) = 2x^2$ and $g(x) = 4x - 3$. Which expression is equivalent to $(f \circ g)(x)$?

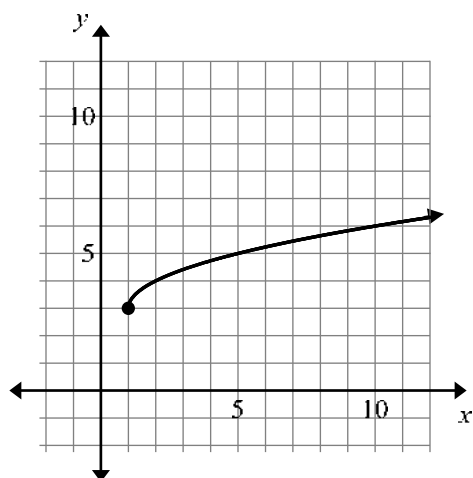
- A. $2x^2 + 4x - 3$
B. $8x^3 - 6x^2$
C. $8x^2 - 3$
D. $32x^2 - 48x + 18$

Algebra II Semester 2 Practice Exam A

7. Which is the inverse of the function $k(x) = 4x^2 - 16$ for $x \geq 0$?

- A. $k^{-1}(x) = \sqrt{\frac{x}{4}} + 4$
- B. $k^{-1}(x) = \frac{\sqrt{x+4}}{2}$
- C. $k^{-1}(x) = \sqrt{x+4}$
- D. $k^{-1}(x) = \frac{\sqrt{x+16}}{4}$

8. What function is represented by this graph?



- A. $y = \sqrt{x+1} + 3$
- B. $y = \sqrt{x+3} + 1$
- C. $y = \sqrt{x-1} + 3$
- D. $y = \sqrt{x+3} - 1$

9. Solve for x : $\sqrt{5x-3} - 12 = -4$

- A. $x = -25$
- B. $x = \frac{11}{5}$
- C. $x = \frac{31}{5}$
- D. $x = \frac{67}{5}$

10. You are trying to determine the height of a regular truncated pyramid that cannot be measured directly. The height h and slant height s of a truncated pyramid are related by the formula $s = \sqrt{h^2 + \frac{1}{4}(b_2 - b_1)^2}$ where b_1 and b_2 are the lengths of the upper and lower bases of the pyramid, respectively. If $s = 5$, $b_1 = 2$, and $b_2 = 4$, what is the height of the pyramid?

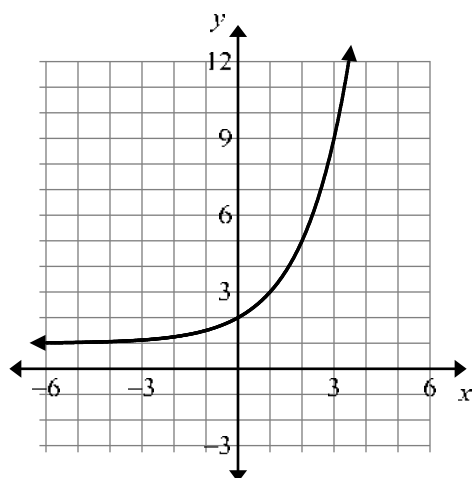
- A. 2
- B. 4
- C. $\sqrt{6}$
- D. $\sqrt{24}$

11. Simplify the expression $[(1-2i)+(1+i)](-3+i)$ where $i = \sqrt{-1}$.

- A. $-7 + 5i$
- B. $-7 - i$
- C. $-5 + 5i$
- D. $5 - 5i$

Algebra II Semester 2 Practice Exam A

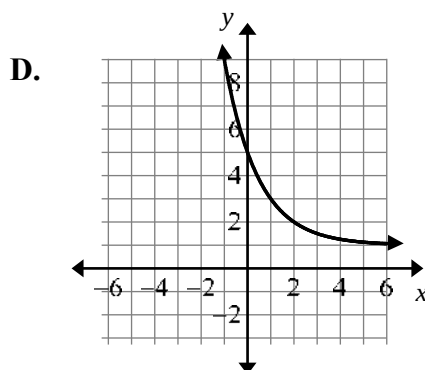
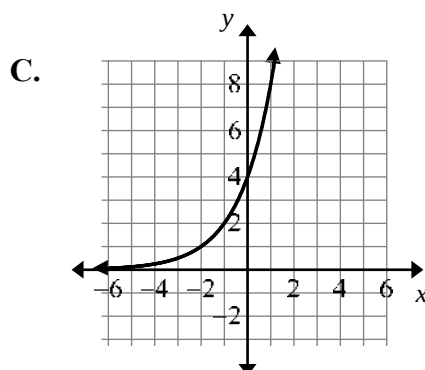
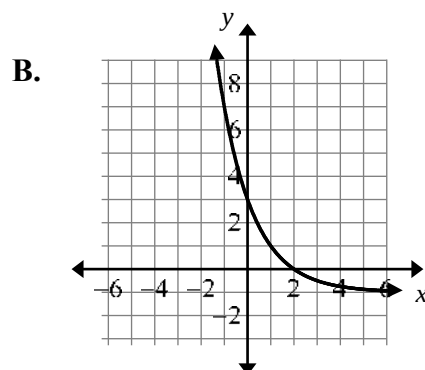
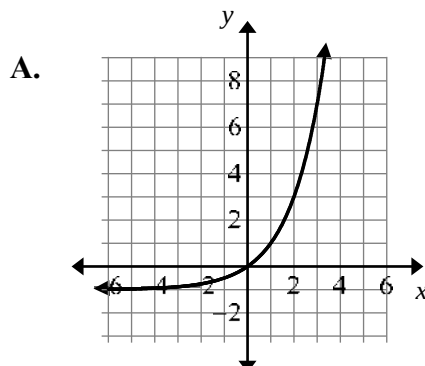
12. What function is represented by the graph?



- A. $y = 1 + 2^x$
- B. $y = 1 + 2x$
- C. $y = 1 + 2x^2$
- D. $y = 1 + 2x^3$

13. Which graph represents

$$f(x) = 4\left(\frac{1}{2}\right)^x - 1?$$



Algebra II Semester 2 Practice Exam A

14. Which expression is equivalent to $\log_3 64$?

- A. $3\log 64$
- B. $64\log 3$
- C. $\frac{\log 3}{\log 64}$
- D. $\frac{\log 64}{\log 3}$

15. Which expression is equivalent to $3\log 4 + \log 6 - \log 8$?

- A. $\log 48$
- B. $\log 1728$
- C. $3\log 2$
- D. $3\log 3$

16. Solve the equation for x : $\ln(2x - 1) = 2$

- A. $x = \frac{2+e}{2}$
- B. $x = \frac{2^e+1}{2}$
- C. $x = \frac{2e+1}{2}$
- D. $x = \frac{e^2+1}{2}$

17. Which is a solution for the equation $2^n = 10$?

- A. $n = \log 5$
- B. $n = \log 8$
- C. $n = \frac{10}{\log 2}$
- D. $n = \frac{1}{\log 2}$

18. Solve for x : $2^{x+1} = 8^{x-2}$

- A. $x = \frac{3}{2}$
- B. $x = \frac{7}{2}$
- C. $x = 2$
- D. no solution

19. In the year 1995, about 20 million people used the Internet. Between 1995 and 2001, the number of people who used the internet grew by about 75% each year.

Which function best models the relationship between p , the number of people using the internet (in millions), and t , the number of years since 1995?

- A. $p(t) = 20^{1.75t}$
- B. $p(t) = (0.75)20^t$
- C. $p(t) = 20(1.75)^t$
- D. $p(t) = 20 + 0.75t$

Algebra II Semester 2 Practice Exam A

20. What is the minimum or maximum of the quadratic function

$$q(x) = 8 - x + x^2?$$

A. $q(x) = \frac{35}{4}$

B. $q(x) = \frac{31}{4}$

C. $q(x) = \frac{1}{2}$

D. $q(x) = 8$

21. The value of w varies directly with x and inversely with y . If $w = 5$ when $x = \frac{3}{4}$

and $y = \frac{1}{2}$, what is the value of w when

$x = 4$ and $y = \frac{1}{3}$?

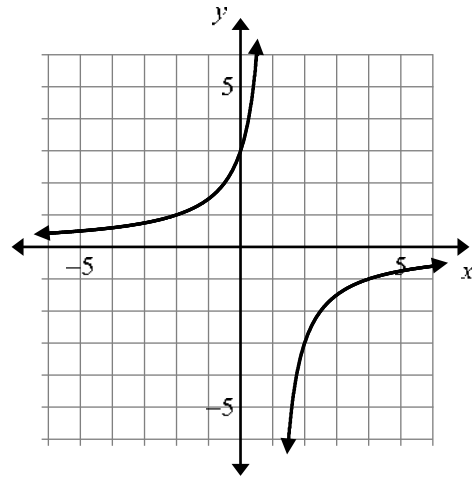
A. 40

B. 90

C. 160

D. $\frac{160}{9}$

22. What function is represented by the graph?



A. $f(x) = -\frac{3}{x}$

B. $f(x) = -\frac{3}{x+1}$

C. $f(x) = -\frac{3}{x-1}$

D. $f(x) = 3 - \frac{3}{x}$

23. Which of the following functions have an asymptote at $y = 1$?

I. $y = \frac{1}{x-1}$

II. $y = \frac{x}{x-1}$

III. $y = \frac{1}{x}$

A. II only

B. I and II only

C. I and III only

D. I, II, and III

Algebra II Semester 2 Practice Exam A

24. Simplify the expression $\frac{\frac{x^2 + 2x - 15}{15}}{\frac{x^2 - 9}{30}}.$

A. $\frac{x-15}{-9}$

B. $\frac{2(x+5)}{x+3}$

C. $\frac{x-9}{15}$

D. $\frac{2(x-3)}{15}$

25. What is the solution set of

$$\frac{3}{x} + \frac{5}{x+2} = 2?$$

A. $\{-1, 3\}$

B. $\{0, 4\}$

C. $\left\{-\frac{1}{2}\right\}$

D. $\{-1\}$

26. What is the solution set of

$$\frac{x+5}{x-2} - \frac{5}{x+2} = \frac{28}{x^2-4}?$$

A. $\{-8, 2\}$

B. $\{-8\}$

C. $\{-4, 2\}$

D. $\{-4\}$

27. The harmonic mean h of a set of n numbers $\{x_1, x_2, x_3, \dots, x_n\}$ is given by the formula

$$\frac{1}{h} = \frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}.$$

What is the harmonic mean of two positive numbers that differ by one if the smaller number is k ?

A. $h = \frac{k}{2}$

B. $h = \frac{2k+1}{2}$

C. $h = \frac{k(k+1)}{2k+1}$

D. $h = \frac{2k+1}{k^2+k}$

28. Given the system of linear equations:

$$2x + 3y = 5$$

$$-x + 4y = 6$$

Which expression below shows the solution to the system using matrices?

A. $\begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{11} \begin{bmatrix} 4 & -3 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 5 \\ 6 \end{bmatrix}$

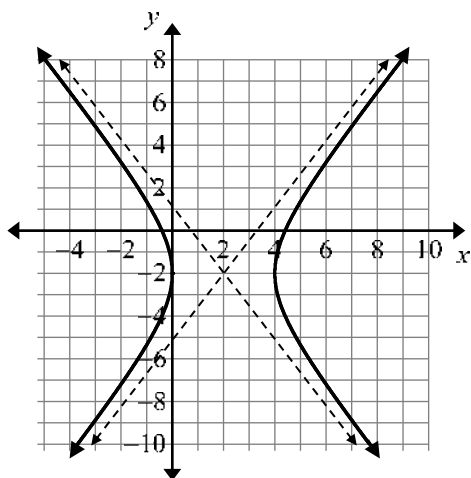
B. $\begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 2 & 3 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} 5 \\ 6 \end{bmatrix}$

C. $\begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{11} \begin{bmatrix} -2 & -1 \\ 3 & -4 \end{bmatrix} \begin{bmatrix} 5 \\ 6 \end{bmatrix}$

D. $\begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{5} \begin{bmatrix} -4 & 3 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} 5 \\ 6 \end{bmatrix}$

Algebra II Semester 2 Practice Exam A

29. What is the equation of the graph below?



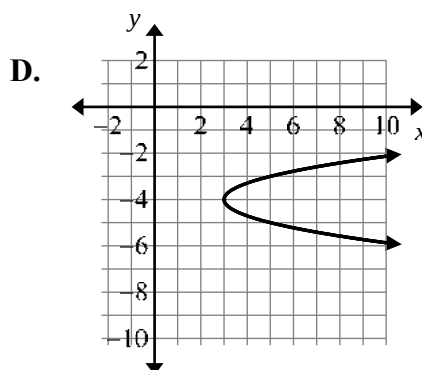
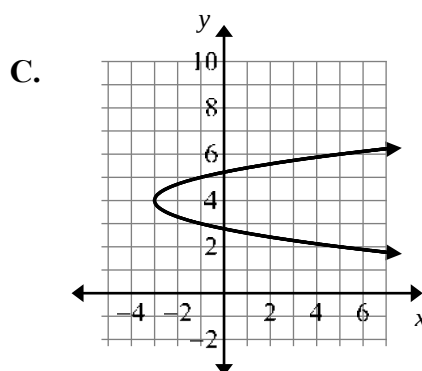
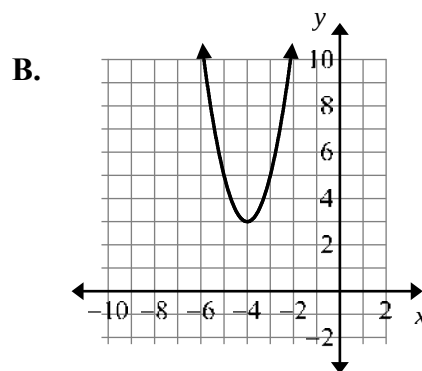
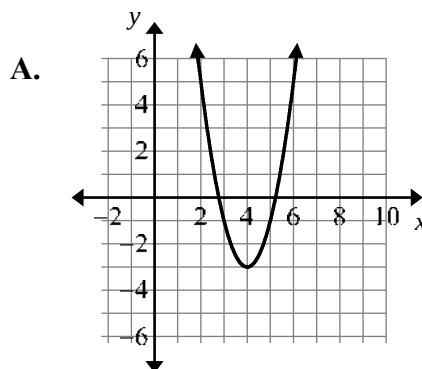
A. $\frac{(x-2)^2}{4} - \frac{(y+2)^2}{9} = 1$

B. $\frac{(x+2)^2}{4} - \frac{(y-2)^2}{9} = 1$

C. $\frac{(y+2)^2}{9} - \frac{(x-2)^2}{4} = 1$

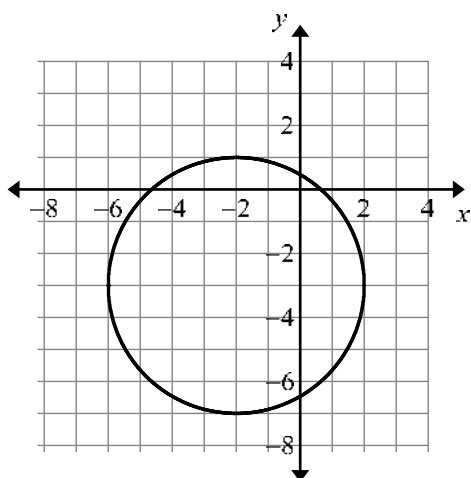
D. $\frac{(y+2)^2}{4} - \frac{(x-2)^2}{9} = 1$

30. Which graph best represents the equation $x - 3 = 2(y + 4)^2$?



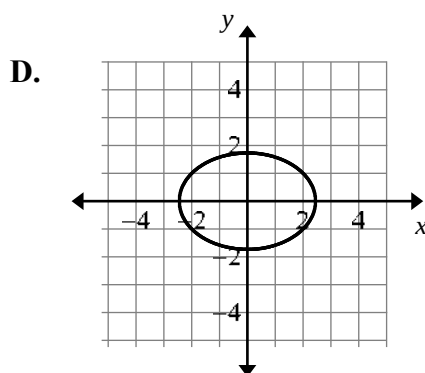
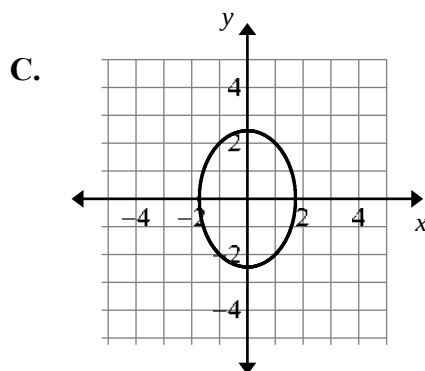
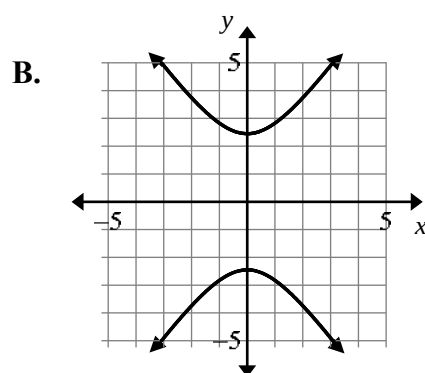
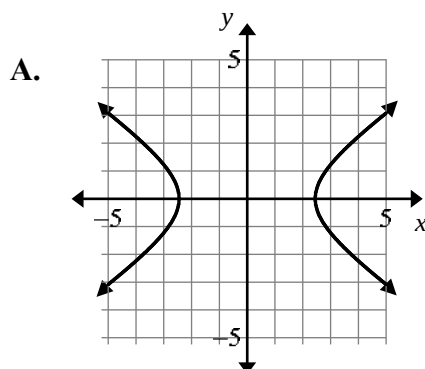
Algebra II Semester 2 Practice Exam A

31. What is the equation for the graph below?



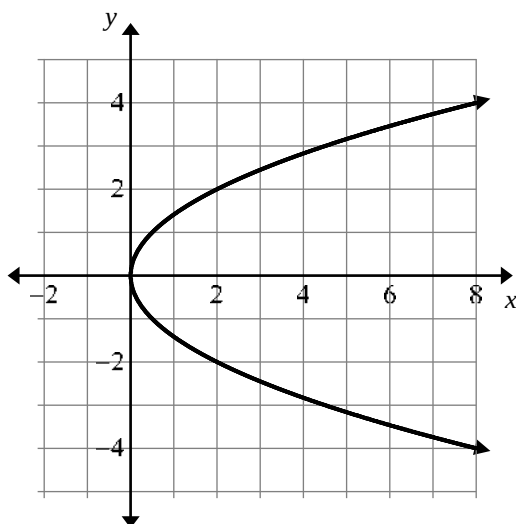
- A. $(x+2)^2 + (y+3)^2 = 4$
- B. $(x-2)^2 + (y-3)^2 = 4$
- C. $(x+2)^2 + (y+3)^2 = 16$
- D. $(x-2)^2 + (y-3)^2 = 16$

32. Which graph best represents the graph of $6y^2 + 3x^2 = 18$?



Algebra II Semester 2 Practice Exam A

33. What equation represents the graph below?

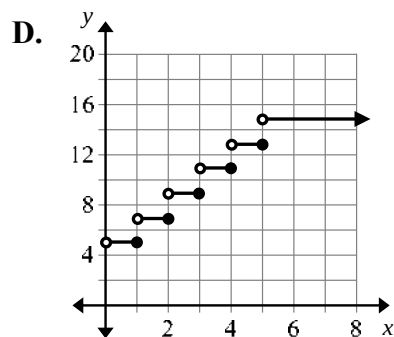
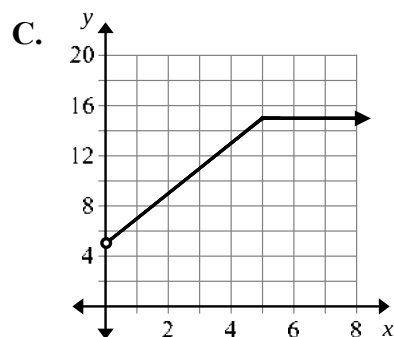
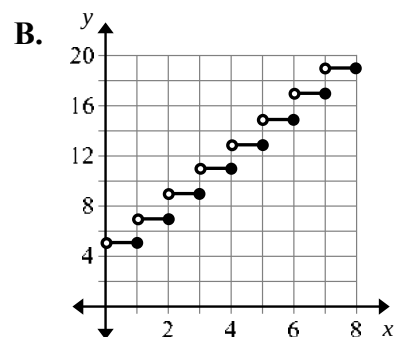
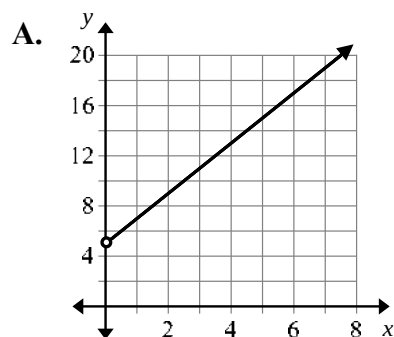


- A. $y = \frac{1}{2}x^2$
- B. $x = \frac{1}{2}y^2$
- C. $y = \sqrt{2x}$
- D. $x = \sqrt{2y}$

34. Which is the equation of a parabola?

- A. $16(x+4)^2 + 4(y+2)^2 = 64$
- B. $y^2 - 16x = 0$
- C. $x^2 + (y-2)^2 = 81$
- D. $\frac{x^2}{25} - \frac{y^2}{25} = 1$

35. The charge for parking at an airport is \$5 for time up to one hour, plus \$2 for each additional hour (or portion of an hour) up to a maximum of \$15. What graph represents this situation?



Algebra II Semester 2 Practice Exam A

36. Expand the expression $\sum_{n=1}^6 -n^2 + 5$.

- A. $-1 - 4 - 9 - 16 - 25 - 36 + 5$
- B. $4 + 1 - 4 - 11 - 20 - 31$
- C. $4 + 3 + 2 + 1 + 0 - 1$
- D. $6 + 9 + 14 + 21 + 30 + 41$

37. What is the series $\frac{4}{7} + \frac{5}{8} + \frac{6}{9} + \frac{7}{10} \dots$ when written in summation notation?

- A. $\sum_{i=4}^{\infty} \frac{i}{i+3}$
- B. $\sum_{i=4}^{\infty} \frac{i+3}{i+6}$
- C. $\sum_{i=4}^{\infty} \frac{4i}{4i-1}$
- D. $\sum_{i=4}^{\infty} \frac{2i-1}{3i-2}$

38. What equation is the sum of the 8 terms in the series $1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \dots + \frac{1}{2187}$?

- A. $S_8 = \frac{8}{2} \left(1 + \frac{1}{2187} \right)$
- B. $S_8 = \frac{1 - \left(\frac{1}{3} \right)^8}{1 - \frac{1}{3}}$
- C. $S_8 = 1 + \left(\frac{1}{3} \right)^8$
- D. $S_8 = \frac{1}{1 - \frac{1}{3}}$

39. Figure 1 shows a square with an area of 9 square units. In Figure 2, the first square has been divided into nine smaller congruent squares and the middle one removed. In Figure 3, each of the squares is again subdivided and the middle square removed. If the pattern continues, what is the area of Figure 6?



Figure 1

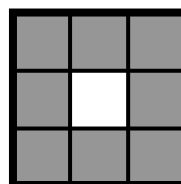


Figure 2

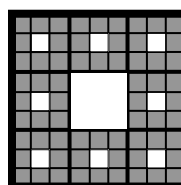


Figure 3

- A. $9 - \left(\frac{1}{9} \right)^5$
- B. $9 - \left(\frac{1}{9} \right)^6$
- C. $9 \cdot \left(1 - \frac{1}{9} \right)^5$
- D. $9 \cdot \left(1 - \frac{1}{9} \right)^6$

Algebra II Semester 2 Practice Exam A

40. What is the sum of the geometric series

$$\sum_{n=1}^{\infty} 3 \left(\frac{3}{4} \right)^{n-1} ?$$

- A. 3
- B. 4
- C. 9
- D. 12

41. Which is a formula of a geometric sequence when $g_1 = 3$ and $g_6 = -96$?

- A. $g_n = -3(1 - (-2)^n)$
- B. $g_n = \frac{3}{1 - (-2)^{n-1}}$
- C. $g_n = -3 \cdot 2^n$
- D. $g_n = 3 \cdot (-2)^{n-1}$

42. Which is a formula of the arithmetic sequence $4, \frac{11}{2}, 7, \frac{19}{2}, \dots$?

- A. $a_n = \frac{5}{2} + \frac{3}{2}n$
- B. $a_n = 4 + \frac{3}{2}n$
- C. $a_n = 4 \cdot \left(\frac{11}{8} \right)^{n-1}$
- D. $a_n = \frac{n}{2} \left(4 + \frac{19}{2} \right)$

43. Which is a recursive rule for the sequence 2, 4, 7, 11, 16, ...?

- A. $t_1 = 2, t_n = 2t_{n-1}$
- B. $t_1 = 2, t_n = t_{n-1} + 2$
- C. $t_1 = 2, t_n = t_{n-1} + n$
- D. $t_1 = 2, t_n = 3t_{n-1} - n$

44. In the 1980's, the standard configuration for a Wyoming license plate was a small number between 1 and 23, inclusive, followed by four digits with repetition allowed.

For example:



How many license plates were possible for the entire state of Wyoming in the 1980's?

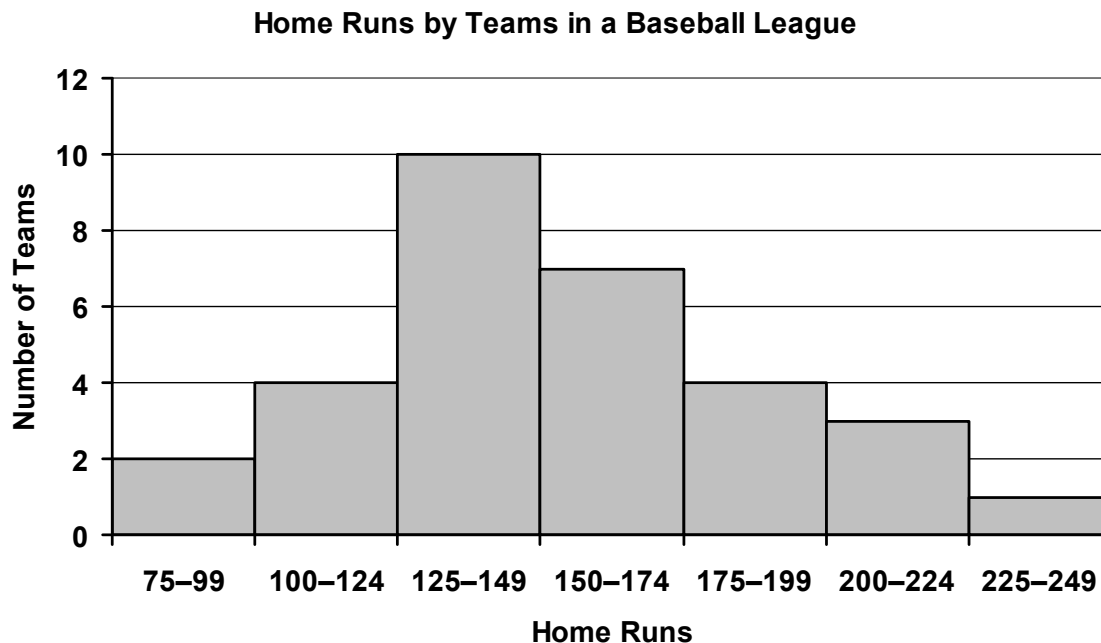
- A. 116,000
- B. 230,000
- C. 300,000
- D. 1,000,000

Algebra II Semester 2 Practice Exam A

45. A student will randomly choose four digits from the set $\{1, 2, 3, 4, 5, 6\}$, without replacement, and arrange them in the order they were chosen. How many different four-digit numbers can be made in this way?
- A. 15
B. 24
C. 360
D. 720
46. Which is a term of the expanded form of $(5 - 2a)^4$?
- A. $1000a$
B. $600a^2$
C. $-40a^3$
D. $-16a^4$
47. What is the coefficient of x^3y in the expansion of $(2x + y)^4$?
- A. 4
B. 8
C. 24
D. 32
48. A cooler contains 8 cans of cola, 6 cans of ginger ale, 4 cans of root beer, and 2 cans of orange soda. If a person reaches in the cooler and pulls out two cans at random, what is the probability that both cans will be ginger ale?
- A. $\frac{1}{10}$
B. $\frac{1}{30}$
C. $\frac{3}{38}$
D. $\frac{3}{40}$
49. Which of these measures are greatly influenced by extreme values?
- I. mean
II. median
III. interquartile range
IV. range
- A. IV only
B. I and IV only
C. II and III only
D. I, III, and IV only

Algebra II Semester 2 Practice Exam A

50. The graph below shows the number of home runs hit by the 31 teams in a baseball league.



Which value could be the median number of home runs?

- A. 177
- B. 162
- C. 147
- D. 112

Algebra II Semester 2 Practice Exam A

Free Response

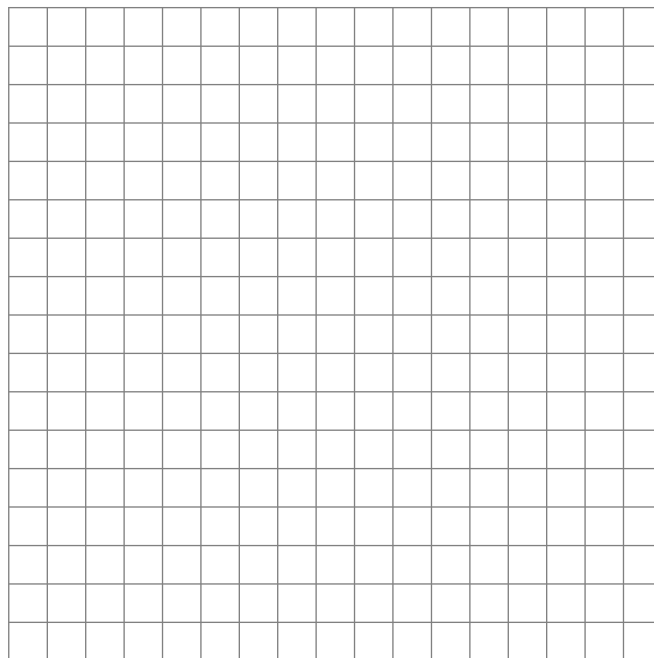
1. Use the functions $p(x) = x^3 - 8$ and $q(x) = \sqrt[3]{x+8}$ to answer the questions below.

A. What are the domain and range of $p(x)$?

B. Sketch the graph of $y = q(x)$ on the grid provided.

C. Are $p(x)$ and $q(x)$ inverse functions of each other? Provide evidence why or why not.

D. Let $r(x) = 2x - 1$. Find $p(r(x))$.



2. The half-life of the radioactive isotope Barium-139 is 83 minutes. A scientist has 20 grams of Barium-139 at time $t = 0$.

A. Write a function that expresses the quantity of Barium-139 as a function of t , the number of minutes after $t = 0$.

The half-life of Sodium-24 is 15 hours. The amount of Sodium-24 after t hours can be expressed as $A(t) \approx A_0 e^{-0.05t}$, where A_0 is the initial amount of the isotope.

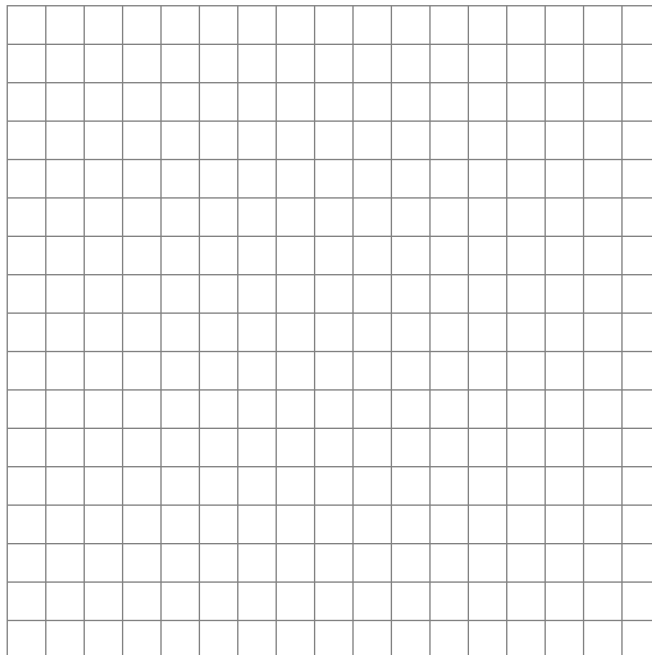
B. If $A_0 = 10$ grams, what amount, in grams, of Sodium-24 would remain after 100 hours? Leave your answer in terms of e .

C. Write an equation for $t(A)$, the time required for 10 grams of Sodium-24 to decay to an amount A .

Algebra II Semester 2 Practice Exam A Free Response

3. Use the function $f(x) = \frac{x^2 - 3x + 2}{2x}$ to answer the questions below.

- A. Sketch the graph of $y = f(x)$ on the grid provided. Be certain to note any asymptotes and intercepts on the graph.



- B. Solve $\frac{15}{x} + \frac{9x-7}{x+2} = 9$.



Spring Final Review

Factor each completely.

1) $4a^3 + 28a^2 + 24a$

2) $a^3 - a^2$

3) $n^2 - 3n$

4) $k^2 - k - 42$

5) $k^2 + 6k - 7$

6) $6x^4 + 30x^3 - 84x^2$

Solve each equation by factoring.

7) $x^2 = -x + 6$

8) $n^2 = -28 - 11n$

9) $m^2 - 6m = 0$

10) $n^2 + 14n = -48$

$$11) 2m^2 + 17m = -35$$

$$12) 3m^2 = 10 - 13m$$

13) Write each of the following as a quadratic functions:

$$(2x+3)(x-2)$$

$$(x+2)(3x-5)$$

$$(3x-1)(x+4)$$

Simplify.

$$14) -7\sqrt[3]{48x^6y^5}$$

$$15) -4\sqrt{147xy}$$

$$16) 6\sqrt[4]{405x^7y^5}$$

$$17) -7\sqrt[3]{-56u^7v^2}$$

Solve each equation.

$$18) \log_4 (x+3) - \log_4 x = 4$$

$$19) \log_6 9 - \log_6 2x = \log_6 26$$

$$20) \ln (x+7) + \ln 2 = \ln 55$$

$$21) \log_7 4 + \log_7 (x-6) = 1$$

Rewrite each equation in exponential form.

22) $\log_3 \frac{1}{243} = -5$

23) $\log_{14} 196 = 2$

Rewrite each equation in logarithmic form.

24) $4^2 = 16$

25) $18^2 = 324$

Name each polynomial by degree and number of terms.

26) $-3p^4 - 2p^3 + 10p^2$

27) $-n^2 - 9n$

28) $-3n^3 + 4n + 6$

29) $3x^5 - 6x^4 - 3x^3$

Simplify each expression.

30) $(3 + 5n^4 + 8n^2) - (2n^2 + 3n^4 + 1)$

31) $(4r - 6r^2 + 5r^4) + (4r^3 + r^4 + 5r)$

$$32) (4n^4 - 8n^2 - 4) - (4 - 5n + 7n^2)$$

$$33) (5 + 2a^2 - 8a) + (6a + 1 - 7a^2)$$

Simplify each and state the excluded values.

$$34) \frac{2k^3 + 14k^2}{k^2 + 12k + 35}$$

$$35) \frac{p^2 - 13p + 42}{p^2 - 12p + 35}$$

$$36) \frac{4b^2 - 28b}{b^2 - 17b + 70}$$

$$37) \frac{x^2 - 2x - 80}{x^2 - 19x + 90}$$

Simplify each expression.

$$38) \frac{5}{5v + 10} \div \frac{v + 10}{v^2 + 3v + 2}$$

$$39) \frac{2}{n + 2} \cdot \frac{n^2 + n - 2}{8n^2 - 8n}$$

$$40) \frac{x + 1}{2x^3 + 4x^2} \div \frac{x - 2}{2x^3 + 4x^2}$$

$$41) \frac{6m^3 + 30m^2}{m^2 + 10m + 25} \cdot \frac{m + 5}{m - 6}$$

$$42) \frac{a-1}{3a^2-9a} - \frac{3}{2a}$$

$$43) \frac{2}{k-2} + \frac{3}{k-4}$$

$$44) \frac{2}{n-5} + \frac{2n}{3n-5}$$

$$45) \frac{p+5}{p+6} - \frac{4}{p+1}$$

Solve each equation. Remember to check for extraneous solutions.

$$46) \frac{2}{n-1} = \frac{1}{n^2-n} + \frac{n+1}{n^2-n}$$

$$47) \frac{5}{v^2-6v+8} = \frac{1}{v-2} + \frac{1}{v^2-6v+8}$$

$$48) \frac{4}{m^2-3m} = \frac{1}{m^2-3m} - \frac{3}{m-3}$$

$$49) \frac{4}{a^2-6a} = \frac{2}{a-6} - \frac{1}{a^2-6a}$$

Simplify.

$$50) \frac{6\sqrt{3}}{\sqrt{7}}$$

$$51) -\frac{9}{4\sqrt{3}}$$

$$52) \frac{10\sqrt{3}}{\sqrt{5}}$$

$$53) \frac{\sqrt{8}}{4\sqrt{6}}$$

Solve each equation. Remember to check for extraneous solutions.

$$54) 1 = \sqrt{b+2} + 1$$

$$55) -2 = \sqrt{x+3} - 9$$

$$56) \sqrt{-5-m} = \sqrt{2m+19}$$

$$57) \sqrt{\frac{m}{8}} = \sqrt{9-m}$$

58) The student population at RayPec HS has increased at a rate of 5% each year. If the population is currently 1,900, what will the population be in 10 years?

59) Write out the equations for:

a) y varies directly as x

b) y varies inversely as x

c) y varies directly as x and inversely as z

d) y varies jointly as x and z and inversely as m

- 60) The time it takes to build a sailboat varies inversely as the number of students working on the boat. Last year it took 5 students 195 hours to build a boat. How many hours would it take 15 students to build a boat?
- 61) The cost of hiring a contractor to build a patio varies jointly with the area of the patio and price per square foot. The cost is \$2832 when the area is 80 square feet and the price per square foot is \$2.95. Find the price per square foot when the cost is \$4368 and the area is 112 square feet?
- 62) What is the equation of the line $y = \frac{2}{3}x + 3$ when the y-intercept is shifted down 2 units and the slope is tripled?
- 63) What is the equation of the line $y = \frac{5}{4}x - 2$ when the y-intercept is shifted up 5 units and the slope is doubled?

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Unit 1A Review – Polynomial and Rational Functions

Reviews do NOT cover all material from the lessons but will hopefully remind you of key points. To be prepared, you must study all packets for lessons 1.1 – 1.6.

1. For the function $s(t)$, s is the number of people swimming at the local pool and t is the temperature measured in Fahrenheit. Identify the dependent and independent variables.

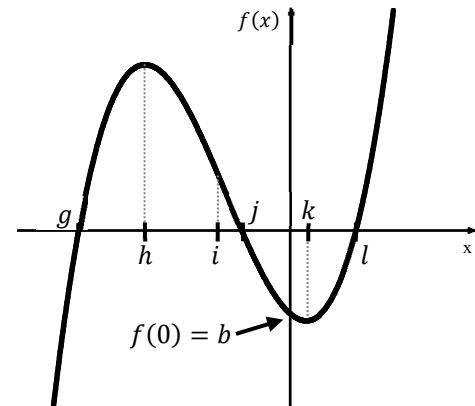
Dep:**Indep:**

2. Let the function f be increasing or decreasing, but not both. State whether the function is increasing or decreasing on the interval $9 < x < 17$ and justify your answer.

x	9	11	13	15	17
$f(x)$	8	12	15	17	18

3. Use the graph to the right to answer the questions below.

- On what interval(s) is the graph concave up?
- On what interval(s) is the graph concave down?
- On what interval(s) is the graph increasing?
- On what interval(s) is the graph decreasing?
- Find the zeros of the function.
- Find the y-intercept of the function.



4. **Calculator active.** Find the average rate of change of the function $w(r) = \sqrt{1 - 4r}$ on the interval $-5 \leq r \leq -1$.

5. **Calculator active.** Using the information in the table below, find the average rate of change for each given interval. Include units.

t seconds	24	60	135	200
$s(t)$ miles	8	1.3	3.5	0.3

a. $135 \leq t \leq 200$

b. $24 \leq t \leq 135$

c. $24 \leq t \leq 200$

6. **Calculator active.** Estimate the rate of change of $f(x) = x^2 - x$ at $x = -1$

7. Mr. Gardener is decreasing the amount of water used on his lawn each month, which causes the height of his grass to also decrease. Does this scenario represent a positive or negative rate of change for the variables of water used and height of the grass?

What is the average rate of change for each function on the given intervals?

8. $y = 7 - 2x$ on $-4 \leq x \leq 1$

9. $y = 3x^2 - 2x + 1$ on $-1 \leq x \leq 2$

What is the rate of change of the average rates of change for each function over consecutive equal-length intervals?

10. $y = 5x + 9$

11. $f(x) = 2x - 5x^2$.

12. The values of a function are given at selected x -values in the table below. The function's concavity does not change. Determine if the function is concave up or concave down. Justify your answer.

x	5	9	13	17	21
$g(x)$	45	20	0	-10	-14

Find the leading coefficient and the degree of each polynomial.

13. $f(x) = x^5 - 2x^2$

L.C. _____ Degree: _____

14. $f(x) = 10 - 3x^2 + 7x^3 - 2x$

L.C. _____ Degree: _____

Let $f(x)$ be a polynomial function with the given values. Are there any guaranteed extrema? If so, state where they occur.

15. $f(0) = -3, f(4) = 0$, and $f(7) = 0$.

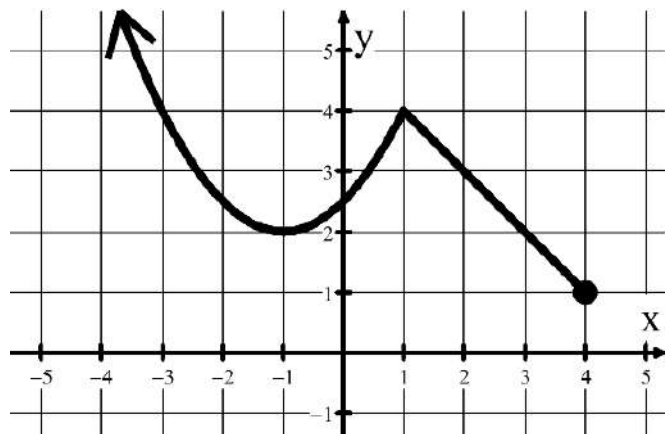
16. $f(-7) = 0, f(-2) = 5, f(0) = 1$, and $f(9) = 0$.

Is there a global maximum or minimum for each function?

17. $f(x) = -5x^6 + 6x^4 - 3x^3 + 1$

18. $f(x) = 2x^5 + x^2 - 6$

Find the following extrema. If there are none, cross it off and write NONE.



19. Absolute **min** of _____ when $x =$

20. Absolute **max** of _____ when $x =$

21. Relative **min**(s) at $x =$

22. Relative **max**(es) at $x =$

For each polynomial function, find the intervals for each condition.

23. $h(x) = x^2 - 8x + 15$. When is $h(x) \leq 0$?

24. $f(x) = -x^3 + 5x^2 + 24x$. When is $f(x) \geq 0$?

25. The degree of a polynomial is 7 with real zeros at $x = -8$, $x = 1$, and $x = 4$. $x = 1$ has a multiplicity of 3. How many non-real zeros does the polynomial have?

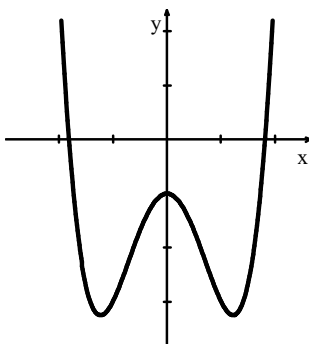
26. $5 - i$ is a non-real zero of a polynomial, find another zero.

27. Find the **degree** of the polynomial from the given input and output values.

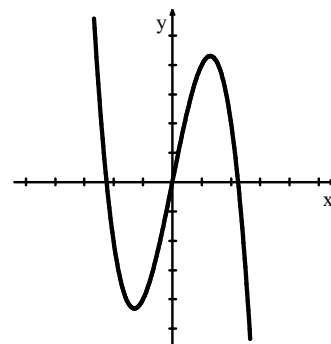
Input	0	1	2	3	4	5	6	7
Output	2	-2	4	20	46	82	128	184

State whether the following graphs represent functions that are even, odd, or neither.

28.



29.



State if the following functions are even, odd, or neither.

30. $f(x) = 4x^7 + 5x^3 - 2x$

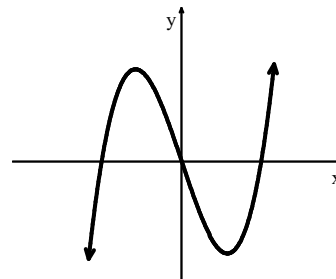
31. $f(x) = 7 - 6x^8 - 3x^2$

Describe the end behavior of each function using limit notation.

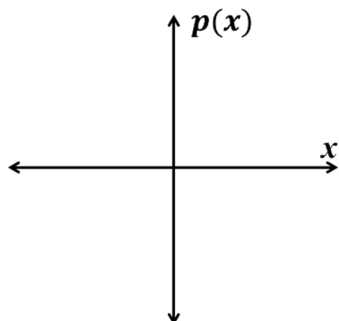
32. $p(x) = -11x^7 - 6x^2 + 4x$

33. $p(x) = -7x^6 + 4x - 8$

34.



35. Sketch the graph of a polynomial function that could match statements $\lim_{x \rightarrow -\infty} p(x) = \infty$ and $\lim_{x \rightarrow \infty} p(x) = \infty$.



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Unit 1-B Review – Polynomial and Rational Functions

Reviews do NOT cover all material from the lessons but will hopefully remind you of key points. To be prepared, you must study all packets for lessons 1.7 – 1.14.

Use the graph of the rational function f to answer the following.

1.

a. Find the domain.

b. Find the zero(s).

c. Find the end behavior of f .

d. Find the limit notation of the vertical asymptote.

e. Find the limit notation of the hole.

Use the rational function to answer the following.

2.

$$g(x) = \frac{x^3 - 2x^2 - 15x}{x^2 + x - 6}$$

a. Find the domain.

f. Find the horizontal asymptote if there is one.

b. Find the zero(s).

g. Find the slant asymptote if there is one.

c. Find the y -intercept.

d. Find the coordinates of hole(s) if there are any.

e. Find the vertical asymptote(s) if there are any.

Use the table of the rational function d to find the following.

3.

t	$d(t)$
-3.1	19
-3.01	199
-3.001	1,999
-3.0001	19,999
-3	undefined
-2.9999	-21,111
-2.999	-2,111
-2.99	-211
-2.9	-21

a. Find $\lim_{t \rightarrow -3^-} d(t) =$

b. Find $\lim_{t \rightarrow -3^+} d(t) =$

c. As t approaches negative three from the left the $d(t)$...

d. As t approaches negative three from the right the $d(t)$...

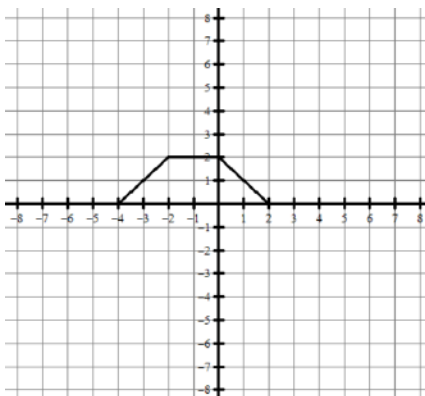
Use the binomial theorem to expand the following.

4. $(x + 3)^5$

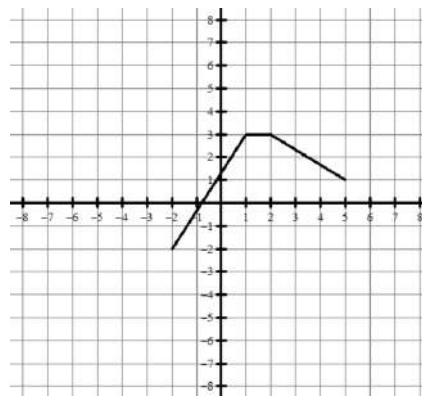
5. Find the 5th term in $(2x - 1)^7$

GRAPHICAL TRANSFORMATIONS. Use the graph of f to graph $g(x)$.

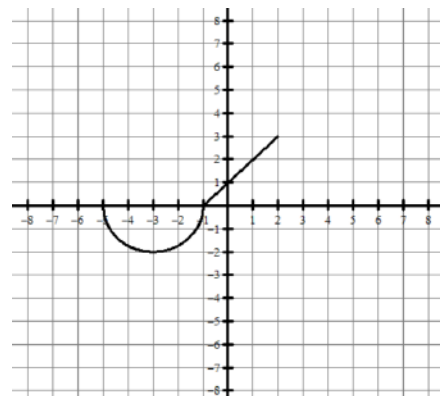
6. $g(x) = -f(x - 2)$



7. $g(x) = 2f(x) - 4$



8. $g(x) = f(2x) + 5$



TRANSFORMATIONS. Use the given information to answer the following.**ALGEBRAICALLY**

9. $f(x) = 6 - 4x^2$
 $g(x) = f(x) + 5$, find $g(x)$.

NUMERICALLY

10. Given the table of values for f .

x	$f(x)$
-6	2
-3	8
2	15
5	-2
8	-13

Let $g(x) = 3f(x) + 2$,
find $g(5)$.

DOMAIN and RANGE

11. Given the graph for f has a domain of $(-5, 3)$ and range of $[-4, 8]$.
Let $g(x) = -f(x + 3) - 2$.
Find the domain and range of $g(x)$.

CALCULATOR ACTIVE. Use regression to answer the following.

12.

x	$f(x)$
14	48
18	50
28	53
33	64
35	70
38	75
41	84
48	105
53	138

- a. The data can be modeled by the function f given by $f(x) = ax^2 + bx + c$. Find the value of a , b and c .
- a. Use your equation to predict the value of $x = 34$.

CALCULATOR ACTIVE. Use the model to answer the questions in context.

13. A diver jumps from a cliff into the water below. The function $h(t) = -16t^2 + 14t + 80$ models the height of the diver over time, where t is time in seconds and h is the height in feet.
- a. What is the restricted domain in this context?
- b. What is the restricted range in this context?

Multiple Choice

14. The rational function f is given by $f(x) = \frac{2x^k(x+2)(x-5)}{3x^5-4x^2-2}$, where k is a positive integer. If the function f has a slant asymptote, what is the value of k ?
- (A) 2
- (B) 3
- (C) 4
- (D) 5
-
15. The rational function g is given by $g(x) = \frac{3x^5-2x^2+9x+5}{x(x-5)^2(x+7)}$. Which of the following describes the end behavior of g ?
- (A) As x increases without bound, $g(x)$ increases without bound, and as x decreases without bound, $g(x)$ increases without bound.
- (B) As x increases without bound, $g(x)$ increases without bound, and as x decreases without bound, $g(x)$ decreases without bound.
- (C) As x increases without bound, $g(x)$ decreases without bound, and as x decreases without bound, $g(x)$ increases without bound.
- (D) As x increases without bound, $g(x)$ decreases without bound, and as x decreases without bound, $g(x)$ decreases without bound.
-

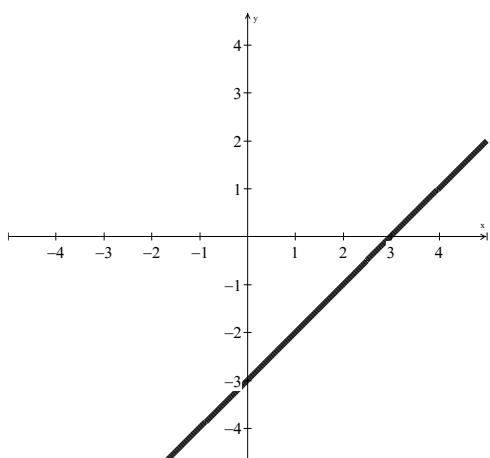
16. The table gives values of $h(x)$ for selected values of x . Which of the following best describes $h(x)$?

x	$h(x)$
1.997	1.489
1.998	1.495
1.999	1.499
2	undefined
2.001	1.501
2.002	1.504
2.003	1.507

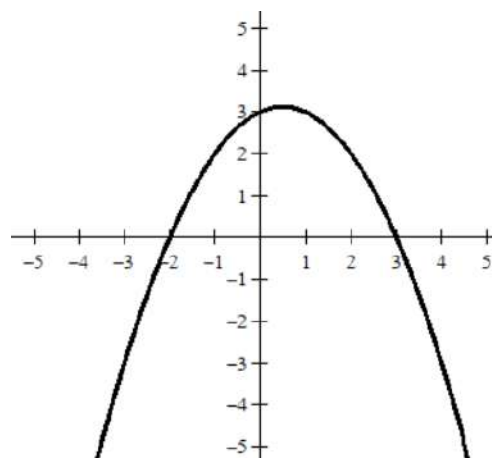
- (A) The graph of h has a hole at $x = 2$
- (B) The graph of h has a vertical asymptote at $x = 2$
- (C) The graph of h has a hole at $x = \frac{3}{2}$
- (D) The graph of h has a vertical asymptote at $x = \frac{3}{2}$

For questions 17 and 18, use the graphs of the polynomials f and g .

Graph of f



Graph of g



17. The function r is defined by $r(x) = \frac{f(x)}{g(x)}$. What is the value of the $r(0)$?
- (A) -2
- (B) 0
- (C) -1
- (D) 3
18. The function r is defined by $r(x) = \frac{f(x)}{g(x)}$. What are all the vertical asymptotes of the graph $y = r(x)$?
- (A) at $x = -2$ only
- (B) at $x = 3$ only
- (C) at $x = -2$ and $x = 3$ only
- (D) There are no vertical asymptotes.

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Unit 2-A Review – Exponential and Logarithmic Functions

Reviews do NOT cover all material from the lessons but will hopefully remind you of key points. To be prepared, you must study all packets for lessons 2.1 – 2.8.

Find an equation that gives the n th term of each sequence. Instead of the initial value use the k th term of the sequence in your equation. k is given for each problem.

1. $\{116, 136, 156, 176, \dots\}$ $k = 2$

2. $\left\{2, \frac{2}{3}, \frac{2}{9}, \frac{2}{27}, \dots\right\}$ $k = 1$

3. For the sequence $\{8, 5, 2, -1, \dots\}$, what is the 315th term?

A function has the following coordinate points. Could the function represent a linear function, exponential function, or neither?

4. $(3, 6), (4, 11), (5, 16)$

5. $(-10, 3), (-9, 6), (-8, 12)$

6. The following function is either linear or exponential. Which is it? Justify your answer.

x	6	11	16
$f(x)$	10	13	16

Is each function linear or exponential. Identify the constant (slope or ratio) that causes the output values to change?

7. $y = -\left(\frac{3}{5}\right)^x$

8. $y - 7 = 6(x - 4)$

9. It is known that y is an exponential function and that it passes through the points $(3, 6)$ and $(9, 42)$. Write an equation for this function.	10. The value $4 \cdot 4 \cdot 4 \cdot 6$ is an output value of an exponential function of the form $f(x) = a \cdot b^x$, where a and b are constants. Write the function along with the input value that represents the output value. $f(x) =$ where $x =$
11. Identify if the exponential function is growth or decay and justify your response. $f(x) = 1.4(0.07)^x$	12. Is the exponential function $f(x) = 8(1.7)^x$ increasing or decreasing?
13. Is the exponential function $f(x) = -(6)^x$ concave up or concave down?	14. Find $\lim_{x \rightarrow -\infty} f(x)$ for the function $f(x) = 4(0.3)^x$
Let $f(x)$ be a function on which a transformation occurs. Let $g(x)$ be a transformation of f. For each problem, name the transformation(s) of f.	
15. $f(x) = 9^x$ and $g(x) = f(x) \cdot 9$	16. $f(x) = 2^x$ and $g(x) = \frac{f(x)}{16}$
17. $f(x) = 7^x$ and $g(x) = (f(x))^{-3}$	18. $f(x) = 7^x$ and $g(x) = -\frac{f(x)}{49}$
Evaluate the function at the given input values.	
19. Let $h(x) = 6 \cdot 4^{x/3}$. Find $h(2)$	20. Let $h(x) = 9 \cdot 2^{x/5}$. Find $h(-1)$

21. Below is a table of values for an exponential function in the form $f(x) = a(b)^x + k$. Write the equation for this exponential function.

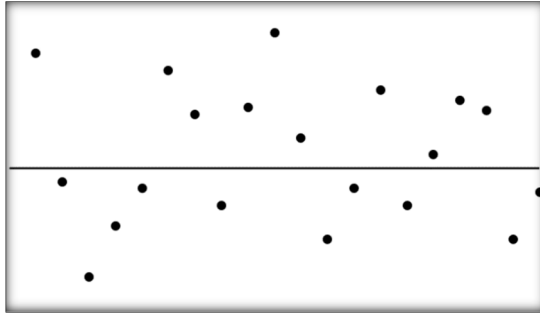
x	0	1	2	3	4
$f(x)$	3	6	18	66	258

22. The table gives the weight of a catfish at selected weeks t during its first eight weeks of life.

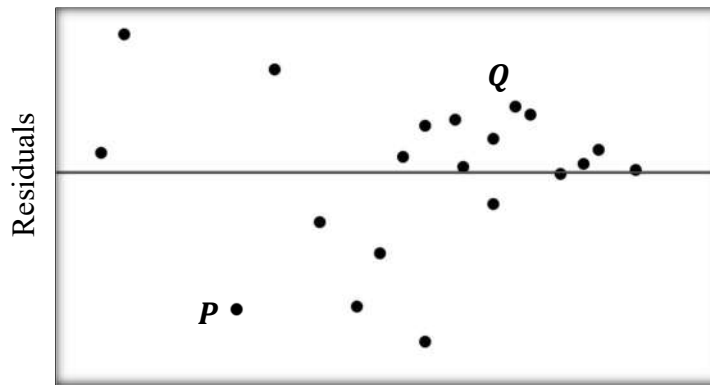
Week	1	3	6	8	9
Weight (pounds)	0.08	0.2	0.467	0.501	0.7

- Use an exponential regression $f(t) = ab^t$ to model these data. Round to three decimals but store the original equation in your calculator.
- According to the model in your calculator, how much will the catfish weigh after 5 weeks?
- According to the model in your calculator, how much will the catfish weigh after 7 weeks?
- If the model continues past 9 weeks, when will the catfish weigh 1 pound?
- Use your model from part *a* to find the error at $t = 8$.
- Is the value predicted an overestimate or underestimate of the actual value?

23. A student creates a model for a data set. The residual plot for their exponential regression is shown. Is the model appropriate? Why or why not?



24. Several students' SAT scores are collected and compared to their GPA from their first year at college. A regression model was calculated with the SAT scores as input values and GPA from the first year at college as the output values. The residual plot for the regression model and data points is below.



- a. The given residual plot has a point labeled P at the coordinate $(1050, -0.633)$. What does point P indicate in the context?
- b. The point labeled Q is at the coordinate $(1420, 0.3)$. What does point Q indicate in the context?

25. Use the tables to find the given values.

a. $g(f(4)) =$

b. $(f \circ g)(0) =$

x	$g(x)$
-5	4
-2	10
0	-2
4	8

x	$f(x)$
-5	5
-2	9
0	0
4	-2

Let $f(x) = 3x^2 - 4x$ and $g(x) = 5 - 3x$.

26. Find $f(g(-1))$.

27. Find $(g \circ f)(1)$.

28. Let $f(x) = 7x - 5$ and $h(x) = f(g(x))$. Fill in the table.

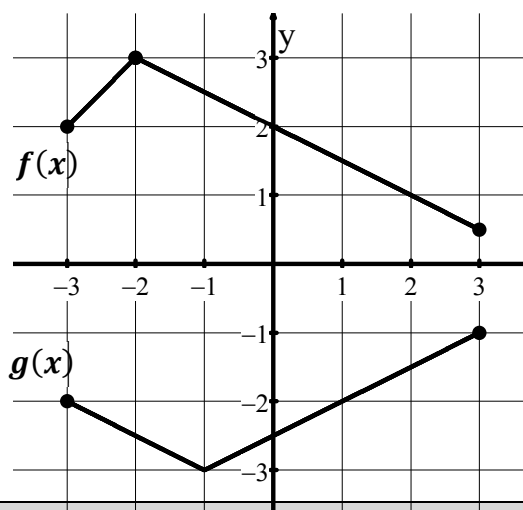
x	$g(x)$	$h(x)$
-5	2	
0	-1	
5	3	

Use the graph to find the following values.

29. Find $f(g(1))$.

30. Find $f(g(-3))$.

31. Find $g(f(2))$.



Let $f(x) = \frac{1}{x-6}$ and $g(x) = 1 - x$.

32. Find $f \circ g$

34. Find $g \circ f$

33. State the domain of $f \circ g$

35. State the domain of $g \circ f$

36. Express h as a composition of two simpler functions f and g where $h(x) = f(g(x))$.

$$h(x) = \sqrt{7 - x^2}$$

$$f(x) =$$

$$g(x) =$$

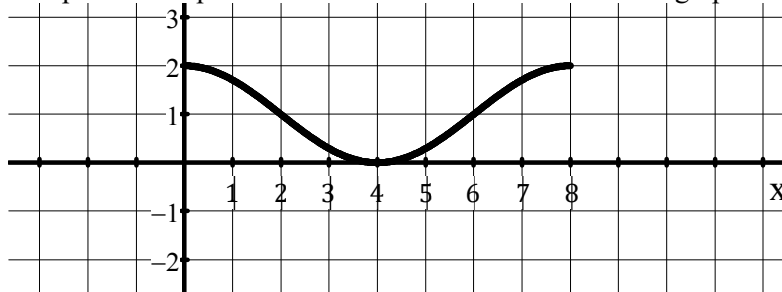
37. Find the inverse of $f(x) = (x - 2)^2 + 1$ for $x \leq 2$.
List the domain and range of $f^{-1}(x)$.

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Unit 3A Review – Trigonometric and Polar Functions

Reviews do NOT cover all material from the lessons but will hopefully remind you of key points. To be prepared, you must study all packets for lessons 3.1 – 3.2.

1. The graph below shows one period of a periodic function. Sketch the rest of the graph on the given axes.



- a. Is the function above increasing, decreasing or both on the interval $20 < x < 24$.

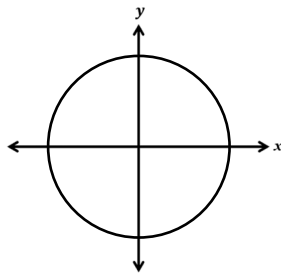
The measurement of an angle in standard position is listed. In which quadrant is the terminal ray?

2. 6.6π

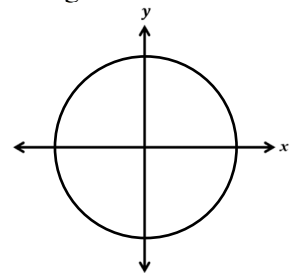
3. -2.3π

Below are various measurements of a circle's radius, an angle within the circle, or the arc subtended by the angle. SKETCH the approximate angle on the axes and find the missing value.

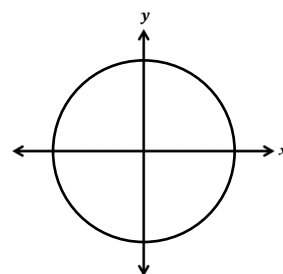
4. Radius is 8.2 and the length of an arc subtended by an angle is 37.63. Find the measure of the angle.



5. Radius is 5.27 and an angle is 0.56π radians. Find the length of the arc subtended by the angle.

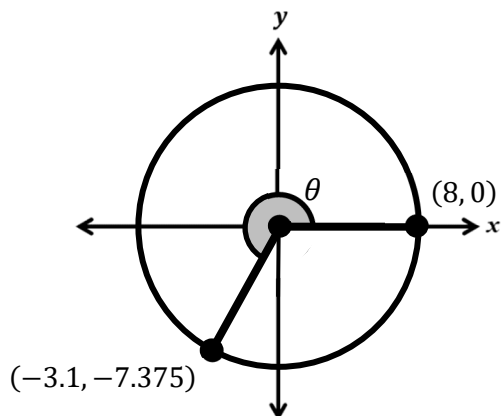


6. An angle is 3.56 radians and the length of an arc subtended by the angle is 5.2. What is the radius of the circle?



Each figure below gives a circle in the xy -plane with center at the origin, and an angle θ in standard position. Find the value of each expression.

7.

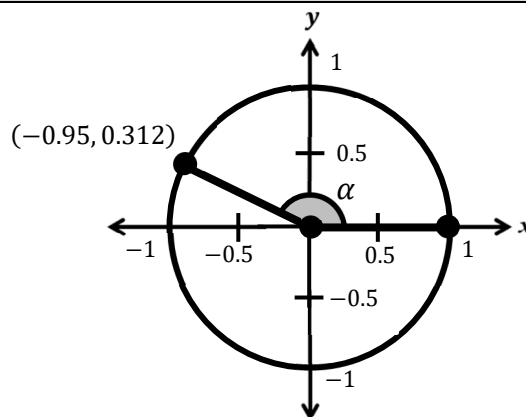


a. $\sin \theta =$

b. $\cos \theta =$

c. $\tan \theta =$

8.



a. $\sin \alpha =$

b. $\cos \alpha =$

c. $\tan \alpha =$

Find the value of each expression. Do not look back at the Unit Circle for help!

9. $\cos \pi$

10. $\sin \frac{\pi}{4}$

11. $\cos \frac{2\pi}{3}$

12. $\cos \frac{7\pi}{6}$

13. $\sin \left(-\frac{\pi}{6}\right)$

14. $\cos \left(-\frac{5\pi}{4}\right)$

For each problem, an angle in standard position in the xy -plane is given in radians. A circle is centered at the origin with the given radius. What are the coordinates of the point of intersection of the terminal ray of the angle and the circle?

15. $\theta = \frac{3\pi}{2}, r = 6$

16. $\theta = \frac{5\pi}{3}, r = 10$

In the xy -plane, the terminal ray of angle θ in standard position intersects a circle of radius r at the given point. What are the values of θ and r , with $0 \leq \theta \leq 2\pi$?

17. $(\sqrt{3}, -1)$

18. $(-6, 6\sqrt{3})$

The function f is given by $f(\theta) = \cos \theta$. Describe the concavity of f on the interval, and if f is increasing or decreasing on the interval.

19. $\frac{\pi}{2} < \theta < \pi$

20. $\frac{3\pi}{2} < \theta < 2\pi$

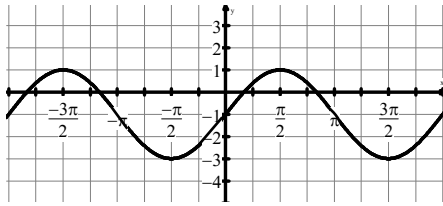
The function f is given by $f(\theta) = \sin \theta$. Describe the concavity of f on the interval, and if f is increasing or decreasing on the interval.

21. $0 < \theta < \frac{\pi}{2}$

22. $\pi < \theta < \frac{3\pi}{2}$

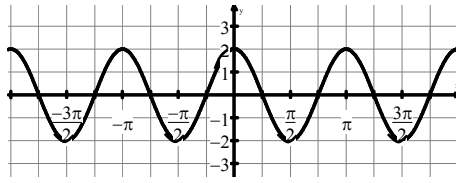
Write the given function for each graph. Use a positive coefficient for all equations.

23. Write a **sine** function.



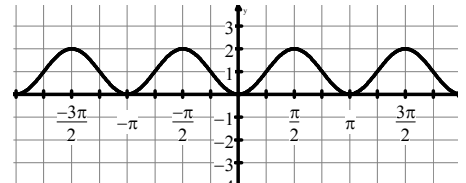
$y =$ _____

24. Write a **cosine** function.



$y =$ _____

25. Write a **sine** function.



$y =$ _____

State the amplitude, period, phase shift, and vertical shift.

26. $y = 3 \cos(5x - \pi)$

Amp: _____ Period: _____

Midline: _____ Freq: _____

Max value: _____ Min value: _____

Phase shift: _____

27. $y = \frac{1}{2} \sin(4x) - 2$

Amp: _____ Period: _____

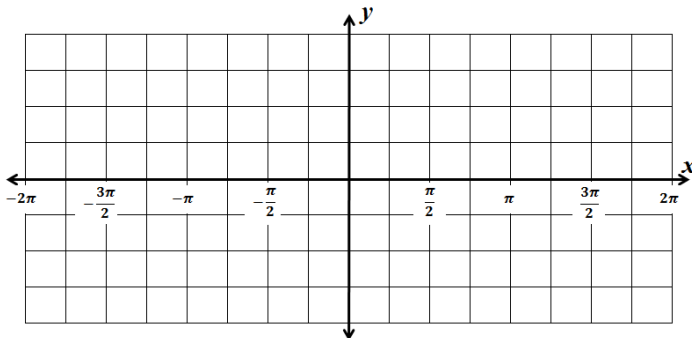
Midline: _____ Freq: _____

Max value: _____ Min value: _____

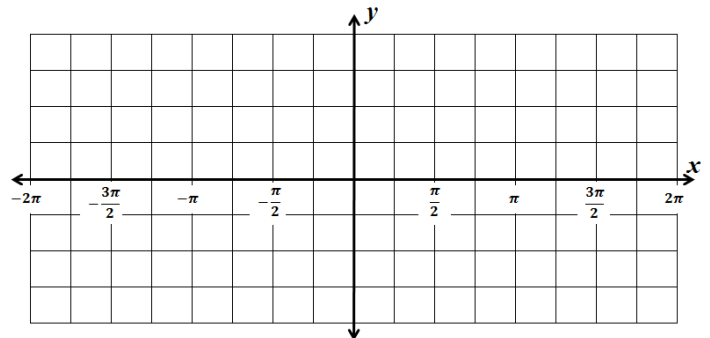
Phase shift: _____

Graph each function on the given set of axes.

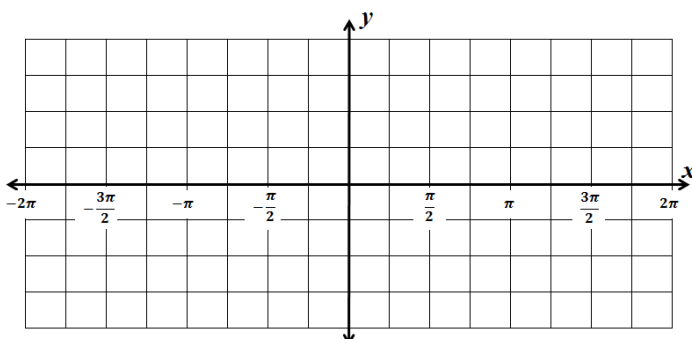
28. $y = 2 \cos x$



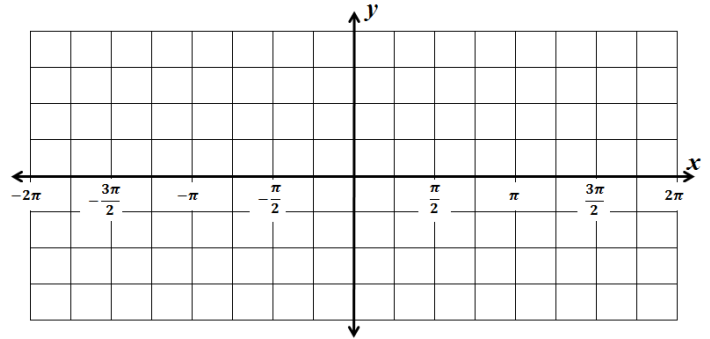
29. $y = 3 \sin 2x - 1$



30. $y = -3 \sin\left(\frac{x}{2} + \pi\right)$



31. $y = 2 \cos(2x - \pi) - 1$



32. Suppose you are riding a Ferris wheel. After everyone is loaded, the wheel starts to turn, and the ride lasts for 105 seconds. Your height h (in feet) above the ground at any time t (in seconds) can be modeled by the equation $h(t) = 50 \sin \left[\frac{\pi}{10} (t - 4) \right] + 65$. You do not need a calculator for any question except the last two.
- What is the period?
 - What does the period represent?
 - What is the frequency?
 - What does the frequency represent?
 - What is your maximum height?
 - What is your minimum height?
 - How many circles will the Ferris Wheel make during the ride?
 - Calculator active.** How high are you when the ride begins?
 - Calculator active.** What is your height when the ride stops?

33. The following data set can be modeled by a sinusoidal function. Use the data to answer each problem below.

x	0	1	2	3	4	5	6	7	8	9
$f(x)$	16	2	-10	-1	18	21	40	52	37	29

x	10	11	12	13	14	15	16	17	18	19
$f(x)$	12	-2	-11	4	17	22	38	49	39	26

- | | | |
|---------------------------------------|---|----------------------------|
| a. Estimate the period and frequency. | b. Estimate the vertical shift (midline). | c. Estimate the amplitude. |
|---------------------------------------|---|----------------------------|

-
- d. Using the above information, create a sinusoidal function model.

-
- e. Using a calculator, find a sinusoidal model from the given data set. Your answer should look similar to your estimate in part d.

Name: _____ Date: _____ Period: _____

Unit 4A Review – Functions Involving Parameters, Vectors, and Matrices

Reviews do NOT cover all material from the lessons but will hopefully remind you of key points. To be prepared, you must study all packets for lessons 4.1 – 4.7.

1. Complete the table of numerical values for the given values of t using the parametric function $f(t) = (1 - t^2, 2t)$.

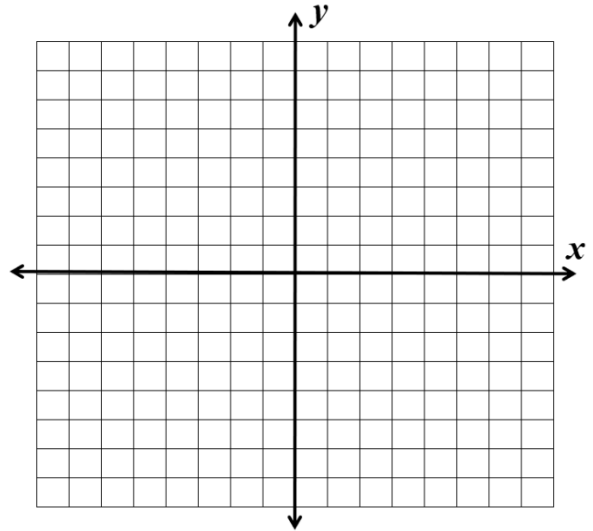
t	-3	-2	-1	0	1	2
x						
y						

2. An object is moving in the xy -plane so that at any time t , the position of the object can be found by evaluating the parametric equations $x(t) = t$ and $y(t) = -(t - 2)^2 + 3$.

- a. Graph the curve represented by the given parametric function for the restricted domain $-1 \leq t \leq 4$.

- b. Find the horizontal relative extrema.

- c. Find the vertical relative extrema.

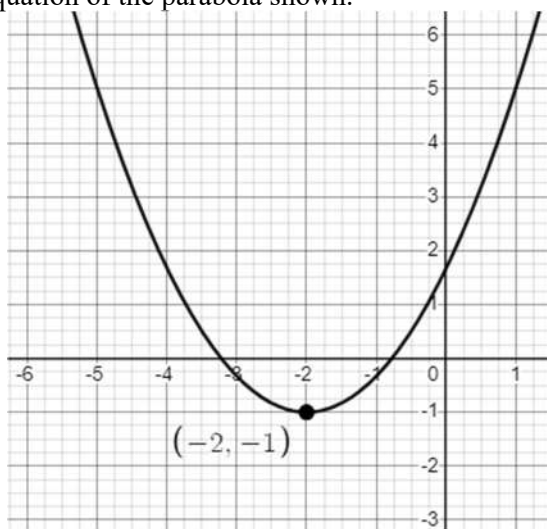


- d. Find the x -intercept(s). Show your work.

- e. Find the y -intercept(s). Show your work.

3. A particles position for a given value of t can be found using the parametric equations $x(t) = t + 1$ and $y(t) = -t^2 + 4$, over the interval $-2 \leq t \leq 0$. Find the following.
- The average rate of change of $x(t)$.
 - The average rate of change of $y(t)$.
 - The slope of the graph between the points on the graph that correspond to $t = -2$ and $t = 0$.
4. Without the use of technology, determine which set of parametric equations will produce the same path as $f(t) = (t^2 + 2t + 1, t + 1)$, but will have a direction of particle motion in the opposite direction?
- $x(t) = -t^2 + 2t + 1, y(t) = -t + 1$
 - $x(t) = t^2 - 2t + 1, y(t) = -t + 1$
 - $x(t) = t^2 - 2t + 1, y(t) = -t - 1$
 - $x(t) = t + 1, y(t) = t^2 + 2t + 1$
5. Find the parametric equations for the circle with the center at $(3, -2)$ and a radius of 2.
6. Find the parametric equations for the linear path of a particle that travels from the point $(-1, 3)$ to the point $(1, 7)$.

7. Find the change of y with respect to x and determine how the two quantities in the implicitly defined function $15x^2 + y^2 - 20 = 0$ vary together on the interval $0 \leq x \leq 1$, and $y \geq 0$.
8. Find the vertex and orientation for the parabola given by $(x - 2) = -\frac{2}{3}(y - 4)^2$.
9. Find the vertex and orientation for the parabola $x^2 - 2x + 2y + 3 = 0$. Show all your work to support your answer.
10. Which of the following is the equation of the parabola shown.



(A) $(x + 1) = \frac{2}{3}(y + 2)^2$

(B) $(x - 1) = -\frac{2}{3}(y + 2)^2$

(C) $(y + 1) = \frac{2}{3}(x + 2)^2$

(D) $(y - 1) = -\frac{2}{3}(x - 2)^2$

11. Use the equation $\frac{(x-2)^2}{4} + \frac{(y-5)^2}{16} = 1$ to find the following of the ellipse.

a. center

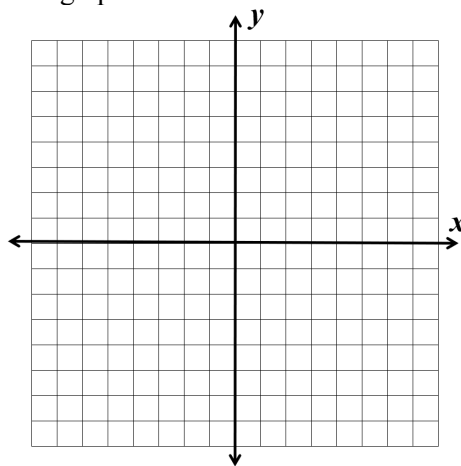
b. horizontal or vertical

c. vertices (ends of major axis)

d. ends of minor axis

e. foci

f. sketch the graph



12. Put the equation of an ellipse into standard form and then identify the center and orientation of the ellipse.
 $x^2 + 25y^2 - 4x - 50y + 4 = 0$.

Center: _____

Orientation: _____

13. Find the parametrization of the conic given by $25(x + 2)^2 + 2(y - 5)^2 = 100$.

14. Use the equation $\frac{(x-1)^2}{16} - \frac{(y-3)^2}{4} = 1$ to find the following of the hyperbola.

a. center

b. horizontal/vertical

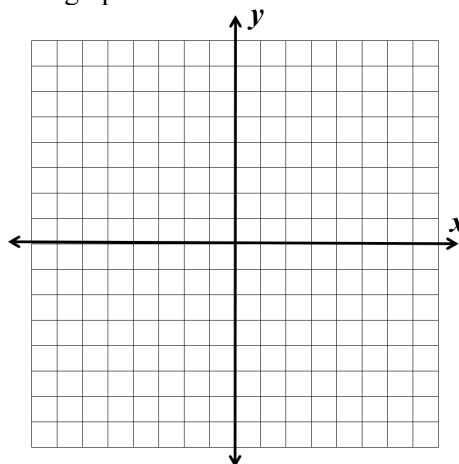
c. Find the length of the transverse axis.

d. Find the length of the conjugate axis.

e. vertices

f. foci

g. sketch the graph



15. Find the parametrization of the conic given by $\frac{(y-4)^2}{49} - \frac{(x-5)^2}{81} = 1$.

16. Which of the following represents the parametric equations $x(t) = t + 1, y(t) = 2t^2 + t + 1$, written in rectangular form?

(A) $y = 2x^2 + 4x + 3$

(B) $y = 2x^2 + 5x + 3$

(C) $y = 2x^2 - 4x + 2$

(D) $y = 2x^2 - 3x + 2$

17. Which of the following represents the parametrization of the conic $x^2 + 2x + 3 - y = 0$?

(A) $(t, (t + 1)^2 + 2)$

(B) $(t, (t + 1)^2 + 3)$

(C) $(t, (t - 1)^2 + 2)$

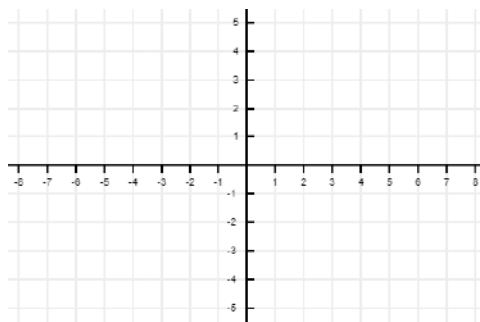
(D) $(t, (t - 1)^2 + 3)$

A river flows directly north with a current that is 12 mph. A ferry boat leaves the west edge of the river and heads 40° north of east at a speed of 30 mph. What is the actual speed and direction of the boat?

7. For the given vector-valued functions,

a. Complete the table and sketch the graph that the endpoints make, $f(t) = \left\langle t - 5, \frac{4}{t} \right\rangle$.

t	x	y
-2		
-1		
0		
1		
2		



b. Find the domains of each of the components of the vector-valued function, then find the domain of the vector-valued function.

8. Describe the motion and find the speed of a particle in motion with the following vector at the given time.

$$v(t) = \langle -2 \cos t, 5 \sin t \rangle, t = \frac{\pi}{3}$$

9. Simplify the following matrices to a single matrix. $-2 \begin{bmatrix} 1 & -4 \\ -3 & 8 \end{bmatrix} + 5 \begin{bmatrix} 0 & 3 \\ -4 & 6 \end{bmatrix}$

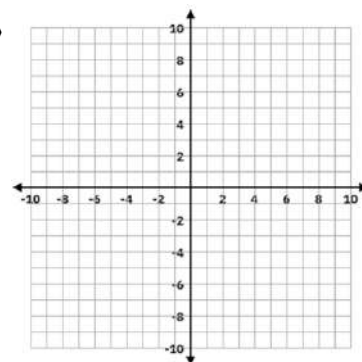
10. Multiply the following matrices. No Calculators. $\begin{bmatrix} 5 & 1 & -3 \\ -3 & 10 & 4 \end{bmatrix} \cdot \begin{bmatrix} 2 & 5 \\ -1 & -2 \\ 0 & 3 \end{bmatrix}$

11. Find the inverse of each matrix if possible.

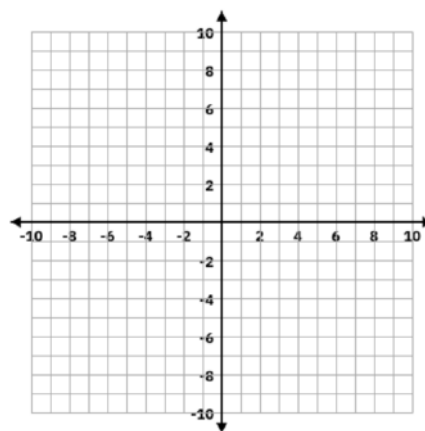
a. $\begin{bmatrix} 5 & 6 \\ 3 & 4 \end{bmatrix}$

b. $\begin{bmatrix} -6 & 4 \\ -3 & 2 \end{bmatrix}$

12. Plot the parallelogram formed by the vectors and then find the area. $\langle -5, 2 \rangle$ and $\langle 8, 7 \rangle$



13. Apply the transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ if $A = \begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix}$.
- a. If $\vec{u} = \langle -4, -3 \rangle$, find $T(\vec{u})$ and sketch both $\vec{u}$ and $T(\vec{u})$.



- b. If $\vec{v} = \langle 5, 1 \rangle$, find $T(\vec{v})$ and sketch both $\vec{v}$ and $T(\vec{v})$.

- c. Describe the transformation that occurs.

- d. What is the general transformation that occurs to $\langle x, y \rangle$.

14. Given the linear transformation, find the associated matrix with that transformation.
 $\langle x, y \rangle$ to $\langle x - 3y, 2x + 4y \rangle$

15. Find the linear transformation given the associated matrix.

$$\begin{bmatrix} -2 & 5 \\ 0 & 2 \end{bmatrix}$$

16. Directions: Find the resulting vector from rotating the given vector by the given angle. $\vec{u} = \langle 4, -6 \rangle$ rotated $\frac{\pi}{2}$ radians counterclockwise.

17. Considering the given transformation, what is the image of the given vector under the transformation. The x-coordinate triples, the y-coordinate doubles and $\vec{u} = \langle 2, -4 \rangle$.

18. Matrix A and B represent the transformations T and U respectively. Find the associated matrix for the composition of the function and then find the vector after the given transformation. Find the associated matrix and $T(U(\vec{v}))$. $A = \begin{bmatrix} 2 & -1 \\ -2 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 1 \\ -2 & 3 \end{bmatrix}$, $\vec{v} = \langle 3, -4 \rangle$

19. Given $\vec{v}$ find the vector $\vec{u}$, that was transformed by matrix A to get $\vec{v}$. $\vec{v} = \langle 6, 4 \rangle$ and $A = \begin{bmatrix} -2 & 3 \\ -2 & 4 \end{bmatrix}$.

20. FedEx did some market research and found that in the southeast United States, they roughly lost about 15% of its customers to UPS from one year to the next. UPS confirmed that they lost about 18% of its customers to FedEx of their customers' from one year to the next. Currently UPS accounts for about 75% of all business in the given area, while FedEx accounts for about 15%.

A) Find a transition matrix.

B) What percent of all business does each company account for after 1 year? 3 years?

C) What will be the eventual long-term distribution of business for the companies?

D) What percent did each company account for last year?

Name: _____ Date: _____ Period: _____

Review

End-of-Unit 2B Review – Exponential and Logarithmic

Lessons 2.9 through 2.15

Reviews do NOT cover all material from the lessons but will hopefully remind you of key points. To be prepared, you must study all packets from Unit 2B.

1. The function f is given by $f(x) = 3\log_2(x) + 8$. Find $f^{-1}(x)$.
2. **Calculator Active:** What is the solution to the equation $4 \cdot 2^{x+5} - 10 = 22$.
3. What are all of the solutions to the inequality: $\ln(x + 10) + \ln 5 \leq \ln(2x + 17)$
4. **Calculator Active:** Mr. Sullivan used to collect football cards. When the Cleveland Browns drafted Johnny Manziel he just knew his rookie card would be worth a lot of money over time! Mr. Sullivan kept track of the value after every game Johnny Manziel played. The data is below.

Games Played	1	3	4	9
Value of Rookie Card (\$)	57.50	37.77	32.74	18.55

- a) What's a logarithmic regression equation that could model this situation?
- b) How many games would Johnny Manziel need to play for his card to be worth \$10?

5. Let a , b , and c be positive constants. What is an equivalent expression to $\log\left(\frac{a^5 b^2}{c^3}\right)$?

6. **Calculator Active:** The function f is given by $f(x) = \log_3(x)$. The function g is given by $g(x) = \log_3(81x)$. Rewrite the function using properties of logarithms and explain any transformations from $f(x)$ to $g(x)$.

7. Using the tables below, determine if the data given is exponential, logarithmic or linear.. Explain your answer.

a.

x	5	7	11	19
y	1	2	3	4

b.

x	1	2	3	4
y	1	2	4	8

c.

x	2	3	4	5
Log y	3	7	11	15

8. The function f is given by $f(x) = \log(3 - x) + 2$. Find the information below for f .

Asymptote:

Domain:

Range:

End Behavior:

Sketch of f :

9. FRQ: Directions: NO CALCULATOR

- Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.
 - Solutions to equations must be real numbers. Determine the exact value of any expression that can be obtained without a calculator. For example, $\log_2 8$ can be evaluated without a calculator.
 - Unless otherwise specified, combine terms using algebraic methods and rules for exponents and logarithms, where applicable. For example, $2x + 3x$, $5^2 \cdot 5^3$, $\frac{x^5}{x^2}$, and $\ln 3 + \ln 5$ should be written in equivalent forms.
 - For each part of the question, show the work that leads to your answers.
-

(A) The functions g and h are given by: $g(x) = \log_2(x^3) + \log_4(x)$ and $h(x) = \frac{4^{x+1}2^{3x}}{2^{2x-3}}$

i. Rewrite $g(x)$ as a single logarithm without negative exponents in any part of the expression.

ii. Rewrite $h(x)$ as an expression as a single base with an exponent.

(B) The function k is given by: $k(x) = 5 + 6 \log_4(x)$. Solve $k(x) = 17$ for values of x in the domain of k .

(C) The function m is given by $m(x) = 2 \ln(x) - \ln(7x - 10)$. Find all input values in the domain of m that yield an output value of 0.

10. Calculator Active

x	2	4	8	16	32
$f(x)$	8	16	24	32	40

Let f be an increasing function defined for $x > 0$. The table above gives values for $f(x)$ at selected values of x . The function g is given by $g(x) = 8.25 - 2^{x-6}$.

(A)

- i. The function h is defined by $h(x) = (g \circ f)(x) = g(f(x))$. Find the value of $h(2)$ as a decimal approximation, or indicate that it is not defined. Show the work that leads to your answer.
- ii. Find the value of $f^{-1}(40)$ or indicate that it is not defined.

(B)

- i. Find all values of x , as decimal approximations, for which $g(x) = 4$ or indicate there are no such values.
- ii. Determine the end behavior of g as x decreases without bound. Express your answer using the mathematical notation of a limit.

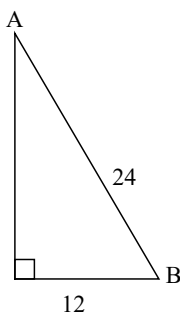
(C)

- i. Use the table of values of $f(x)$ to determine if f is best modeled by a linear, quadratic, exponential, or logarithmic function.
- ii. Give a reason for your answer in part C (i) based on the relationship between the change in the output values of f and the change in the input values of f . Refer to the values in the table in your reasoning.

Pre-Calculus 11 - Final Exam Review**Multiple Choice**

Identify the choice that best completes the statement or answers the question.

- _____ 1. What is the exact sine of $\angle A$?



- | | |
|-------------------------------|-------------------------------|
| A $\frac{1}{\sqrt{3}}$ | C $\frac{2}{\sqrt{3}}$ |
| B $\frac{1}{3}$ | D $\frac{1}{2}$ |

- _____ 2. Determine, to the nearest tenth of a degree the measure of $\angle C$.

- | | |
|-----------------------|------------------------|
| A 25.3° | C 140.7° |
| B 55.5° | D 154.7° |

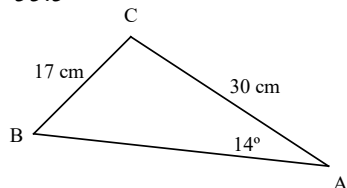


Diagram not drawn to scale.

- _____ 3. What is the reference angle for 200° in standard position?

- | | |
|----------------------|----------------------|
| A 100° | C 20° |
| B 70° | D 110° |

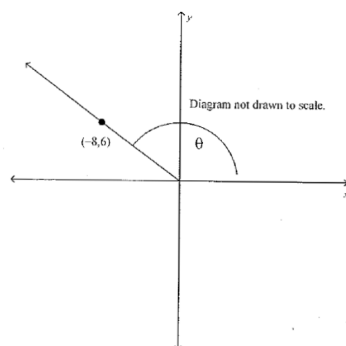
- _____ 4. What are the three other angles in standard position that have a reference angle of 54° ?

- | | |
|--|--|
| A $99^\circ, 144^\circ, 234^\circ$ | C $144^\circ, 234^\circ, 324^\circ$ |
| B $108^\circ, 162^\circ, 216^\circ$ | D $126^\circ, 234^\circ, 306^\circ$ |

- _____ 5. What is the exact value for $\tan 240^\circ$?

- | | |
|-------------------------------|---------------------|
| A $\frac{1}{\sqrt{3}}$ | C 1 |
| B $-\sqrt{3}$ | D $\sqrt{3}$ |

6. The coordinates of point P on the terminal arm of an angle are shown. What are the exact trigonometric ratios



- A $\sin A = -\frac{4}{5}, \cos A = \frac{3}{5}, \tan A = -\frac{4}{3}$
 B $\sin A = \frac{5}{3}, \cos A = -\frac{5}{4}, \tan A = -\frac{3}{4}$
 C $\sin A = \frac{3}{5}, \cos A = -\frac{4}{5}, \tan A = -\frac{3}{4}$
 D $\sin A = \frac{4}{5}, \cos A = -\frac{3}{5}, \tan A = -\frac{3}{4}$

7. An angle is in standard position such that $\cos \theta = \frac{1}{9}$. What are the possible values of θ , to the nearest degree, if $0^\circ \leq \theta \leq 360^\circ$?

- A 6° and 174°
 B 6° and 276°

- C 84° and 264°
 D 84° and 276°

8. Determine the length of x , to the nearest tenth of a centimetre.

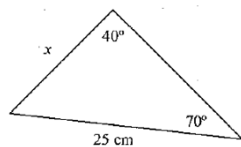


Diagram not drawn to scale.

- A 26.6
 B 36.5

- C 11.2
 D 17.1

9. Determine the measure of x , to the nearest tenth of a degree.

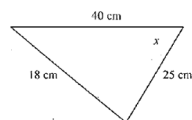
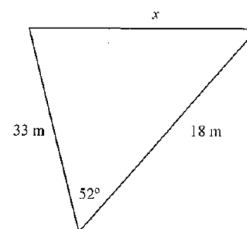


Diagram not drawn to scale.

- A 25.6°
 B 18.1°

- C 136.3°
 D 71.9°

10. Determine the length of x , to the nearest tenth of a metre.



- A 27.7
B 21.8
C 26.1
D 37.6

11. Solve the following triangle, rounding side lengths to the nearest tenth of a unit and angles measures to the nearest degree.

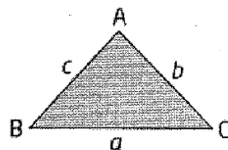


Diagram not drawn to scale.

$$\angle A = 152^\circ, b = 19, a = 23.5$$

- A $\angle B = 22^\circ, \angle C = 6^\circ, c = 5.0$
B $\angle B = 158^\circ, \angle C = 84^\circ, c = 5.0$
C $\angle B = 68^\circ, \angle C = 174^\circ, c = 28.7$
D $\angle B = 35^\circ, \angle C = 7^\circ, c = 28.7$

12. What is the function $y = 2(x - 4)^2 - 2$ written in standard form?

- A $y = 2x^2 - 8x + 30$
B $y = 2x^2 - 8x + 34$
C $y = 2x^2 - 16x + 34$
D $y = 2x^2 - 16x + 30$

13. Which function is **not** quadratic?

- A $f(x) = (6x + 9)\left(\frac{1}{9}x - 9\right)$
B $f(x) = x(x - 9)(6x + 8)$
C $f(x) = 7x^2 + 8$
D $f(x) = 6(x - 9)^2$

14. What is the equation of the quadratic function $y = x^2 - 26x + 41$ in vertex form?

- A $y = -(x + 13)^2 - 210$
B $y = -(x - 13)^2 - 210$
C $y = (x + 13)^2 - 128$
D $y = (x - 13)^2 - 128$

15. Factor $-4x^2 + 68x - 120$ completely.

- A $-4(x - 2)(x - 15)$
B $-4(x + 2)(x + 15)$
C $-4(x + 2)(x - 15)$
D $-4(x - 2)(x + 15)$

16. Solve $-8x^2 + 120x + 432 = 0$.

A $x = 18$ and $x = -3$

C $x = \frac{9}{4}$ and $x = -\frac{3}{8}$

B $x = -18$ and $x = 3$

D $x = -144$ and $x = 24$

17. Determine the roots of the quadratic equation $-5x^2 + 55x = 50$.

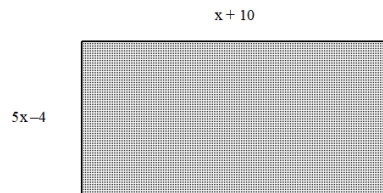
A $x = -10$ and $x = -1$

C $x = 10$ and $x = 1$

B $x = -50$ and $x = -5$

D $x = 2$ and $x = \frac{1}{5}$

18. A rectangle has dimensions $x + 10$ and $5x - 4$, where x is in centimetres. If the area of the rectangle is 72 cm^2 , what is the value of x , to the nearest tenth of a centimetre?



A $x = 2.0$

C $x = 11.2$

B $x = -4.6$

D $x = -11.2$

19. Solve $(x + 1)^2 = 43$.

A $1 + \sqrt{43}$ and $1 - \sqrt{43}$

C $2\sqrt{11}$

B $-1 + \sqrt{43}$ and $-1 - \sqrt{43}$

D $\sqrt{42}$

20. The roots, to the nearest hundredth, of $y = -\frac{1}{2}x^2 - 2x + \frac{7}{10}$ are

A -8.65 and 0.65

C -2.16 and 0.16

B 4.32 and -0.32

D -4.32 and 0.32

21. What does the expression $7\sqrt{7} - 6\sqrt{12} - (4\sqrt{28} + 4\sqrt{3})$ simplify to?

A $15\sqrt{7} - 16\sqrt{3}$

C $-\sqrt{7} - 16\sqrt{3}$

B $15\sqrt{7} + 16\sqrt{3}$

D $-\sqrt{7} + 16\sqrt{3}$

22. Express $\sqrt[5]{64n^{10}m^{15}}$ in simplified form.

A $4n^2m^3(\sqrt[5]{4})$

C $4n^2m^3(\sqrt[5]{2})$

B $2n^3m^2(\sqrt[5]{5})$

D $2n^2m^3(\sqrt[5]{2})$

23. Express $\frac{2\sqrt{21} - 3\sqrt{7}}{\sqrt{7}} + \frac{4\sqrt{3} - 8}{\sqrt{4}}$ in simplest form.

A $6\sqrt{3} - 5$

C $4\sqrt{3} - 7$

B $6\sqrt{21} - 14\sqrt{7}$

D $2\sqrt{21} - 3\sqrt{7} + 4\sqrt{3} - 2$

24. The volume, V , in cubic units of a cylinder is given by $V = \pi r^2 h$, where r is the radius and h is the height, both in the same units. Find the exact radius of a cylinder with a height of 64 cm and a volume of $576\pi \text{ cm}^3$. Express your answer in simplest form.

A $\frac{1}{\sqrt{64}} \text{ cm}$

B 8 cm

C 9 cm

D 3 cm

25. Express $-7\sqrt{6}(-6\sqrt{5} - 2\sqrt{6})$ in simplest form.

A $14\sqrt{6} + 42\sqrt{30}$

B 252

C $42\sqrt{30} + 84$

D $1260 + 14\sqrt{6}$

26. Determine the range of the function $y = \sqrt{x-2}$.

A $\{y \in R\}$

B $\{y \in R, y \geq 0\}$

C $\{x \in R\}$

D $\{x \in R, x \geq -2\}$

27. Solve $\sqrt{4x-5} = 6$

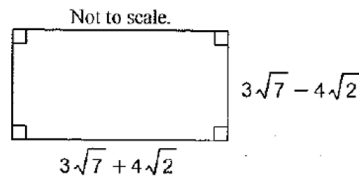
A $x = \frac{121}{16}$

B $x = \frac{11}{16}$

C $x = \frac{121}{4}$

D $x = \frac{11}{4}$

28. Find a simplified expression for the area of this shape.



A $9\sqrt{7} + 16\sqrt{2}$

B $9\sqrt{7} - 16\sqrt{2}$

C 95

D 31

29. What are the restrictions on x if the solution to the equation $-4 - \sqrt{4-x} = 6$ involves real numbers?

A $x \leq 10$

B $x \geq 6$

C $x \leq 4$

D $x \geq 100$

30. The non-permissible value(s) for the rational expression $\frac{12}{x^2-4}$ is/are

A $x \neq \pm 2$

B $x \neq 2\sqrt{3}$

C $x \neq 2$

D $x \neq 4$

___ 38. Express the quotient $\frac{x^2-5x-24}{x^2-11x+24} \div \frac{2x^2+7x+3}{x^2+x-12}$ in simplest form

A $\frac{2x+1}{x+4}$

B $\frac{x+4}{2x+1}$

C $\frac{(x+3)(2x+1)}{(x-3)(x+4)}$

D $\frac{(x-3)(x+4)}{(x+3)(2x+1)}$

___ 39. What is $\frac{5(4x^2-y^2)}{2x^2-15xy-8y^2}$

A $\frac{5(2x+y)}{x-8y}, x \neq -\frac{y}{2}, x \neq 8y$

B $\frac{5(2x+y)}{x+8y}, x \neq -\frac{y}{2}, x \neq -8y$

C $\frac{5(2x-y)}{x+8y}, x \neq \frac{y}{2}, x \neq -8y$

D $\frac{5(2x-y)}{x-8y}, x \neq -\frac{y}{2}, x \neq 8y$

___ 40. Solve the rational equation $\frac{x}{x+1} = \frac{4-x}{x^2-3x-4} + \frac{6}{x-4}$.

A $x = 10$

B $x = 10$ and -1

C $x = -10$

D $x = -10$ and 1

___ 41. Solve $\frac{x^2-12x+32}{-20-x} = \frac{x^2-13x+40}{x-19}$

A $x = 6, -2,$ and 8

B $x = -20, -19,$ and -8

C $x = -6$ and 2

D $x = 16$

___ 42. The number of real roots for the equation $y = -30.6x^2 + 30.7x - 39.8$ is

A 2

B 0

C 1

D impossible to tell

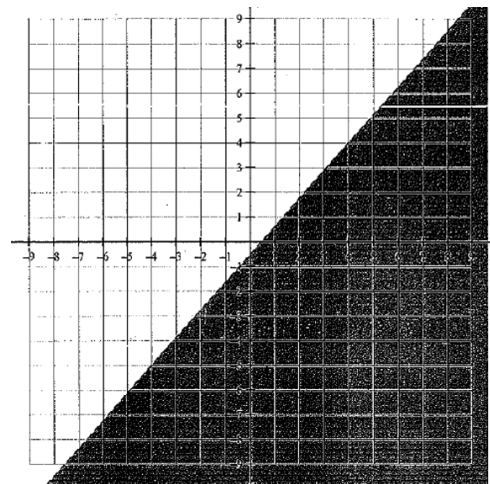
___ 43. Which inequality represents the graph to the right?

A $y > \frac{8}{9}x - 2$

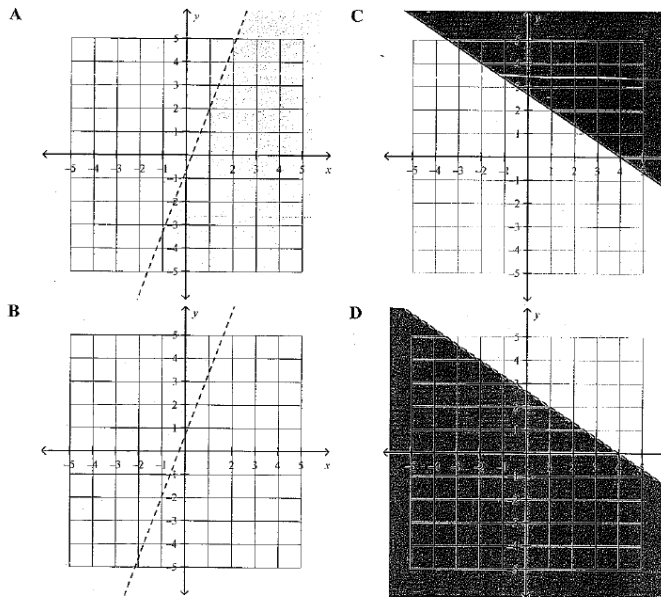
B $y < \frac{8}{9}x - 2$

C $y > \frac{9}{8}x - \frac{1}{2}$

D $7 < \frac{9}{8}x - \frac{1}{2}$



44. Which graph represents the solution to $y < -\frac{2}{3}x + \frac{8}{3}$ is



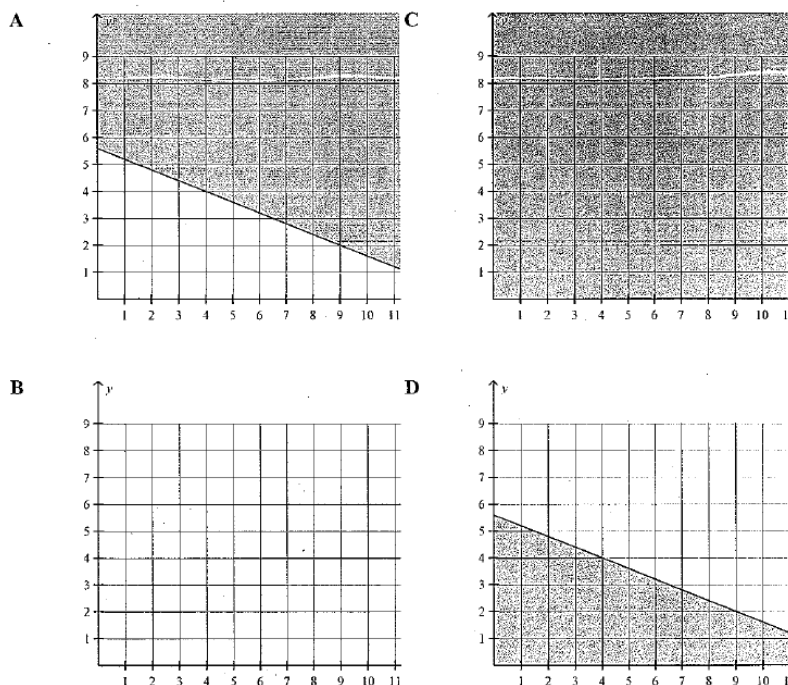
For questions #45-46, consider the following situation: A sports store makes a profit of \$50 on every pair of cross-country skis sold and \$125 on every set of snowshoes sold. The manager's goal is to have a profit of at least \$700 a day from the sales of these two items.

45. If x represents the number of pairs of cross-country skis sold and y represents the number of pairs of snowshoes sold, what inequality models the combination of ski and snowshoe sales that will meet or exceed the daily profit goal?

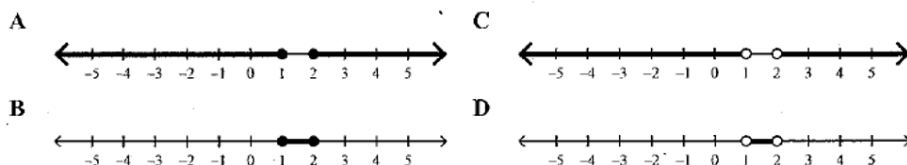
- A $50y + 125x \leq 700$
 B $50y + 125x > 700$

- C $50y + 125x \geq 700$
 D $50y - 125x < 700$

46. Which graph represents the combinations of ski and snowshoe sales that will meet or exceed this daily goal?



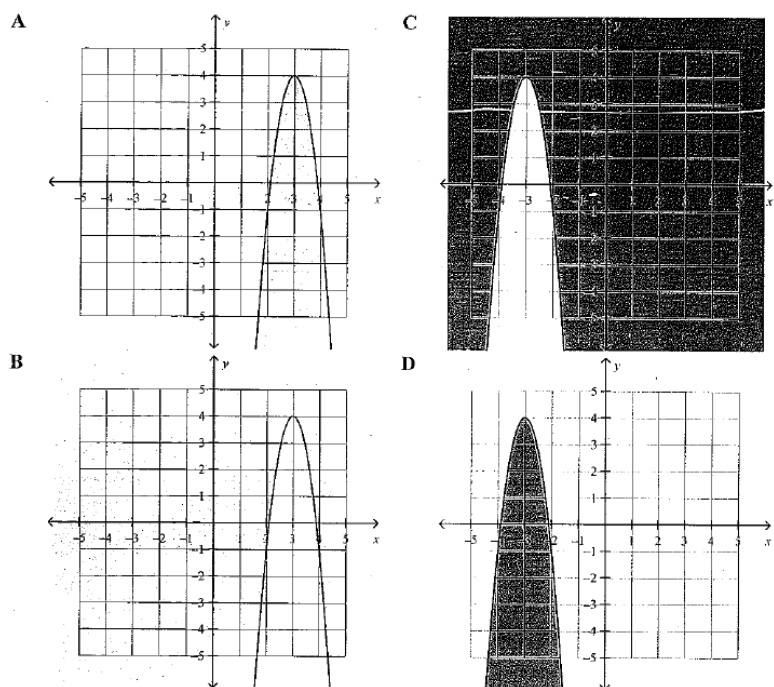
47. Which graph represents the solution to the inequality $2x^2 - 6x + 4 \geq 0$?



48. The solution set to the inequality $-2x^2 + 8x - 6 > 0$?

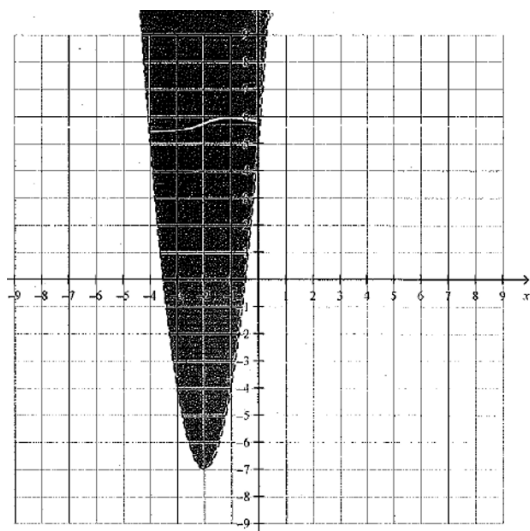
- A $\{x | 1 < x < 3, x \in \mathbb{R}\}$ C $\{x | x < 1, x > 3, x \in \mathbb{R}\}$
 B $\{x | -3 < x < -1, x \in \mathbb{R}\}$ D $\{x | x < -3, x > -1, x \in \mathbb{R}\}$

49. Which graph represents the solution to the inequality $y \leq -5(x + 3)^2 + 4$?



50. Which quadratic inequality is represented by the graph shown below?

- A $y > -3(x + 2)^2 - 7$
 B $y > 3(x + 2)^2 - 7$
 C $y > -3(x - 7)^2 - 2$
 D $y \leq 3(x - 7)^2 - 2$



Short Answer

- The point $(-5, 7)$ is located on the terminal arm of $\angle A$ in standard position.
 - Determine the primary trigonometric ratios for $\angle A$.
 - Determine the measure of $\angle A$ to the nearest degree.
- If $\angle \theta$ is a point on the unit circle with coordinates $\left(-\frac{\sqrt{3}}{3}, \frac{\sqrt{6}}{3}\right)$. Determine the measure of $\angle \theta$ to the nearest degree.
- Use your knowledge of special triangles and the unit circle to determine the exact primary trigonometric ratios for an angle of 300° . Make sure to rationalize any denominators.
- In $\triangle ABC$, $BC = 10$ cm, $AB = 12$ cm, and $\angle A = 45^\circ$. Determine the measure of $\angle ABC$.
- Solve the following $\triangle ABC$ if $BC = 8$ cm, $AC = 12$ cm, and $AB = 15$ cm.
- Express the quadratic function $y = -3x^2 + 12x - 10$ in vertex form.
- Find the value of k that makes the expression $x^2 + 22x + k$ a perfect square trinomial.
- Determine the number of real roots for the equation $3x^2 = 8x - 4$. Then, find the roots of the equation by
 - using the quadratic formula
 - factoring
- Simplify each expression.
 - $\sqrt{162}$
 - $5\sqrt{2} - 2\sqrt{5} + \sqrt{125} - \sqrt{8}$
 - $\sqrt{3}(4\sqrt{6} + 2\sqrt{3})$
 - $\sqrt{2}(2\sqrt{2} + 2) - 3(5\sqrt{2} + 1)$
- Rationalize the denominator in each expression. Express each answer in simplest form.
 - $\frac{2\sqrt{3} + 4}{\sqrt{3}}$
 - $\frac{5\sqrt{3} + \sqrt{2}}{2\sqrt{3} - \sqrt{2}}$
 - $\frac{2\sqrt{a} - 3\sqrt{b}}{2\sqrt{a} + 3\sqrt{b}}$

Simplify each expression and state any non-permissible values.

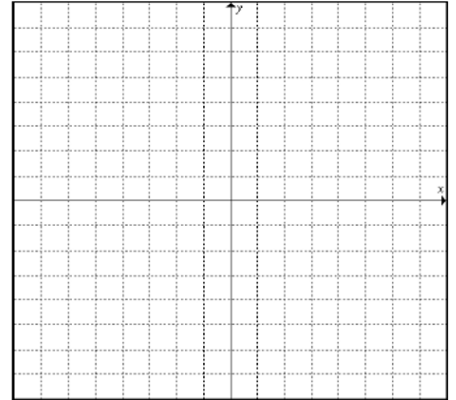
11. $\frac{2x^2 + x - 1}{6x^2 - x - 2} \times \frac{2x^2 + 3x + 1}{2x^2 + 5x - 3} \div \frac{x^2 + 2x + 1}{x + 3}$

12. $\frac{x}{x^2 - 3x - 4} - \frac{4}{x + 1}$

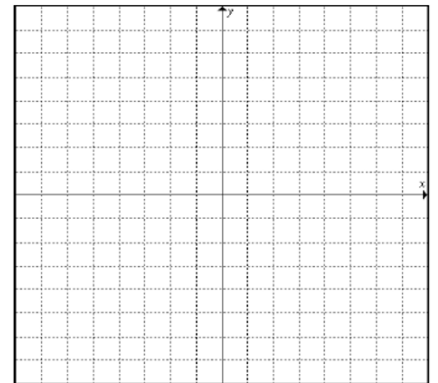
13. Find the x -intercepts of the quadratic function $y = 3x^2 - 10x + 6$. Express your answers as exact values.

14. Solve $4 - \sqrt{4 + x^2} = x$.

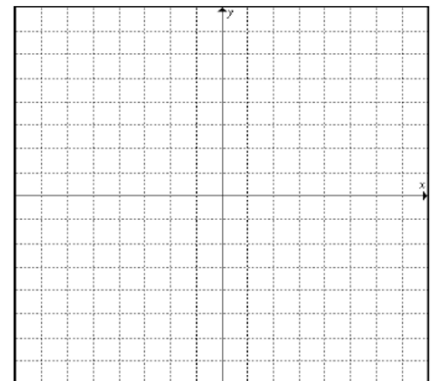
15. Solve. $1 + \frac{2x}{x+4} = \frac{3}{x-1}$
16. Solve the quadratic equation $x^2 = -\frac{7}{2}x - 3$, by factoring.
17. Solve the quadratic equation $2x^2 - 10x - 20 = 4x^2 + 6x$, by completing the square.
18. Solve the quadratic equation $x^2 + 4x - 1 = 0$, by using the Quadratic Formula.
19. Put the quadratic function $y = 2x^2 - 12x + 16$ into factored form. Then, graph the function, making sure to include the vertex, the coordinates of the x & y intercepts and at least one other point.



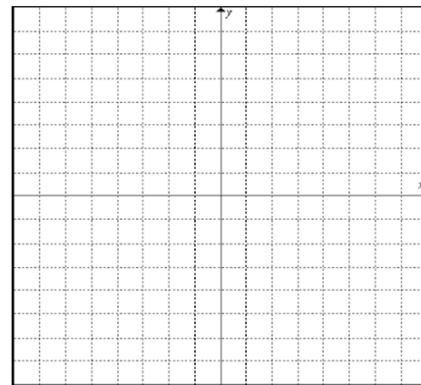
20. Put the quadratic function $y = -\frac{1}{2}x^2 - 2x - 1$ into vertex form. Then, graph the function, making sure to include the vertex, the axis of symmetry, and at least one other point. Determine the domain and range of the function.



21. Graph the linear inequality $2x - 3y < -6$.



22. Graph the quadratic inequality $2x^2 + 4x - 6 \leq y$.



23. A ball is thrown straight up from a bridge over a river and falls into the water. The height, h , in metres, of the ball above the water t seconds after being thrown is approximately modelled by the relation $h = -5t^2 + 10t + 35$.

- What is the maximum height of the ball above the water?
- How long does it take for the ball to reach the maximum height?
- After how many seconds does the ball hit the water?
- How high is the bridge above the river?

24. A uniform border on a framed photo has an area four times that of the photo. What are the outside dimensions of the border if the dimensions of the photo are 30 cm by 20 cm?

25. Jerry jogged 9 km in one hour. He covered the last 4 km at a speed that was 2 km/h slower than his speed over the first 5 km. What was his speed over the first 5 km?



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①

UNIT 1: Absolute Value and Radicals (CH. 1)

1) Arrange from least to Greatest

a) $-|4-7|$, $|-(4-7)|$, $-|5-(-3)|$, $-|4|-|-7|$

-3 , 3 , -8 , -11

③ ④ ② ①

$-|4|-|-7|$, $-|5-(-3)|$, $-|4-7|$, $|-(4-7)|$

2) Evaluate each expression without a calculator

(a) $\sqrt[3]{\frac{27}{8}}$ = $\frac{3}{2}$

(b) $36^{3/2}$ = $\sqrt{36^3}$ = 6^3 = 216

(c) $\sqrt[4]{(x-4)^4}$ = $|x-4|$

↑ absolute value!

(this is because we can only have a positive square root)

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(2)

3) Simplify each radical expression

a) $-3\sqrt{48x^2} + 7\sqrt{75x^2}$

$$-3(4x)\sqrt{3} + 7(5x)\sqrt{3}$$

$$= -12x\sqrt{3} + 35x\sqrt{3} = \boxed{23x\sqrt{3}}$$

b) $\frac{1}{\sqrt{3}}$

$$\frac{1}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} \rightarrow \boxed{\frac{\sqrt{3}}{3}}$$

c) $\frac{1}{\sqrt{14}-2} \times \frac{(\sqrt{14}+2)}{(\sqrt{14}+2)} \rightarrow \text{multiply by difference of squares}$

$$= \frac{\sqrt{14}-2}{14-4} = \boxed{\frac{\sqrt{14}-2}{10}}$$

d) $\sqrt{x^2+4x+4} - \sqrt{x^2+12x-36}$

$$\sqrt{(x+2)^2} - \sqrt{(x-6)^2}$$

$$= |x+2| - |x-6| = x+2 - (x-6) = 8$$

$$\boxed{= 8}$$

e) $\frac{2}{3}\sqrt[3]{54x} + \frac{1}{4}\sqrt[3]{128x} \rightarrow \frac{2}{3}\sqrt[3]{27 \cdot 2x} + \frac{1}{4}\sqrt[3]{64 \cdot 2x}$

$$\frac{2}{3}(3)\sqrt[3]{2x} + \frac{1}{4}(4)\sqrt[3]{2x}$$

$$= 2\sqrt[3]{2x} + 1\sqrt[3]{2x}$$

$$\boxed{= 3\sqrt[3]{2x}} \quad x \geq 0$$

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4) Find the Product and Simplify:

3

(a) $(\sqrt{x+2} + 3)^2$ $(\sqrt{x+2} + 3)(\sqrt{x+2} + 3)$

$$= x+2 + 6\sqrt{x+2} + 9$$

$$= x + 6\sqrt{x+2} + 11$$

$$x \geq -2$$

otherwise
the solution
is NOT real

(b) $\left(\frac{4-\sqrt{32}}{4}\right)^2$

$$\left(\frac{4-4\sqrt{2}}{4}\right)^2 = (1-\sqrt{2})^2 = (1-\sqrt{2})(1-\sqrt{2})$$

$$= 1 - 2\sqrt{2} + 2$$

$$= 3 - 2\sqrt{2}$$

5) Determine the restrictions, solve, and check solutions for extraneous roots.

a. $(\sqrt{10-3x})^2 = (\sqrt{2x+20})^2$

$$10-3x = 2x+20$$

$$-10 = 5x$$

$$x = -2$$

✓ okay as it meets restrictions
(not extraneous)

Restrictions

$$10-3x \geq 0$$

$$2x+20 \geq 0$$

$$-3x \geq -10$$

$$2x \geq -20$$

$$x \leq \frac{10}{3}$$

$$x \geq -10$$

b. $(-\sqrt{x+2})^2 = (2)^2$

$$x+2 = 4$$

$$x = 2 \rightarrow \text{extraneous}$$

$$x+2 \geq 0$$

$$x \geq -2$$

c. $(\sqrt{x+9})^2 = (\sqrt{1-x})^2$

$$x \geq -9, x \leq -1$$

because

$$-\sqrt{4} \neq 2$$

$\rightarrow$ no solution

$$x+9 = 1-x$$

$$2x = -8$$

$$x = -4 \rightarrow \checkmark$$

not extraneous

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UNIT 2: Rational Expressions (CH. 2)

1) Simplify the below rational expressions and state restrictions

a. $\frac{2x^2+x-6}{x^2+4x-5} \cdot \frac{x^3-3x^2+2x}{4x^2-6x}$

$$= \frac{(\cancel{2x-3})(x+2)}{(x+5)(\cancel{x-1})} \cdot \frac{\cancel{x}(x-2)(\cancel{x-1})}{\cancel{2x}(2x-3)}$$

$$= \frac{(x+2)(x-2)}{2(x+5)} \quad x \neq -5$$

b. $\frac{x^2-14x+49}{x^2-49} \div \frac{3x-21}{x+7}$

$$\frac{(x-7)(x-7)}{(x+7)(x-7)} \times \frac{(x+7)}{3(x-7)} = \frac{(\cancel{x-7})(\cancel{x-7})(\cancel{x+7})}{3(\cancel{x+7})(\cancel{x-7})(\cancel{x-7})} = \boxed{\frac{1}{3}}$$

c. $\frac{\left(\frac{x^2-1}{x}\right)}{\frac{(x-1)^2}{x}}$

$$\frac{(\cancel{x-1})(x+1)}{\cancel{x}} \times \frac{\cancel{x}}{(\cancel{x-1})(x-1)}$$

$$= \frac{x+1}{x-1} \quad x \neq 1$$

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2) Simplify and solve for x in the following equations.

$$a) \frac{1}{x-2} + \frac{3}{x+3} = \frac{4}{x^2+x-6}$$

$(x+3)(x-2)$

common denominator is $(x+3)(x-2)$

$$x \neq 2, -3$$

$$\frac{1(x+3)(x-2)}{x-2} + \frac{3(x-2)(x+3)}{x+3} = \frac{4(x+3)(x-2)}{(x+3)(x-2)}$$

$$x+3 + 3(x-2) = 4$$

$$x+3 + 3x - 6 = 4$$

$$4x - 3 = 4$$

$$4x = 7$$

$$x = 7/4$$

$$b) \frac{6}{x+2} - \frac{3}{x^2+x-2} = \frac{x}{x^2+3x+2}$$

$(x+2)(x-1) \quad (x+2)(x+1)$

common denominator

$$(x+2)(x+1)(x-1)$$

$$x \neq -2, -1, 1$$

$$6(x+1)(x-1) - 3(x+1) = x(x-1)$$

$$6x^2 - 6 - 3x - 3 = x^2 - x$$

$$6x^2 - 9 - 3x = x^2 - x$$

$$5x^2 - 2x - 9 = 0$$

$$x = -1.16, 1.56$$

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3. Solve the following word problems:

- a) **DISTANCE Problem:** Ed is a runner and he runs a 8 km loop every day. The first 4 km, he runs at 12km/hr. He runs much slower on the way home. If it takes him 1 hour in total to run the loop, how fast is he running for the last 4 km?

6



	D	R	t
There	4 km	12 km/hr	$\frac{4}{12}$
Back	4 km	x	$\frac{4}{x}$

$$\frac{4}{12} + \frac{4}{x} = 1$$

$$4x + 48 = 12x$$

$$48 = 8x$$

$$x = 6$$

Ed runs
6 km/hr
For the last 4 km

- b) **WORK Problem:** It takes Louise 2 hours to paint a room and it takes Pete 8 hours to paint the same room. How long does it take them if they paint the room together?

$$\frac{1}{2} + \frac{1}{8} = \frac{1}{x}$$

$$4x + x = 8$$

$$5x = 8$$

$$x = 8/5 \text{ hrs}$$

It takes $8/5$ (1.6) hours
to paint together

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- c) The sum of a number and it's reciprocal is $\frac{10}{3}$, what is the number?

(7)

$$\left[x + \frac{1}{x} = \frac{10}{3} \right] \times 3x$$

UNIT 3: Trigonometry (CH 3)

$$3x^2 + 3 = 10x$$

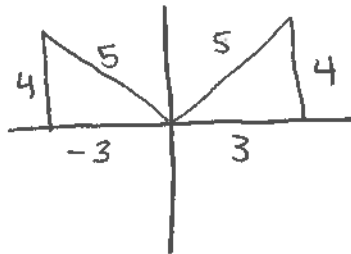
$$3x^2 - 10x + 3 = 0$$

$$(3x-1)(x-3) = 0$$

$$x = \frac{1}{3} \text{ and } 3$$

1. Given the following trigonometric ratios, draw a triangle, find the missing side and use this to solve for the two missing trigonometric ratios ($\sin \theta$, $\cos \theta$ or $\tan \theta$). HINT: there should be two answers for each. (3 marks each)

a) $\sin \theta = \frac{4}{5}$



$$\cos \theta = \pm \frac{3}{5}$$

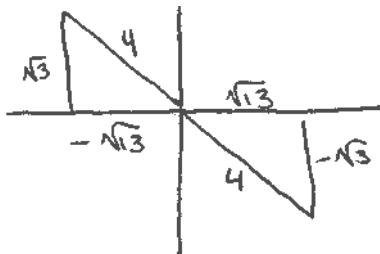
$$\tan \theta = \pm \frac{4}{3}$$

Answer:

$$\cos \theta = \pm \frac{3}{5}$$

$$\tan \theta = \pm \frac{4}{3}$$

b) $\tan \theta = -\frac{\sqrt{3}}{\sqrt{13}}$



Answer:

$$\cos \theta = \pm \frac{\sqrt{13}}{4}$$

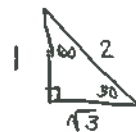
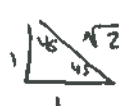
$$\sin \theta = \pm \frac{\sqrt{3}}{4}$$

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2. Evaluate exactly (without a calculator) (1 mark each)



a) $\sin 270^\circ = \boxed{-1}$ $\sin \theta = \frac{y}{r}$ $\cos \theta = \frac{x}{r}$ $\tan \theta = \frac{y}{x}$	b) $\cos(-45^\circ) = \boxed{\frac{1}{\sqrt{2}}}$
c) $\tan 240^\circ = \boxed{\sqrt{3}}$	d) $\frac{\sin 135^\circ}{\cos(-225^\circ)} = \boxed{-1}$

3. Find all θ for $0^\circ \leq \theta \leq 360^\circ$ that satisfy the given equation. There could be more than one answer! Use the unit circle and/or special triangles! (1.5 marks each)

a) $\cos \theta = 0$ $\theta = 90^\circ, 270^\circ$	b) $\tan \theta = -1$ $\theta = 135^\circ, 315^\circ$
c) $\sin \theta = \frac{\sqrt{3}}{2}$ $\theta = 60^\circ, 120^\circ$	d) $\cos \theta = 0.5$ $\theta = 60^\circ, 300^\circ$

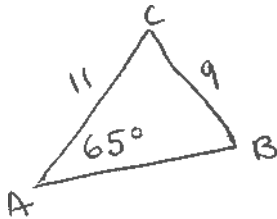
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4. Use the sine law to solve the following triangles. In the case where there is no triangle, write "no solution." In the case where there is two triangles, solve for both. (2 marks each)

a) $\angle A = 65^\circ, a = 9, b = 11$



ASS

$$\frac{\sin 65}{9} = \frac{\sin B}{11}$$

$$\sin B = 1.11$$

NO TRIANGLE

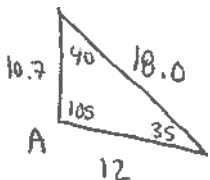
Answer:

NO
TRIANGLE

b) $\angle A = 105^\circ, \angle B = 35^\circ, c = 12\text{cm}$

$$\angle C = 40^\circ$$

AAS



$$\frac{\sin 40}{12} = \frac{\sin 35}{b}$$

$$b = 10.7$$

$$\frac{\sin 40}{12} = \frac{\sin 105}{a}$$

Answer:

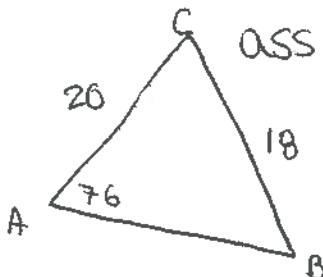
$$b = 10.7$$

$$\angle C = 40^\circ$$

$$a = 18.0$$

c) $\angle A = 76^\circ, a = 25, b = 20$

*change maybe?



ASS

$$\frac{\sin 76}{25} = \frac{\sin B}{20}$$

$$\angle B = 50.1^\circ$$

OR ~~129~~ NO

$$\angle B = 50.1^\circ$$

$$\frac{\sin 53.9}{c} = \frac{\sin 76}{25} \rightarrow c = 20.8$$

Answer:

$$\angle B = 50.1^\circ$$

$$\angle C = 53.9^\circ$$

$$c = 20.8$$

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5. Determine whether or not you need to use the Sine Law or the Cosine Law to solve the triangle. Then, solve the triangle (watch for no solution and 2 solutions!) (3marks each)

a) $a=8, b=10, c=15$

SSS

$$\rightarrow 8^2 = 10^2 + 15^2 - 2(10)(15)\cos A$$

$$8^2 - 10^2 - 15^2 = -300 \cos A$$

$$-261 = -300 \cos A$$

$$0.87 = \cos A$$

$$\angle A = 29.5^\circ$$

$$\frac{\sin 29.5}{8} = \frac{\sin B}{10} \rightarrow \angle B = 38.0^\circ$$

Answer:

$$\angle A = 29.5^\circ$$

$$\angle B = 38.0^\circ$$

$$\angle C = 112.5^\circ$$

b) $\angle A = 25^\circ, a=9, b=20$

ASS

$$\frac{\sin 25}{9} = \frac{\sin B}{20} \Rightarrow \angle B = 69.9^\circ \text{ or } 110.1^\circ \text{ } \left. \begin{array}{l} \Delta 1 \\ \Delta 2 \end{array} \right\} \text{ambiguous case!}$$

$$\Delta 1 \quad \frac{\sin 85.1}{c} = \frac{\sin 25}{9} \quad \text{if } \angle B = 69.9^\circ$$

$$\angle C = 85.1^\circ$$

$$c = 21.2$$

$$\Delta 2 \quad \text{if } \angle B = 110.1^\circ$$

$$\angle C = 44.9^\circ$$

$$c = 15.0$$

Ambiguous Case

Answer:

$$\Delta 1:$$

$$\angle B = 69.9^\circ$$

$$\angle C = 85.1^\circ$$

$$c = 21.2$$

$$\Delta 2:$$

$$\angle B = 110.1^\circ$$

$$\angle C = 44.9^\circ$$

$$c = 15.0$$

c) $a=14, b=12, \angle C=35^\circ$

SSA

$$c^2 = 14^2 + 12^2 - 2(12)(14)\cos 35$$

$$c = 8.05$$

$$\angle B \rightarrow \frac{\sin B}{12} = \frac{\sin 35}{8.05}$$

$$\angle B = 59.4^\circ$$

$$\angle A = 85.6^\circ$$

Answer:

$$\angle A = 85.6^\circ$$

$$\angle B = 59.4^\circ$$

$$c = 8.05$$

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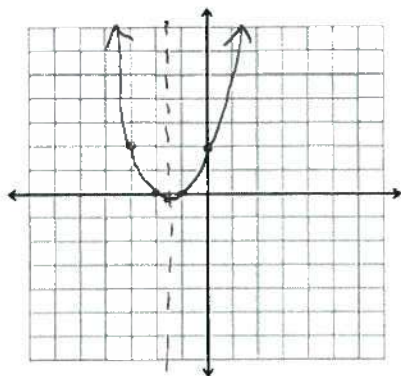
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UNIT 4: Factoring and Quadratic Functions (Ch.5)

1. Sketch the graph of the quadratic function. Identify the vertex, axis of symmetry, and x-intercept(s), domain, range and state the maximum or minimum value.

(a) $f(x) = x^2 + 3x + 2$



$$f(x) = (x+2)(x+1)$$

x-int @ $x = -2, x = -1$

$(-2, 0) (-1, 0)$

y-int @ $y = 2 \rightarrow (0, 2)$

Vertex @ $(-1.5, -0.25)$

$(-3/2, -1/4)$

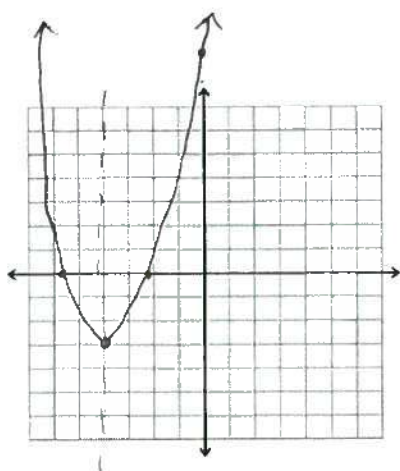
Domain: $x \in \mathbb{R}$

Range: $y \geq -1/4$

Min value: $y = -1/4$

Axis of Symmetry: $x = -3/2$

(b) $f(x) = (x+4)^2 - 3$



Vertex: $(-4, -3)$

y-int: $y = 13 \quad (0, 13)$

x-int: $(-2.27, 0) (-5.73, 0)$

Axis of Symmetry $x = -4$

Min value: $y = -3$

Domain: $x \in \mathbb{R}$

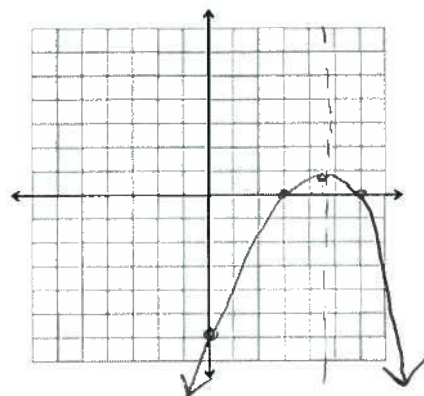
Range: $y \geq -3$

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(c) $m(x) = -\frac{1}{3}x^2 + 3x - 6$



y-int: $(0, -6)$

x-int: $(3, 0)$ & $(6, 0)$

Vertex: $(\frac{9}{2}, \frac{3}{4})$

Max Value: $y = \frac{3}{4}$

Axis of Symmetry: $x = \frac{9}{2}$

Domain: $x \in \mathbb{R}$

Range: $y \leq \frac{3}{4}$

2. Write the standard form of the equation of the parabola that has the indicated vertex and whose graph passes through the given point.

- a) Vertex: $(-2, 5)$; Point: $(0, 9)$

$$f(x) = a(x+2)^2 + 5$$

$$9 = a(0+2)^2 + 5$$

$$9 = 4a + 5$$

$$4 = 4a$$

$$a = 1$$

$$f(x) = 1(x+2)^2 + 5$$

- b) Vertex: $(\frac{5}{2}, -\frac{3}{4})$; x-intercept = $-2 \rightarrow (-2, 0)$

$$f(x) = a(x - \frac{5}{2})^2 - \frac{3}{4}$$

$$f(x) = \frac{1}{27}(x - \frac{5}{2})^2 - \frac{3}{4}$$

$\rightarrow$ Plug in $(-2, 0)$ $0 = a(-2 - \frac{5}{2})^2 - \frac{3}{4}$

$$\frac{3}{4} = 20.25a \rightarrow a = \frac{1}{27}$$

3. Write the standard form of the quadratic function that passes through the following 3 points. $(0, 2), (6, 2), (-8, 4)$

Axis of symmetry = $\frac{6-0}{2} = 3$

vertex = $(3, k)$

$$f(x) = \frac{1}{56}(x-3)^2 + \frac{103}{56}$$

$$f(x) = a(x-3)^2 + k$$

$$2 = a(-3)^2 + k$$

$$k = 2 - 9a \quad k = \frac{103}{56}$$

$$f(x) = a(x-3)^2 + 2 - 9a$$

$$4 = a(-8-3)^2 + 2 - 9a$$

$$4 = 112a + 2 \quad a = \frac{1}{56}$$

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4. What is the maximum area of a rectangle that can be constructed with a perimeter of 64 cm?



$$2l + 2w = 64$$

(13)

$$l = \frac{64 - 2w}{2} \rightarrow l = 32 - w$$

$$\text{Area} \rightarrow A(l) = w(32 - w)$$

$$A(l) = -w^2 + 32w$$

$$A(l) = -1(w^2 - 32w + 256) + 256$$

$$= -1(w - 16)^2 + 256$$

The maximum area is 256 cm²

UNIT 5: Solving Quadratic Equations (CH. 6)

max area occurs when $w = 16$

- 1) Solve the following Quadratic Equations (use the method of your choice)

a. $(x + 13)^2 = 25$

$$x + 13 = \pm 5$$

$$x = 5 - 13 = -8$$

$$x = -5 - 13 = -18$$

$$x = -8, -18$$

b. $(2x + 3)^2 - 27 = 0$

$$(2x + 3)^2 = 27$$

$$2x + 3 = \pm \sqrt{27}$$

$$2x + 3 = \pm 3\sqrt{3}$$

$$2x = -3 \pm 3\sqrt{3}$$

$$x$$

c. $(x - 7)^2 = (x + 3)^2$

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d. $\frac{1}{8}x^2 - x - 16 = 0$

$$\frac{1}{8}(x^2 - 8x + \underline{\quad}) - 16 = 0$$

$$\frac{1}{8}(x^2 - 8x + 16) - 16 - 2 = 0$$

$$\frac{1}{8}(x-4)^2 - 18 = 0$$

$$(x-4)^2 = 144$$

$$\sqrt{(x-4)^2} = \sqrt{144}$$

$$x-4 = \pm 12$$

$$x = 4 \pm 12$$

$$x = 16, -8$$

e. $3x^2 + 24x + 16 = 0$

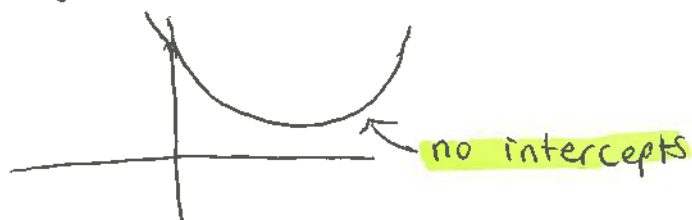
$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-24 \pm \sqrt{24^2 - 4(3)(16)}}{2(3)} = \frac{-24 \pm \sqrt{384}}{6} = \frac{-24 \pm 8\sqrt{6}}{6}$$

$$x = -0.73, -7.3$$

← OR →

$$= \frac{-12 \pm 4\sqrt{6}}{3}$$

f. $\frac{1}{4}x^2 - 2x + 7 = 0$

Solve by graphing: NO SOLUTION

g. $12x - 9x^2 = -3$

$$-9x^2 + 12x + 3 = 0$$

$$-3(3x^2 + 4x + 1) = 0$$

$$-3(3x+1)(x+1) = 0$$

$$x = -\frac{1}{3}, -1$$

h. $25x^2 + 80x + 61 = 0$

$$x = \frac{-80 \pm \sqrt{80^2 - 4(25)(61)}}{2(25)} = \frac{-80 \pm \sqrt{360}}{50} = \frac{-80 \pm 10\sqrt{3}}{50}$$

$$x = \frac{-8 \pm \sqrt{3}}{5} = -1.25, -1.94$$

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i. $3x + 4 = 2x^2 - 7$

$2x^2 - 3x - 11 = 0$

* Solved by graphing

$x = -1.71, 3.21$

a. $2x^2 - 3x = 4x + 12$

$2x^2 - 7x - 12 = 0$

$x = 4.76, -1.26$

2. Brian decides to start training for swimming in a river. The current in the river is 4 km/hr. If he swims upstream 2 km and then back downstream to where he started in 3 hours, what is his swimming speed?

X = Brian's swimming speed

total time = 3 = time_{up} + time_{down}



	R	D	t
Up	$x - 4$	2 km	$\frac{2}{x - 4}$
Down	$x + 4$	2 km	$\frac{2}{x + 4}$

$$\frac{2}{x + 4} + \frac{2}{x - 4} = 3$$

$$\hookrightarrow 2(x - 4) + 2(x + 4) = 3(x - 4)(x + 4)$$

$$2x - 8 + 2x + 8 = 3(x^2 - 16)$$

$$4x = 3x^2 - 48$$

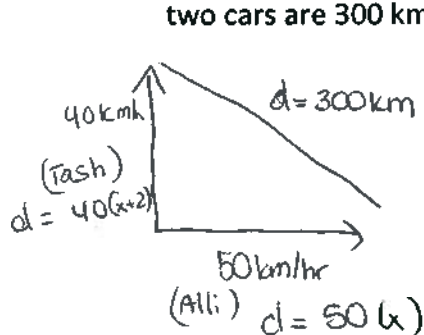
$$3x^2 - 4x - 48 = 0$$

$x = 4.7 \text{ km/hr}$

Brian's swimming speed is 4.7 km/hr

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3. Natasha leaves school at 3pm and she drives north at 40 km/hr. 2 hours later (at 5pm), Alli leaves and she drives East at 50 km/hr. How long does it take before the two cars are 300 km apart?



$$t_{\text{Alli}} = x$$

$$t_{\text{Natasha}} = x + 2$$



Pythagoras
Theorem

$$(40(x+2))^2 + (50x)^2 = 300^2$$

$$(40x+80)^2 + 2500x^2 = 90000$$

$$1600x^2 + 6400x + 6400 + 2500x^2 = 90000$$

$$4100x^2 + 6400x - 83600 = 0$$

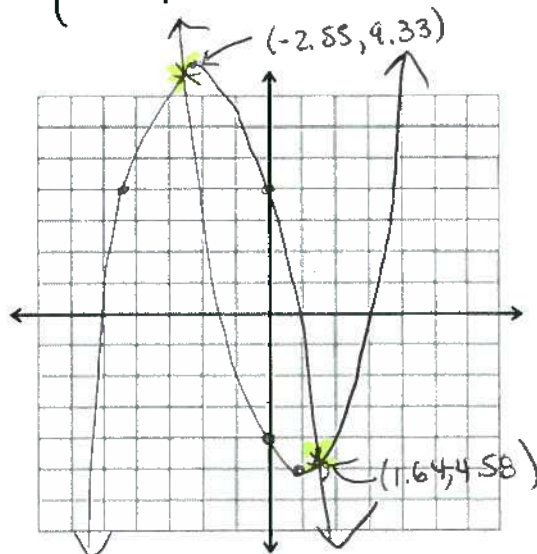
$$41x^2 + 64x - 836 = 0$$

$$x = 3.80 \text{ hrs} \quad \text{total time} =$$

$$= 5.8 \text{ hrs}$$

1. Solve the system of equations and inequalities by graphing. If doing on calculator, sketch an accurate graph.

a)
$$\begin{cases} y = x^2 - 2x - 4 \\ y = -\frac{3}{4}x^2 - 4x + 4 \end{cases}$$



$$(-2.55, 9.32)$$

$$(1.64, -4.58)$$

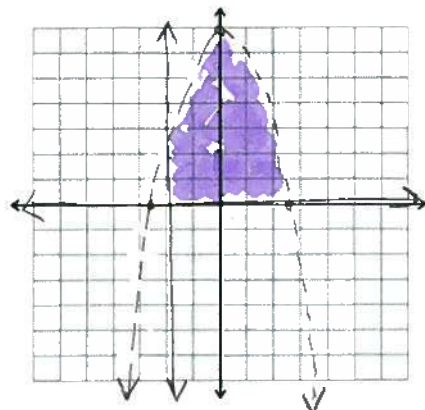
It takes
5.8 hrs for
the cars to
be 300 km apart

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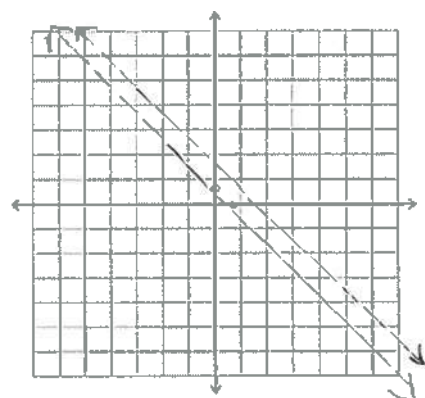
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b)
$$\begin{cases} x^2 + y \leq 7 \\ x \geq -2 \\ y \geq 0 \end{cases}$$



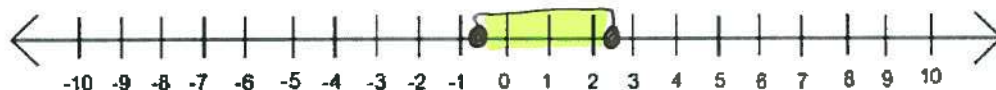
c)
$$\begin{cases} 2x + y > 2 \\ 6x + 3y < 2 \end{cases} \quad y < \frac{2}{3} - \frac{6x}{3}$$



NO SOLUTION

d) $-2x^2 + 3x + 4 \geq 0$

$$x = \frac{-3 \pm \sqrt{9 - 4(4)(-2)}}{2(-2)} = \frac{-3 \pm \sqrt{41}}{-4} = -0.85 \text{ or } 2.35$$



Check $x=0$
 $4 \geq 0$ yes

* NOTICE FILLED IN CIRCLE BEC

$-0.85 \leq x \leq 2.35$

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2. Solve the following systems algebraically:

$$a) \begin{cases} y = -x^2 - 4x + 5 \\ y = -3x + 7 \end{cases}$$

$$-x^2 - 4x + 5 = -3x + 7$$

$$x^2 + x + 2 = 0$$

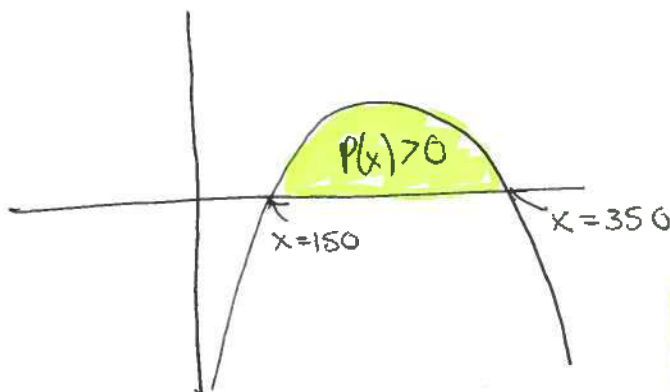
$$x = \frac{-1 \pm \sqrt{1 - 4(1)(2)}}{2(1)} \leftarrow \text{NO SOLUTION}$$

$$b) \begin{cases} (-7x + 6y = -4)^{x^2} \\ 14x - 12y = 8 \end{cases} \quad \begin{matrix} -14x + 12y = -8 \\ 14x - 12y = 8 \end{matrix} \quad \left. \begin{matrix} \\ \end{matrix} \right\} \begin{matrix} \text{Infinite \# of} \\ \text{solutions (same line)} \end{matrix}$$

- a) The profit for a construction company is $P(x) = -0.1x^2 + 50x - 5250$, where x is the total number of hours worked by the employees in a week. What total hours worked by the employees will produce a profit for the company?

Graphing calculator $P(x) > 0$

$$-0.1x^2 + 50x - 5250 > 0$$



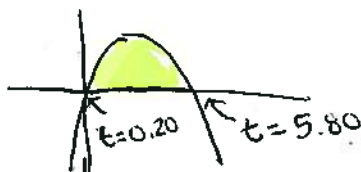
$$150 < x < 350$$

The construction company makes a profit when the employees work between 150 and 350 hours

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- b) The height in metres of a ball thrown upward from a building is $h(t) = -4.9t^2 + 29.4t + 24.3$, where "t" is the time in seconds after releasing the ball. During what time interval will the ball be above 30 meters? Graphing calculator



$$30 < -4.9t^2 + 29.4t + 24.3$$

$$0 < -4.9t^2 + 29.4t - 5.7$$

$$\rightarrow 0.2 < t < 5.8$$

the ball will be above 30 meters

UNIT 7: Absolute Value, Rational and Reciprocal Functions (CH. 4)

between 0.2 and

5.8 seconds

1. Write the absolute value function as a piecewise function:

a) $g(x) = -4|x + 2| + 3$

$$g(x) = \begin{cases} -4(x+2) + 3, & x \geq -2 \\ +4(x+2) + 3, & x < -2 \end{cases}$$

$$g(x) = \begin{cases} -4x - 5, & x \geq -2 \\ 4x + 11, & x < -2 \end{cases}$$

b) $f(x) = \frac{1}{3}|2x - 7| + 9$

$$f(x) = \begin{cases} \frac{1}{3}(2x - 7) + 9, & x \geq \frac{7}{2} \\ -\frac{1}{3}(2x - 7) + 9, & x < \frac{7}{2} \end{cases}$$

$$\left. \begin{array}{l} 2x - 7 \geq 0 \\ 2x \geq 7 \\ x \geq \frac{7}{2} \end{array} \right\} \begin{array}{l} 2x - 7 < 0 \\ x < \frac{7}{2} \end{array}$$

$$f(x) = \begin{cases} \frac{2x}{3} - \frac{7}{3} + \frac{27}{3}, & x \geq \frac{7}{2} \\ -\frac{2x}{3} + \frac{7}{3} + \frac{27}{3}, & x < \frac{7}{2} \end{cases}$$

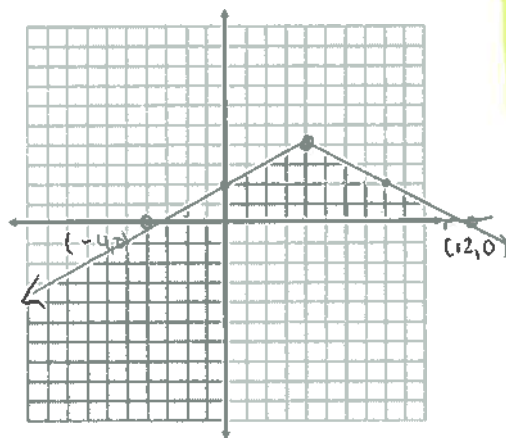
$$\Rightarrow f(x) = \begin{cases} \frac{2x}{3} + \frac{20}{3}, & x \geq \frac{7}{2} \\ -\frac{2x}{3} + \frac{34}{3}, & x < \frac{7}{2} \end{cases}$$

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2. Graph the absolute value function. State the vertex, intercepts (x and y), domain and range.

a) $h(x) = -\frac{1}{2}|x - 4| + 4$



Vertex (4, 4)
 y-intercept (0, 2)
 x-intercept (-4, 0) (12, 0)
 Domain: $x \in \mathbb{R}$
 Range: $y \leq 4$

3. Solve the following absolute value functions:

a) $|x+3| = -3x$

① $x+3 = -3x$

$3 = -4x$

$x = \frac{3}{-4}$

✓
 yes
 $2.25 = 2.25$

② $x+3 = 3x$

$3 = 2x$

$x = \frac{3}{2}$

X No → extraneous
 ↓

$|3/2 + 3| \neq -3(3/2)$

$4.5 \neq -4.5$

ONLY $x = -3/4$

b) $|x-3| = |2x+4|$

case ① → $x-3 = 2x+4$

$-7 = x$

check: $|-10| = |-10|$ ✓ yes

case ② → $-x+3 = 2x+4$

$-1 = 3x$
 $x = -1/3$

check: $|-10/3| = |-10/3|$ → yes

case ③ → $x-3 = -2x-4$

$3x = -1$

$x = -1/3$ → yes!

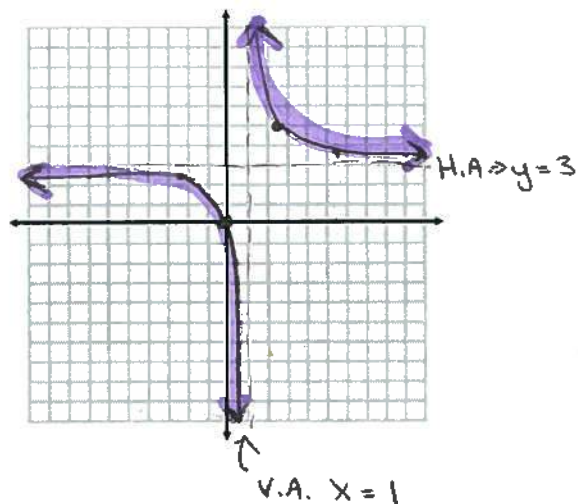
$x = -7, -1/3$

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4. Sketch the following graphs and label both the horizontal and vertical asymptotes. State the domain of each function.

a) $f(x) = \frac{3x}{x-1}$



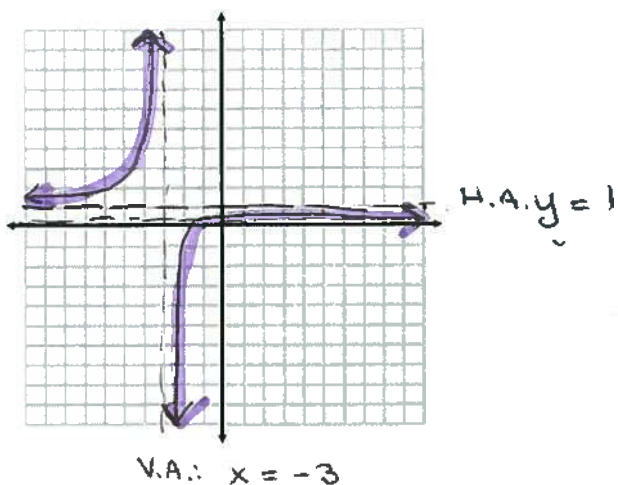
$$V.A. \Rightarrow x = 1$$

$$H.A. \Rightarrow y = 3$$

b) $h(x) = \frac{x^2-1}{x^2+2x-3} = \frac{(x-1)(x+1)}{(x+3)(x-1)} = \frac{x+1}{x+3}$

$$H.A. \rightarrow y = 1$$

$$V.A. \rightarrow x = -3$$

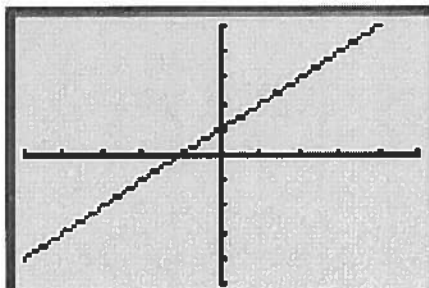


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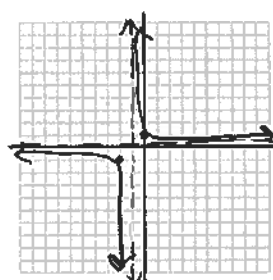
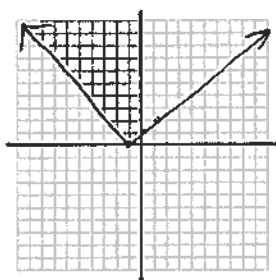
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5. Given the following graph of $y = f(x)$, sketch the graph of $y = \frac{1}{f(x)}$ and the graph of $y = |f(x)|$

a)



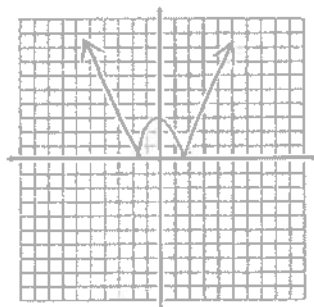
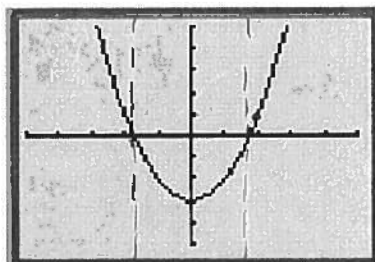
$$y = |f(x)|$$



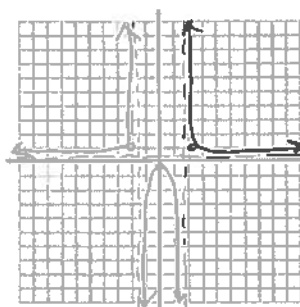
$$\text{H.A. } y = 0$$

$$\text{V.A. } x = -1$$

b)



$$y = |f(x)|$$



$$\text{H.A. } y = 0$$

$$\text{V.A. } \left. \begin{array}{l} x \approx -1.8 \\ x \approx 1.8 \end{array} \right\} \text{approx}$$

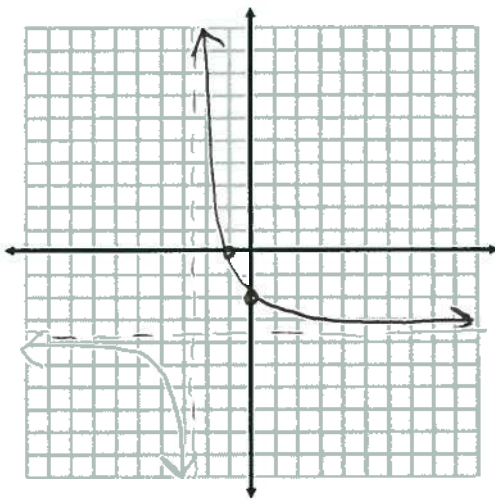
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6. Sketch a graph of a rational function that has the following characteristics.

a)

x-intercept	-1
y-intercept	-2
Vertical Asymptote	$x = -2.5$
Horizontal Asymptote	$y = -3.5$



UNIT 8: Sequences and Series (CH. 8)

1. Given the series defined by $\sum_{k=2}^{15} 16 \left(\frac{1}{2}\right)^{k-1}$, determine the common ratio, the number of terms and the sum.

$$\begin{aligned} a_1 &= 16 \left(\frac{1}{2}\right)^2 \\ a_1 &= 8 \end{aligned}$$

$$\begin{aligned} a_2 &= 16 \left(\frac{1}{2}\right)^{3-1} \\ a_2 &= 16 \left(\frac{1}{4}\right) = 4 \end{aligned}$$

$$r = \frac{a_2}{a_1} = \frac{4}{8} = \frac{1}{2}$$

$$\text{number of terms} = 15 - 2 + 1 = 14$$

$$S_n = \frac{a(1-r^n)}{1-r} = \frac{8(1-(\frac{1}{2})^{14})}{1-\frac{1}{2}} =$$

$$S_{14} = 15.999$$

Name: _____

2. Find the 14
- th
- term of the sequence {3, 11, 19, ...}

$d = 8$

$a = 1$

$n = 14$

$$t_n = a + (n-1)d$$

$$t_{14} = 1 + (14-1)8$$

$$t_{14} = 1 + 13 \times 8$$

$$t_{14} = 105$$

3. Three consecutive terms of a geometric sequence are 2.5,
- $y+3$
- and 9.6. Find the value of
- y
- .

$$\frac{y+3}{2.5} = \frac{9.6}{y+3}$$

$$y^2 + 6y + 9 = (9.6)(2.5)$$

$$y^2 + 6y + 9 = 24$$

$$y^2 + 6y - 15 = 0 \rightarrow \text{solved by graphing / reject -ve ans}$$

$$y = 1.9$$

or
 -7.9

$$(y+3)(y+3) = (9.6)(2.5)$$

4. Compute the sum of the first 8 terms in the sequence {1, -3, 9, ...}

$$r = \frac{-3}{1} = -3$$

$$n = 8$$

$$a = 1$$

$$S_n = \frac{a(1-r^n)}{1-r}$$

$$= \frac{1(1-(-3)^8)}{1-(-3)}$$

$$S_8 = -1640$$

5. How can you tell whether or not an infinite geometric series has a finite or an infinite sum?

$$|r| < 1 \rightarrow$$

$$-1 < r < 1$$

and

$$r \neq 0$$

The common difference must be between -1 and 1 and cannot = 0

6. Find the 18
- th
- term in an arithmetic sequence who's 2
- nd
- term is 11 and who's 8
- th
- term is 41.

$$t_2 = 11 = a + (2-1)d$$

$$t_8 = 41 = a + (8-1)d$$

$$\rightarrow 11 = a + d$$

$$- (41 = a + 7d)$$

$$- 30 = -6d$$

$$d = 5$$

plug in to find t_{18}

$$a = 6$$

$$t_{18} = 6 + (18-1)5$$

$$t_{18} = 6 + 17(5)$$

$$t_{18} = 91$$

Name: _____

7. Find the 9th term in a geometric sequence who's first term is 6 and who's 4th term is $-\frac{3}{4}$.

$$t_1 = a = 6$$

$$t_4 = ar^{n-1}$$

$$t_4 = 6(r)^{4-1} \rightarrow * t_4 = -\frac{3}{4}$$

$$-\frac{3}{4} = 6(r)^3$$

$$-\frac{3}{24} = r^3 \quad -\frac{1}{8} = r^3 \rightarrow \boxed{r = -\frac{1}{2}}$$

$$t_9 = 6\left(-\frac{1}{2}\right)^{9-1}$$

$$t_9 = \frac{3}{128}$$

8. Calculate the sum of the infinite geometric series given by

$$a = 8\left(-\frac{1}{2}\right)^{2-1}$$

$$(k=2)$$

$$= 8\left(-\frac{1}{2}\right) = -4$$

$$\sum_{k=2}^{\infty} 8\left(-\frac{1}{2}\right)^{k-1}$$

$$a_2 = 8\left(-\frac{1}{2}\right)^{3-1} = 8\left(\frac{1}{4}\right) = 2$$

$$(k=3)$$

$$r = \frac{2}{-4} = -\frac{1}{2}$$

$$S_{\infty} = \frac{a}{1-r} = \frac{-4}{1-(-\frac{1}{2})} = -1.6$$

$$\boxed{S_{\infty} = -\frac{8}{5}}$$

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Formulas

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \rightarrow \text{quadratic formula}$$

$$x = -\frac{b}{2a}$$

$$\left. \begin{array}{l} x = -\frac{b}{2a} \\ y = c - \frac{b^2}{4a} \end{array} \right\} \begin{array}{l} \text{vertex Formula} \\ \text{vertex} = (x, y) \\ = \left(-\frac{b}{2a}, c - \frac{b^2}{4a}\right) \end{array}$$

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c} \left\} \text{sine Law}\right.$$

$$a^2 = b^2 + c^2 - 2bc \cos A \left\} \text{cosine law}\right.$$

$$t_n = a + (n-1)d \left\} \begin{array}{l} n^{\text{th}} \text{ term of} \\ \text{arithmetic sequence/series} \end{array}\right.$$

$$S_n = \frac{n}{2}(2a + (n-1)d) \left\} \begin{array}{l} \text{sum of arithmetic} \\ \text{sequence/series} \end{array}\right.$$

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$$t_n = ar^{n-1} \quad \left. \vphantom{t_n = ar^{n-1}} \right\} \begin{array}{l} n^{\text{th}} \text{ term of geometric} \\ \text{sequence / series} \end{array}$$

$$\begin{array}{l} * \\ \text{Finite} \end{array} \quad S_n = \frac{a(1-r^n)}{1-r} \quad \left. \vphantom{S_n = \frac{a(1-r^n)}{1-r}} \right\} \begin{array}{l} \text{sum of geometric (finite)} \\ \text{sequence / series} \end{array}$$

$$\begin{array}{l} * \\ \text{infinite} \end{array} \quad S_n = \frac{a}{1-r} \quad \left. \vphantom{S_n = \frac{a}{1-r}} \right\} \begin{array}{l} \text{sum of infinite} \\ \text{geometric sequence / series} \end{array}$$

Name: _____ Date: _____ Period: _____

Unit 3-B Review – Trigonometric Functions

Reviews do NOT cover all material from the lessons but will hopefully remind you of key points. To be prepared, you must study all packets for lessons 3.8 – 3.15.

Write an equation that represents all asymptotes of the graph of f in the xy -plane.

1. Let $f(\theta) = \tan\left(\frac{\theta}{6}\right)$.

In the xy -plane, the angle θ is in standard position. What is the slope of the terminal ray of the angle?

2. $\theta = \frac{7\pi}{4}$

Evaluate.

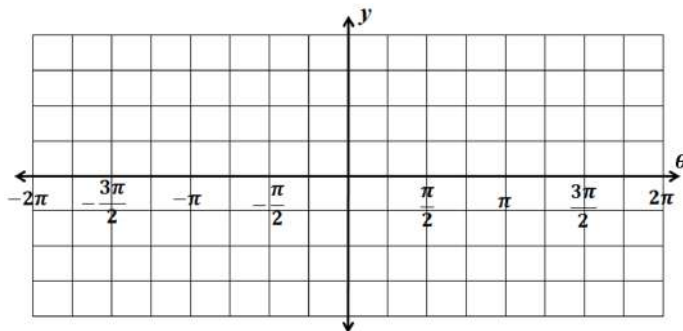
3. $\tan \frac{\pi}{6}$

4. $\tan \frac{3\pi}{2}$

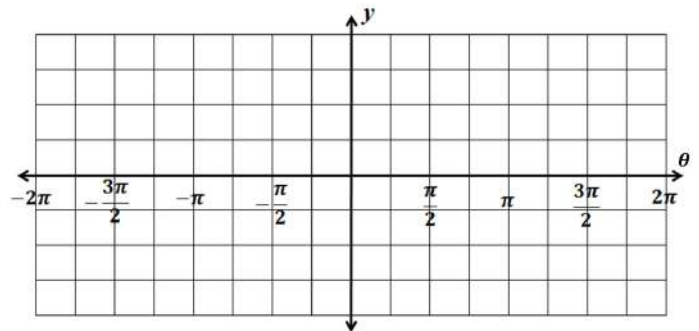
5. $\tan \frac{4\pi}{3}$

Graph each trig function.

6. $g(\theta) = \tan\left(\theta + \frac{\pi}{4}\right) - 1$

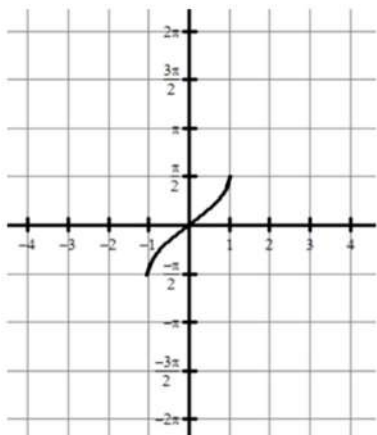


7. $g(\theta) = -2 \tan(2\theta) + 3$

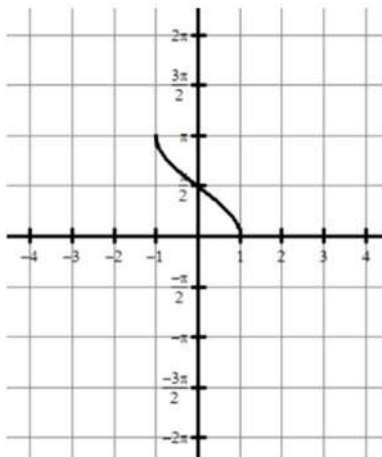


The parent function is shown below. Use the parent function to graph $g(x)$.

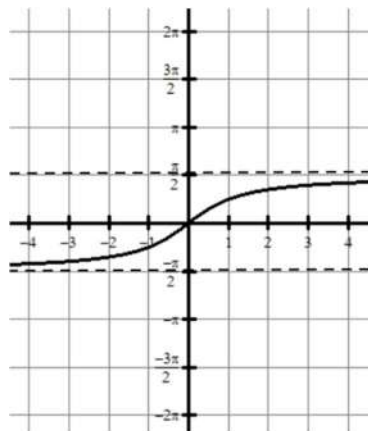
8. $g(x) = 3\sin^{-1}(-x) + \frac{\pi}{2}$



9. $g(x) = 2\cos^{-1}(x - 3)$



10. $g(x) = -\tan^{-1}(x) - \pi$



Find the inverse of each function and list the domain and range of $f^{-1}(x)$.

11. $f(x) = 2\sin x - 5$ for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$

Solve each equation for $0 \leq \theta \leq 2\pi$. Find the exact value(s) using the unit circle

12. $2\sin \theta + 5 = 4$

13. $\cos^2 \theta + \cos \theta = 0$

Solve each equation. Find ALL approximate value(s) using a calculator.

14. $5 + 5\sin^2 x = 8$

15. $\tan^2 x - 3\tan x = 18$

Evaluate the following expressions. Use exact values.

16. $\sec\left(\frac{\pi}{3}\right)$

17. $\csc\left(-\frac{\pi}{2}\right)$

18. $\cot\left(\frac{5\pi}{6}\right)$

Evaluate the following expressions. Use approximate values from calculator.

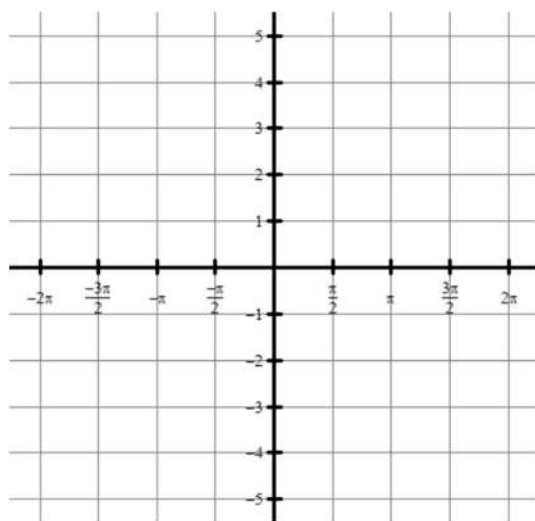
19. $\csc(1.84)$

20. $\sec\left(\frac{3\pi}{5}\right)$

21. $\cot\left(\frac{4\pi}{9}\right)$

Graph the following. State the range and all vertical asymptotes.

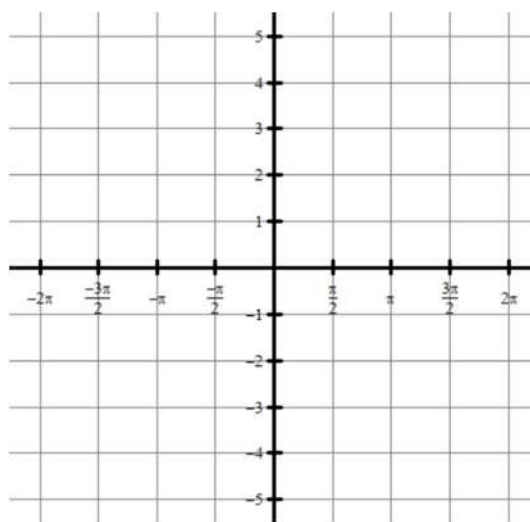
22. $f(x) = 2\csc\left(x + \frac{\pi}{2}\right) - 1$



Range:

Vertical Asymptotes:

23. $f(x) = 3\sec\left(\frac{1}{2}x\right)$



Range:

Vertical Asymptotes:

Use trig identities to write each expression in terms of a single trig identity.

24. $\cos^2 x \sec x$

25. $(1 - \sin^2 x) \csc^2 x$

26. $\frac{1}{1 + \cot^2 x}$

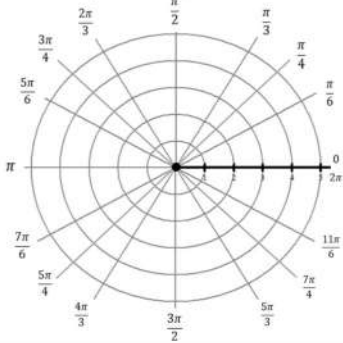
Find the exact value of the sum or difference.

27. $\sin\left(\frac{3\pi}{4} - \frac{5\pi}{6}\right)$

28. $\cos\left(\frac{\pi}{2} + \frac{2\pi}{3}\right)$

Polar Coordinates

29. Plot $\left(-3, \frac{2\pi}{3}\right)$



30. Convert from polar to rectangular.

$\left(4, \frac{5\pi}{6}\right)$

31. Convert from rectangular to polar where $0 \leq \theta \leq 2\pi$.

$(2, -4)$

32. Convert from rectangular complex to polar.

$-3 - 5i$

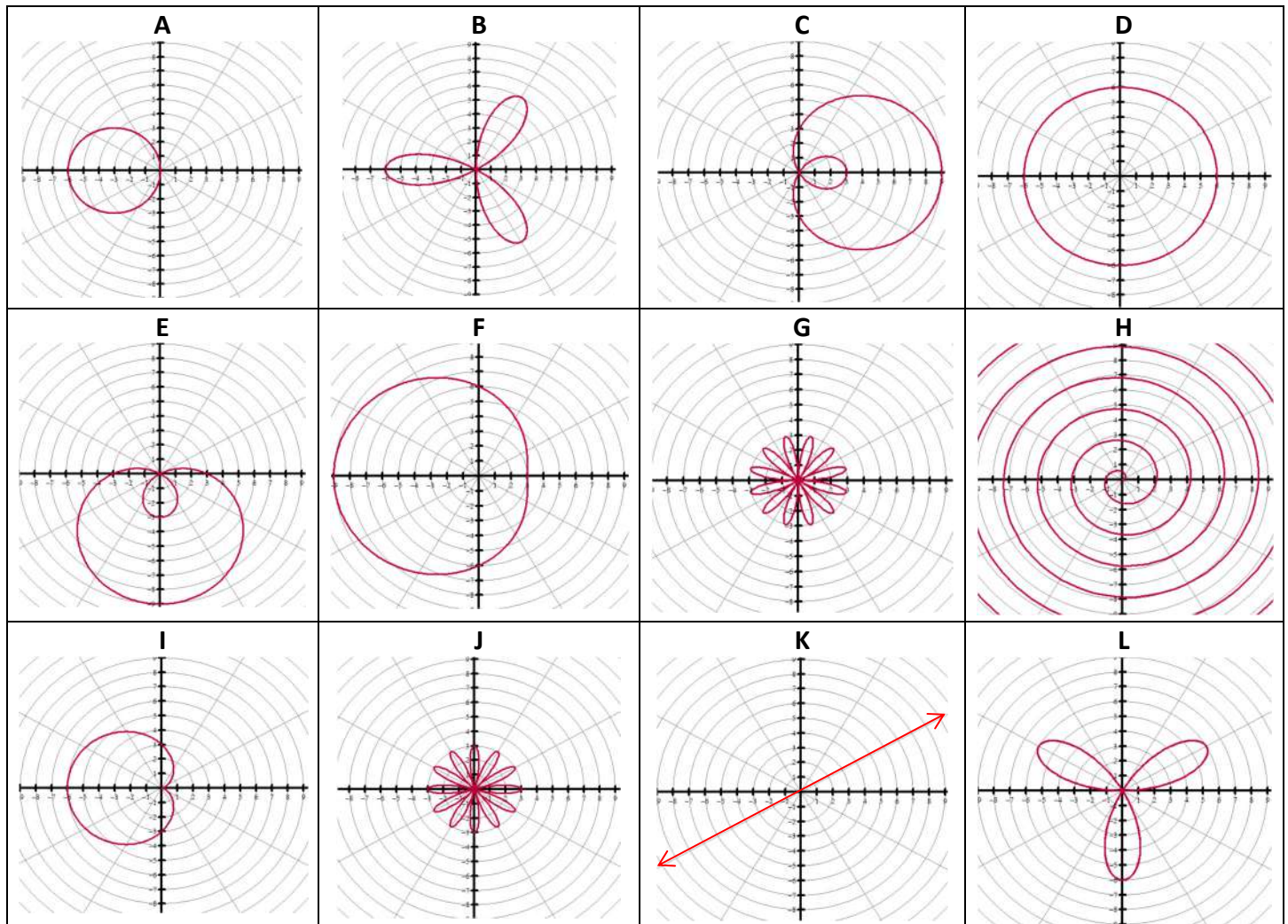
33. Convert from polar complex to rectangular.

$6 \left[\cos\left(\frac{3\pi}{4}\right) + i \sin\left(\frac{3\pi}{4}\right) \right]$

Match the equation to its graph below.

34. $r = 3 + 6 \cos(\theta)$ matches graph: _____	35. $r = 3 - 6 \sin(\theta)$ matches graph: _____	36. $r = 3 - 3 \cos(\theta)$ matches graph: _____
37. $r = 6 - 3 \cos(\theta)$ matches graph: _____	38. $r = -6 \cos(\theta)$ matches graph: _____	39. $r = 6 \sin(3\theta)$ matches graph: _____
40. $r = 3 \cos(6\theta)$ matches graph: _____	41. $r = \frac{\theta}{3}$ matches graph: _____	42. $r = -6 \cos(3\theta)$ matches graph: _____
43. $r = 3 \sin(6\theta)$ matches graph: _____	44. $r = 6$ matches graph: _____	45. $\theta = \frac{\pi}{6}$ matches graph: _____

GRAPHS:



Use the table of selected values for the polar function $r = f(\theta)$ to answer the following.

46.

- a. Is f increasing or decreasing on the interval $0 \leq \theta \leq \frac{\pi}{2}$?
- b. Is the distance between $f(\theta)$ and the pole is increasing or decreasing on the interval $0 \leq \theta \leq \frac{\pi}{2}$?
- c. Find the average rate of change of f between $\theta = \frac{\pi}{8}$ and $\theta = \frac{3\pi}{8}$.

θ	r
0	-5
$\frac{\pi}{8}$	-4.619
$\frac{\pi}{4}$	-3.536
$\frac{3\pi}{8}$	-1.913
$\frac{\pi}{2}$	0

- d. Estimate the value of $f\left(\frac{\pi}{4}\right)$ using an average rate of change.

Multiple Choice

47. The function f is given by $f(x) = 4 \sin(x) + 1$. For what values of x where $0 \leq x \leq 2\pi$ is $f(x) \leq -1$?
- (A) $\frac{7\pi}{6} \leq x \leq \frac{11\pi}{6}$
- (B) $\frac{4\pi}{3} \leq x \leq \frac{5\pi}{3}$
- (C) $0 \leq x \leq \frac{7\pi}{6}$ and $\frac{11\pi}{6} \leq x \leq 2\pi$
- (D) $0 \leq x \leq \frac{4\pi}{3}$ and $\frac{5\pi}{3} \leq x \leq 2\pi$